

SAARTHI

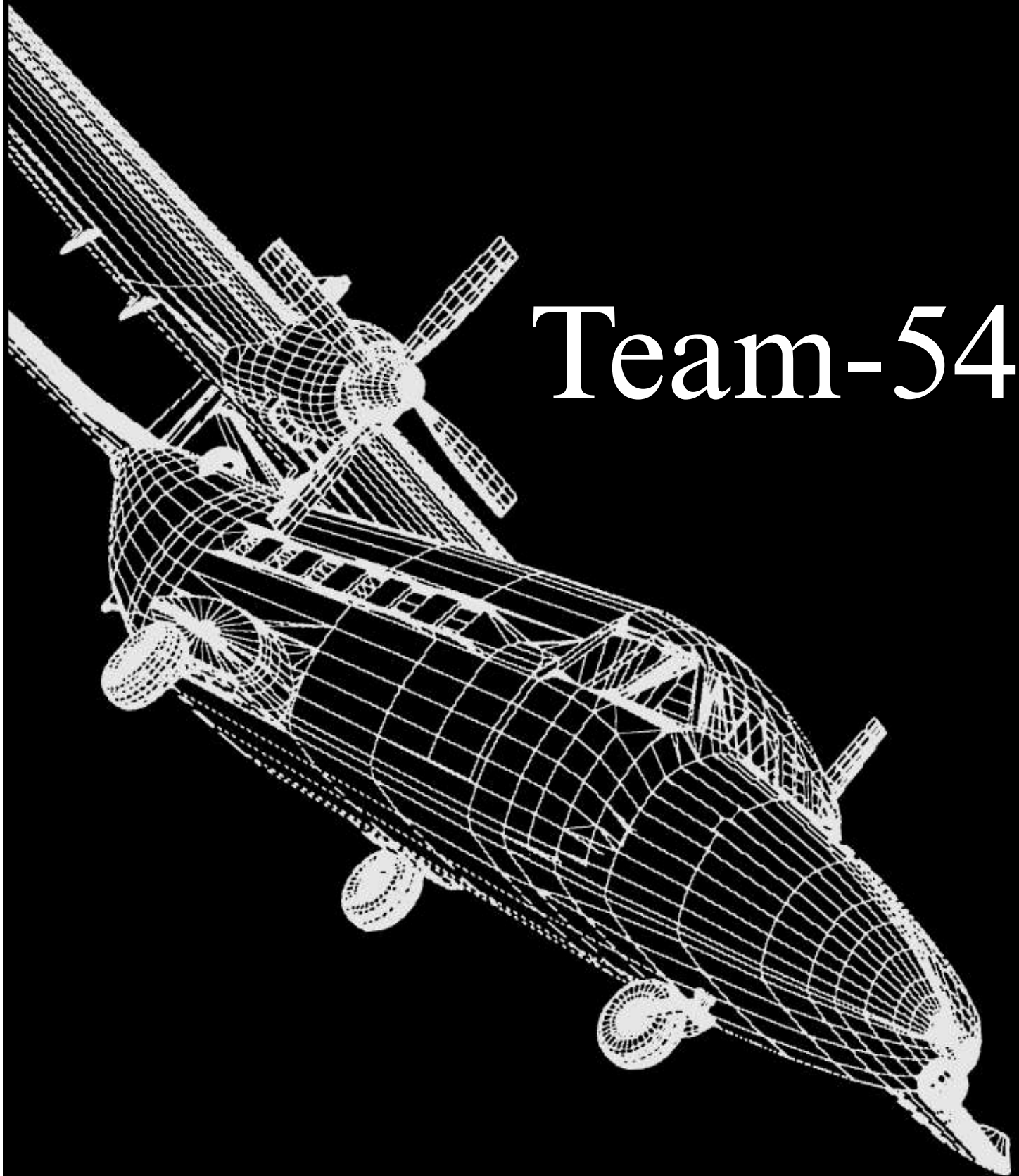
*Synchronized Aerodynamic Augmentation
using Rotational Thrust for High-Lift Integration*

STOL Aircraft for LAT



Inter IIT
Tech Meet 14.0

Team-54



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1 Executive Summary

1.1 Overview of the Design Approach

The design approach involves a comprehensive analysis of aerodynamic properties to ensure optimal performance for the target flight conditions. The workflow has a bilateral approach working on wing and propellers simultaneously, starting in either one with mathematical modelling, followed by design and CFD analysis.

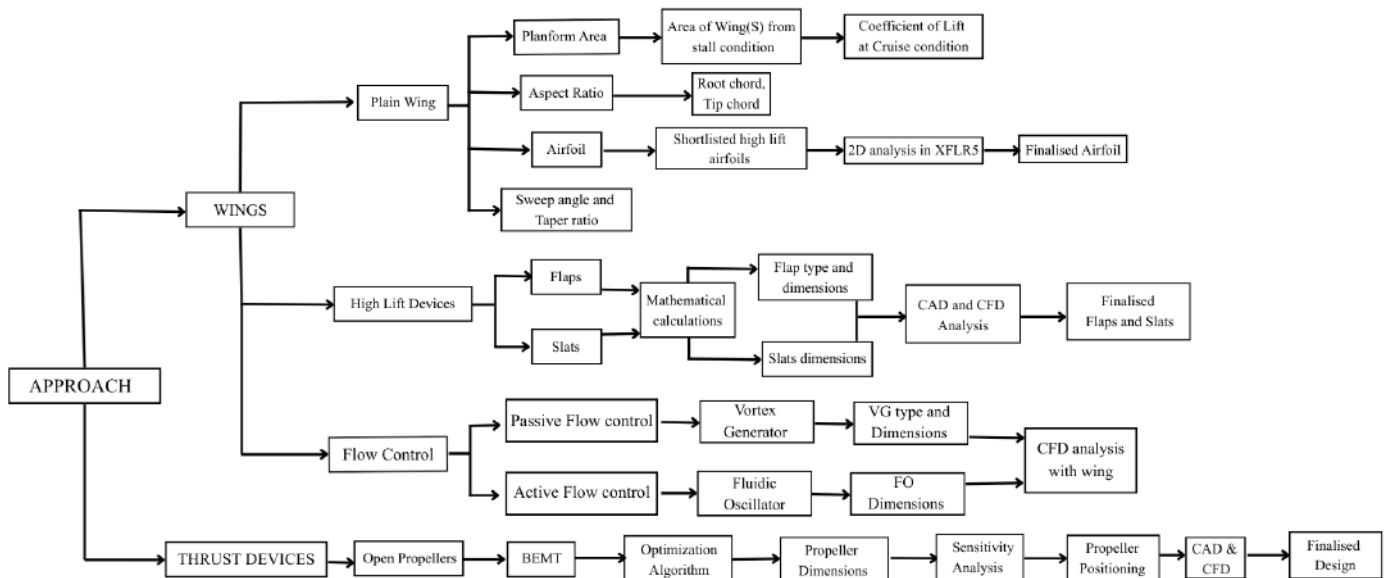


Figure 1.1: Design Process Flowchart

1.2 Key Findings

1.2.1 Wing

- Wing sizing is done first and planform area, taper ratio, sweep angle and chord and span lengths have been calculated using mathematical models.
- Aspect ratio study has been done using mathematically models from Prandtl's lifting line theory and thin airfoil theory followed by CFD to get to a final value of 15. Airfoil used is **NACA 4412**.

Wing Parameter	Value
Planform Area (S)	0.1663 m ²
Span (b)	1.58 m
Aspect ratio	15
Taper ratio (λ)	0.45
Mean aerodynamic chord $\bar{c}$	0.11034 m
Root chord	0.14523 m
Tip chord	0.06535 m
Quarter-chord sweep $\Lambda_{c/4}$	0°

1.2.2 High Lift Devices

- Slats and single-slotted flaps were designed to increase the lift during takeoff. A combination has been used such that flow separation has been delayed and the maximum lift coefficient also increased.

- 2D CFD simulation was done to determine the optimal position and deflection of the slats and flaps which are listed in the table below.

Slat Parameter	Value
c_s/c	0.16
Slat span	0.97b
Δx	0.102c
Δy	0.03c
Deflection δ_s	5°

Flap Parameter	Value
c_f/c	0.256
Deflection (δ_f)	30°
Flap span	0.59b
Overhang	0.03c
Gap	0.02c

1.2.3 Propeller

- The Genetic Algorithm optimization returns results specific to the parameter bounds.

Parameter	Value
Number of Blades	8
Diameter	0.0722 m
Chord (Root)	0.0117 m
Chord (Tip)	0.0025 m
Optimized Blade AoA	3.63° (uniform)
RPM	54267
Generated Thrust	6.250 N
Power Required	237.654 W
Tip Mach Number	0.601
Slipstream Velocity V_{slip}	53.751 m/s
ΔL_{raw}	20.986296 N
$\Delta L_{penalized}$	20.986288 N

r/R	Pitch Angle [deg]
0.100	69.113
0.196	42.027
0.291	30.782
0.406	23.746
0.502	20.163
0.598	17.629
0.694	15.711
0.809	13.887
0.904	12.573
1.000	11.067

- The NACA 4412 airfoil is identified as the optimal choice based on our requirements.
- An elliptical chord distribution provides a more uniform lift distribution compared to a linear chord distribution.
- Final propeller hub offsets are $x/c = -0.25$ and $z/c = -0.18$, where negative x denotes placement 0.25 chords upstream of the wing leading edge and negative z denotes placement 0.18 chords below the wing plane.
- The CFD analysis of the optimized propeller parameters obtained from Genetic algorithm showed a deviation of 7.2% ie. 6.45 N from the theoretical thrust prediction.
- The total thrust is **77.4 N** which eventually leads to **T/W** ratio of **0.529**

1.2.4 Passive Flow Control: Vortex Generators

Vortex generators offer higher maximum lift, better post-stall and high-lift C_L/C_D , and only minor geometric intrusion. The includes the geometric parameters h , L , D , d , b , t , and β , which are defined and described in the vortex-generators section with proper diagram.

Parameter	Value
Type	Trapezoidal (Counter-rotating)
Height, h	1.6 mm
Length of major base, L	4.8 mm
Inter-vane spacing, D	25.2 mm
Intra-vane spacing, d	5.6 mm
Length of minor base, b	0.96 mm
Thickness, t	0.4 mm
Inflow angle, β	15°

1.2.5 Active Flow Control: Fluidic Oscillators

- Fluidic oscillators (FOs) provide strong, sweeping jets without any moving parts and can energize large areas with small number of active devices. 28 of these are distributed along each half-span.

- The geometric and dynamic parameters corresponding to the fluidic oscillators are described below:

Fluidic Oscillator Parameter	Value
Fluidic oscillator width, w	5.175 mm
Outlet nozzle width, w_{out}	3.2 mm
Inlet nozzle width, w_i	0.723 mm
Inlet chamber width, w_c	0.723 mm
Outlet throttle width, w_t	0.723 mm
Fluidic oscillator height, h	0.361 mm
Inlet nozzle feedback channel width, h_{ft}	0.872 mm
Diffuser angle of outlet nozzle, θ_o	107°
Distance between inlet of mixing chamber and throat, L	5.64 mm

1.3 Proposed Design

The proposed design is shown below with the CFD results are shown in the tables below.



Figure 1.2: Proposed Full Wing Design

1.4 Performance Achievements

Configuration	C_L	C_D	C_L/C_D
Wing with flap and slat	1.76193	0.23315	7.56
Wing with flap, slat, VG	1.9144	0.2402	7.97
With flap, slat, VG, Propellers	5.72	1.575	3.632
With flap, slat, VG, Propellers, FO	6.747	1.533	4.401

The CFD values of the final proposed design of wing is as follows

Performance Parameter	Value
$C_{L_{max}}$	6.747
C_L/C_D at takeoff	4.401

The maximum lift coefficient achieved at takeoff is 6.747 with a lift-to-drag ratio of 4.401.

2 Problem Understanding

2.1 Background

Short Takeoff and Landing (STOL) aircraft historically used powered-lift and high-lift systems to reduce takeoff distance and improve low-speed performance. NASA's YC-14/YC-15 showed the potential of thrust-augmented lift and inspired later designs. Building on this legacy, LAT Aerospace aims to develop a next-generation, powered-lift fixed-wing concept for India that achieves very high lift without mechanically complex tilt-wing/tilt-rotor solutions.

2.2 Problem Statement

A fixed wing employing an integrated thrust-producing device is to be designed to generate very high lift coefficients (target envelope $C_L \in [6, 8]$). The design is to be validated through rigorous numerical analysis (CFD) with wind-tunnel testing performed if practical to further substantiate the results. Aerodynamic characterization of the thrust device is to be included, and shifts in the aerodynamic centre as well as structural integrity under high-lift conditions are to be addressed. The aim is to integrate aerodynamics, propulsion, controls, and numerical optimization while remaining practical and efficient for STOL operations.

2.3 Key Requirements

- Lift coefficient target: $C_L = 6.5$ (envelope 6–8).
- Performance: $L/D > 5$ with flaps; $L/D > 20$ without flaps.
- Minimum lift: 15 kg at $V_\infty = 20$ m/s.
- Operating speeds: $V_\infty = 20$ m/s (takeoff mode, flaps deployed); 80 m/s (cruise, flaps retracted).
- Cruise Conditions: $V_{cr} = 80$ m/s, $h_{cr} = 10000$ ft = 3.048 km
- Thrust envelope: $0.5 < T/W < 0.65$ for $W = 15$ kg.
- Constraints: powered lift only from the thrust plant; no tilt-wing, no tilt-rotor, no VTOL.

Parameters	Value
g	9.80665 kg/m ³
h	10000 ft (3.048 km)
dT/dh	-6.5 km/°C
T_{TO}	288.15 K
ρ_{TO}	1.225 kg/m ³
T_{cr}	268.34 K
ρ_{cr}	0.904 kg/m ³
V_{TO}	20 m/s
V_{stall}	16.67 m/s
V_{cr}	80 m/s
W	15 kg

Table 2.1: Parameters Derived from the Problem Statement

3 Validation and Mesh Convergence Studies

3.1 CFD Validation Study

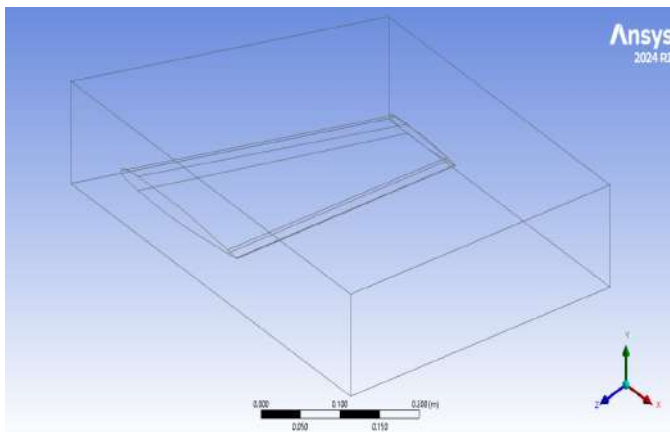
To establish the accuracy and reliability of the CFD simulations, a separate validation case was first simulated using a numerical approach and boundary conditions that remained consistent with those employed in the subsequent simulations. The validation involved simulating transonic flow over the Onera M6 wing, as documented in the NPARC Validation Archive. The results from this case were compared with the experimental data provided in the archive before proceeding with the subsequent simulations. All simulations are performed using ANSYS Fluent academic version 2025 R1.

3.1.1 Geometry and Case Setup

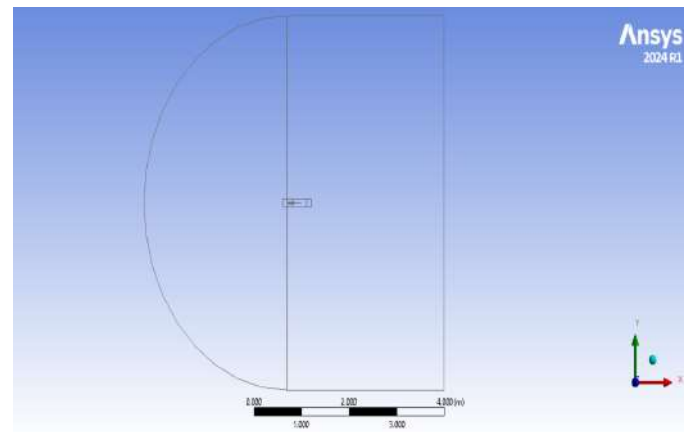
The Onera M6 wing is a swept, semi-span wing with no twist. It uses a symmetric airfoil using the Onera D section. The geometric parameters of the wing are summarized in the table below, accompanied by an image illustrating the overall geometry and the computational domain:

Table 3.1: Onera M6 Wing Geometry and Case Data

(a) Geometry Parameters		(b) Case Data	
Parameter	Value	Parameter	Value
Mean Aerodynamic Chord, $\bar{c}$	0.155 m	Mach Number, M	0.8395
Span, b	0.289 m	Freestream Velocity (m/s), U_∞	285.8
Aspect Ratio	3.8	Density (kg/m^3), ρ	4.3076
Taper Ratio	0.562	Viscosity (kg/ms), μ	1.627×10^{-5}
Leading-edge Sweep	30.0°	Reynolds Number, Re	11.72×10^6
Trailing-edge Sweep	15.8°	Angle of Attack, α	3.06°



(a) Geometry of Onera Wing



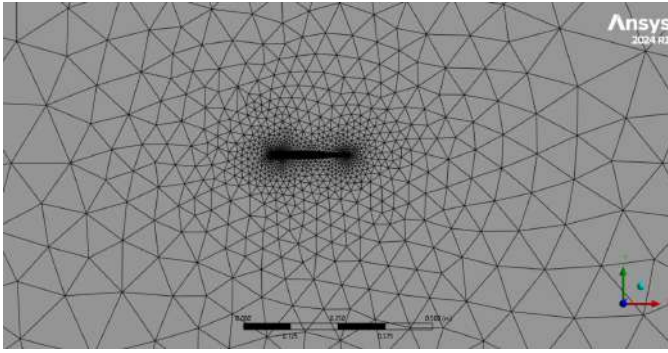
(b) Computational Domain

Figure 3.1

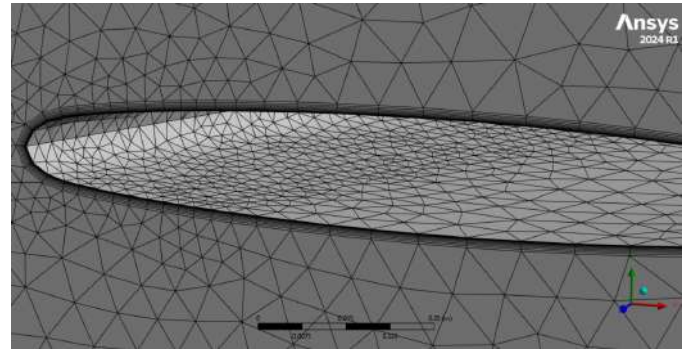
The simulation conditions are also shown in the above table. The calculated Reynolds number is approximately 11 million , and therefore, the Spalart–Allmaras one-equation turbulence model was employed with default model constants. Owing to the geometric and flow symmetry about the root chord, only half of the wing was simulated.

3.1.2 Mesh Details

An unstructured mesh primarily composed of tetrahedral elements was created, comprising close to 100,000 nodes and 353,000 elements. Near the wing surface, structured prism layers were generated to accurately resolve the boundary layer, with a total thickness of 1.8 mm distributed across 10 layers.



(a) Generated Mesh



(b) Inflation layers close to the wing

Figure 3.2

3.1.3 Results

Figure 3.3 shows the pressure coefficient distribution along the wing surface at different spanwise locations ($y/b = 0.99, 0.90, 0.80, 0.65, 0.44, 0.20$). The computed results match the experimental data closely, with a mean absolute error of less than 5%, demonstrating that the simulation approach is accurate and dependable. This confirms that the same method can be used in future studies to produce reliable results.

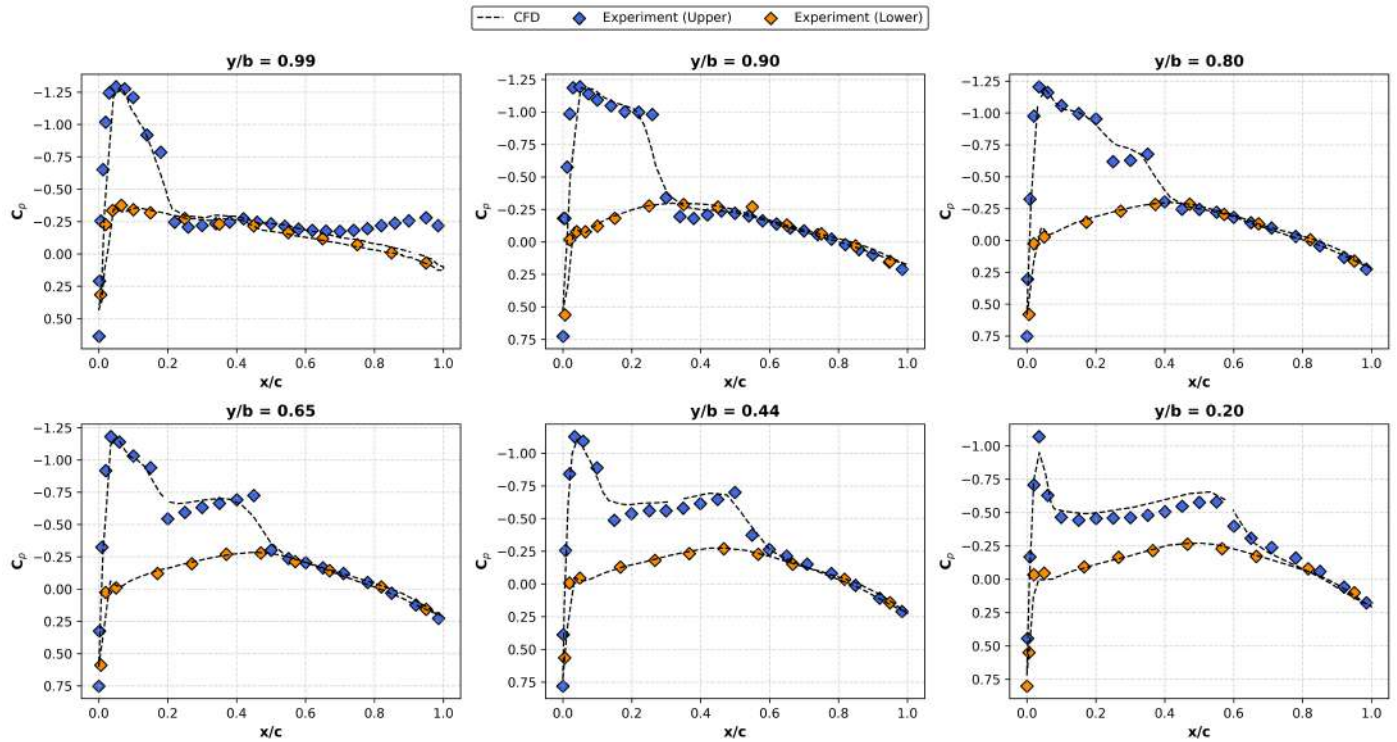


Figure 3.3: Pressure coefficient distribution at various spanwise locations on the wing.

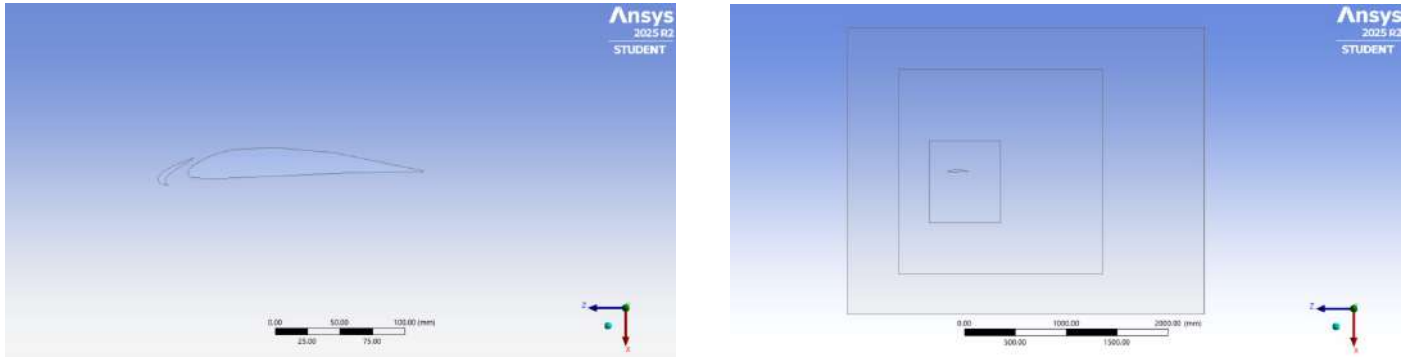
3.2 Mesh Convergence Study

The effect of mesh resolution on the computed results is evaluated in the present study to ensure solution reliability. This analysis helps identify a grid that provides accurate results while optimizing the trade-off between accuracy and computation time.

3.2.1 Geometry and Case Setup

The 2D NACA 4412 airfoil with both forward lift-enhancing device is selected, as it represents the main focus of the 2D simulations described in the following sections. The flap is deflected by 30° , and the angle of attack is 10° , as illustrated in the

accompanying figure.



The calculations are carried out under the takeoff conditions, which are detailed below:

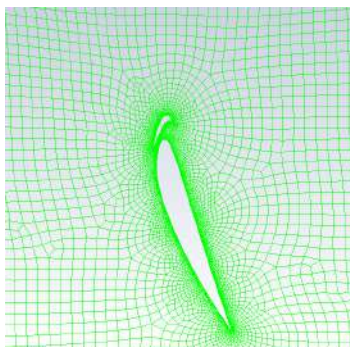
- Fluid density: $\rho = 1.225 \text{ kg/m}^3$
- Fluid viscosity: $\mu = 1.789 \times 10^{-5} \text{ kg/ms}$
- Freestream velocity: $U_\infty = 20 \text{ m/s}$
- Characteristic Length: $L = 0.19 \text{ m}$
- Reynolds number: $Re = \frac{\rho U_\infty L}{\mu} \approx 2 \times 10^5$
- Turbulence model: k- ω Shear Stress Transport (SST)

3.2.2 Mesh Details

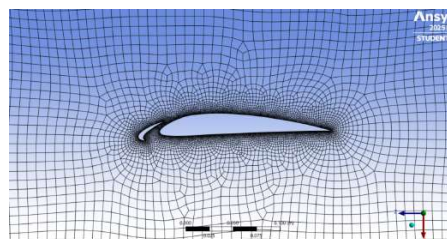
Three different meshes have been generated, with each successive mesh containing nearly twice the number of elements as the previous one, while preserving similar refinement near the boundaries. The mesh details are summarized in the adjoining table.

Table 3.2: Mesh Parameters

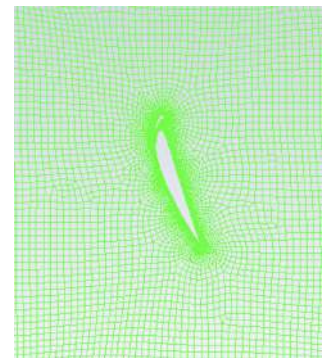
Case	No. of Elements	No. of Nodes	Average Cell Size
Mesh-1	16542	17614	0.112
Mesh-2	33805	34397	0.054
Mesh-3	66914	70754	0.022



(a) Mesh-1



(b) Mesh-2

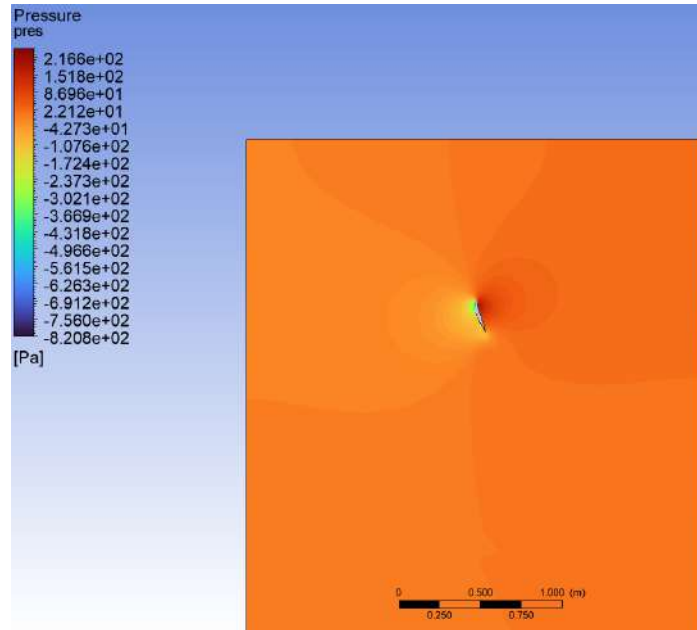


(c) Mesh-3

Figure 3.5: Meshes used in the convergence study

3.2.3 Results

The chord-wise distribution of the pressure coefficient is plotted for all the three meshes. The comparison reveals that between meshes 2 and 3, the values are found to have minimally changed. Hence, mesh-2 is employed for further computations to optimize the computational efficiency.



(a) Mesh-1



(b) Mesh-2



(c) Mesh-3

Figure 3.6: Pressure Distributions

4 Wing Sizing

The wing design process is rigorous, iterative, and involves analyzing multiple couplings of effects with their tradeoffs. To approach this problem in an organized fashion, we divided the process into various stages starting first with the design and analysis of the clean wing, which includes geometric parameters, such as planform area and Oswald factor, efficiency factor, aspect ratio, taper ratio, sweep and twist. Then, there is the quest for the right airfoil for our application. This is followed by flap and slat selection, then determining the optimum position of those for take-off conditions to maximize the Lift while preserving the minimum requirement for the lift-to-drag ratio.

4.1 Estimation of Planform Area

To estimate the planform area, we will use the stall condition during takeoff. This condition will give us an estimate of the minimum planform area required for our wing. Using Eq. 17.5 and Eq. 17.4,

$$\begin{aligned} C_{L_{max}} &= 6.5 \\ C_{L_{TO,max}} &= 0.8 \times 6.5 = 5.2 \\ W &= \frac{1}{2} \rho_{TO} V_{stall}^2 S C_{L_{TO,max}} \\ \Rightarrow S &= 0.1663 \text{ m}^2 \end{aligned}$$

4.2 Estimation of Aerodynamic Coefficients

Using the given planform area and the employing trim condition during cruise, the cruise lift coefficient is

$$\begin{aligned} C_{L_{cr}} &= \frac{W}{\frac{1}{2} \rho_{cr} V_{cr}^2 S} \\ \Rightarrow C_{L_{cr}} &= 0.3058 \end{aligned}$$

The minimum requirement for the lift-to-drag ratio during flaps retracted during cruise gives the maximum drag coefficient during cruise as

$$\begin{aligned} \frac{C_L}{C_D} &\geq 20 \\ \Rightarrow C_{D_{cr,max}} &= 0.01529 \end{aligned}$$

Similarly, at takeoff the maximum drag coefficient can be found as

$$\begin{aligned} \frac{C_L}{C_D} &\geq 5 \\ \Rightarrow C_{D_{TO,min}} &= 0.7222 \end{aligned}$$

4.3 Sweep Angle

Sweep angle is a characteristic feature in high speed wings. The primary goal of adding sweep to the wing is to reduce drag during high Mach numbers by delaying the onset of shock formation. For our application, the wings will remain in low subsonic even at cruise velocity. Therefore, we have chosen the quarter-chord sweep angle to be zero. Moreover, sweep angle reduces the $C_{L_{max}}$ of the wing and increases the structural load which serve as additional reasons to not have swept wings for our design requirement.

$$\Rightarrow \Lambda_{c/4} = 0^\circ$$

4.4 Estimation of Taper Ratio

The taper ratio influences the spanwise lift distribution of the wing. The ideal is an elliptical lift distribution as it gives the minimum lift-induced drag. It can be achieved through a geometrically elliptic wing planform shape. However, the problem with having an elliptic wing with constant geometric and aerodynamic washout is that the entire wing stalls at once. An added difficulty is in the manufacturing of an elliptic planform.

On the other hand, a rectangular wing has forgiving stall characteristics where the stalling starts from the root and moves outboard, making the tip the last part of the wing to stall. However, due to the lower sectional lift towards the tips, there is larger unused lift potential and lower aerodynamic efficiency.

Therefore, a straight tapered wing is a logical starting point for early wing design since it has better stall characteristics than the elliptic wing while producing lower lift-induced drag compared to the rectangular wing.

According to Ref. [24], for a zero quarter-chord sweep angle, the optimum taper ratio is 0.45 because it produces a lift distribution very close to the elliptic lift distribution.

According to Ref. [10], for any given sweep angle there exists an optimum taper ratio that could be approximated by the curve

$$\lambda_{opt} = 0.45e^{-0.0357\Lambda_c/4} \quad (4.1)$$

Using this equation too, we get the same value for taper ratio.

$$\Rightarrow \lambda = 0.45$$

4.5 Estimation of Oswald Efficiency Factor

The Oswald efficiency factor is an important parameter to estimate accurately, especially during the early stages of design. There are several ways to estimate the Oswald factor as seen in Ref. [6]. According to Hörner in Ref. [23], the wing's theoretical Oswald factor is given as a function of both the taper ratio and the aspect ratio.

$$e_{theo} = \frac{1}{1 + f(\lambda)AR} \quad (4.2)$$

The function of taper ratio is presented by Hörner in Ref. [23] as

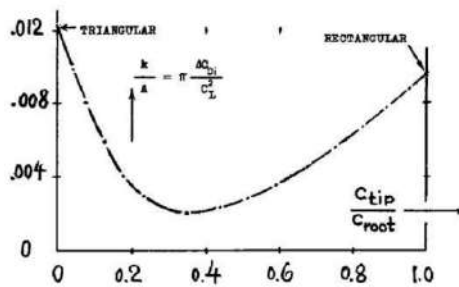


Figure 4.1: $f(\lambda)$ for Oswald Factor Estimation

In Ref. [6], the authors have approximated the above function as

$$f(\lambda) = 0.0524\lambda^4 - 0.15\lambda^3 + 0.1659\lambda^2 - 0.0706\lambda + 0.0119 \quad (4.3)$$

Adding a translation to ensure $\lambda_{opt} = 0.45$ is maintained, Nita-Scholz have given the final theoretical Oswald factor for the wing

including sweep correction term as

$$\Delta\lambda = -0.357 + 0.45e^{0.0357\Lambda_{c/4}} \quad (4.4)$$

$$e_{theo} = \left(\frac{1}{1 + f(\lambda - \Delta\lambda)AR} \right)^{\cos\Lambda_{c/4}} \quad (4.5)$$

For an aspect ratio of 15, the value of Oswald factor using above formula comes out as

$$e_{theo} = 0.9728$$

This is a theoretical estimate which is quite high and very close to the elliptic case.

The Oswald factor using Ref. [24] and [22], the formula for straight wings of high aspect ratio is given as

$$e = 1.78(1 - 0.045AR^{0.68}) - 0.64 \quad (4.6)$$

$$e = 0.6349, \text{ for } AR = 15 \quad (4.7)$$

For our problem, we have taken an intermediate value of **0.85** as a safe estimate for the Oswald efficiency factor, which is close to what is common for unswept tapered wings.

$$\implies e = 0.85 \quad (4.8)$$

5 Airfoil Selection

Airfoil choice really shapes how the whole wing performs in different situations. It involves weighing things like the lift coefficient needed for the design, the Reynolds number range, and how it stalls against downsides in drag buildup, pitching moment, and the depth needed for structure. Picking an airfoil is not about finding some perfect shape that works everywhere. It is more about the best trade-off that fits the mission you have in mind. That way, you get stable handling that you can predict across all flight conditions.

We need to figure out the Reynolds number for cruise and takeoff before we pick the airfoil. Our aspect ratio goal falls somewhere between 9 and 15.

$$AR = 12 \quad (\text{Assume})$$

$$\bar{c} = 0.1177 \text{ m}$$

$$Re_{cr} = \frac{0.904 \times 80 \times 0.1177}{1.692 \times 10^{-5}} = 5.03 \times 10^5$$

$$Re_{TO} = \frac{1.225 \times 20 \times 0.1177}{1.789 \times 10^{-5}} = 1.61 \times 10^5$$

We must also consider the fact that we are using powered-lift wings and the slipstream velocity that will be directed onto the wing will be around twice the freestream velocity. Therefore, the wing will face a Reynolds number in the wide range of 100,000 to 1 million.

A set of high-lift airfoils in this range of Reynolds number was shortlisted for analysis. An optimum selection of airfoil should satisfy the following criteria throughout the Re range:

- High $C_{l_{max}}$
- High C_{l_0}
- High C_l/C_d
- High stall angle and smooth stall characteristics
- Low C_d at zero angle of attack
- Wide C_d bucket
- C_d remains low as angle of attack increases
- C_m at zero angle of attack should not be too negative to avoid trim drag

While selecting the airfoil, we must also be aware that for the 3D wing, the lift coefficient will be lower and the drag coefficient will be larger than for the 2D airfoil. Table 5.1 and Table 5.2 present the aerodynamically efficient high-lift airfoils selected for analysis. Each of the following airfoils has a C_{l_0} much greater than 0.3058, C_l/C_d much greater than 5 at low Re and 20 at high Re , and reasonable t/c to ensure sufficient structural stiffness and to avoid warping.

- AG35 is ruled out since it has a sudden drop in lift after stall for $Re = 100,000$. It also has a small drag bucket and is very thin compared to other airfoils.
- The Eppler airfoils have small stall angle for $Re = 100,000$. It has smaller $C_{l_{max}}$ and C_l/C_d compared to other airfoils at all Reynolds numbers. Thus, these can be disqualified from the selection procedure.
- Selig S1223 has high C_{l_0} and $C_{l_{max}}$. However, the C_d values are very high due to the high camber. The drag bucket is extremely small and irregular at different Reynolds number. It also has a highly negative C_m at zero angle of attack. Airfoils having high camber have such large negative pitching moments. The aircraft must counteract this using other actuators or control surfaces such as downforce from tails which results in increase of trim drag.
- Clark-Y airfoil has slight irregularity in its stall behaviour. It also has small C_l/C_d and throughout our Reynolds number range.

Airfoil	C_{l_0}	$C_d _{\alpha=0}$	$C_l/C_d _{\alpha=0}$	$C_{l_{max}}$	Stall Angle	Stall Behaviour	Drag Bucket (C_l)	Drag Bucket (α)	$C_m _{\alpha=0}$
NACA 4412	0.4335	0.01835	23.62	1.3316	9.25°	Post-stall recovery	-0.4 to 1.3	0° to 10°	-0.1041
AG35	0.4515	0.01330	33.95	1.2350	9.25°	Sudden Drop	-0.1 to 1.0	-3.5° to 5°	-0.0577
MH113	0.5531	0.03581	15.45	1.8234	14.00°	Smooth	0.2 to 1.8	-4° to 10°	-0.1540
Clark-Y	0.3674	0.01875	19.59	1.3694	12.5°	Slightly Irregular	-0.4 to 1.4	-4° to 10°	-0.0908
Eppler 193	0.4114	0.01614	25.49	1.1845	8.25°	Post-stall recovery	0.0 to 1.2	-3° to 8°	-0.0939
Eppler 387	0.4171	0.01557	26.79	1.1859	8.00°	Post-stall recovery	0.2 to 1.2	-3° to 8°	-0.0940
Eppler 397	0.3549	0.03544	10.01	1.4734	10.50°	Post-stall recovery	-0.2 to 1.4	-5° to 10°	-0.1284
S1223	1.1729	0.02337	50.19	2.0814	9.75°	Irregular, Sudden Drop	0.9 to 1.1	-2° to 0°	-0.2688
SD7032	0.3898	0.01739	22.42	1.4474	13.25°	Smooth Drop	0.0 to 1.2	-2° to 7°	-0.0941
GOE 549	0.5124	0.01906	26.88	1.5304	12.25°	Irregular	0.3 to 0.6	-2.5° to 0°	-0.1061

Table 5.1: High-lift airfoils data at $Re = 100,000$

Airfoil	C_{l_0}	$C_d _{\alpha=0}$	$C_l/C_d _{\alpha=0}$	$C_{l_{max}}$	Stall Angle	Stall Behaviour	Drag Bucket (C_l)	Drag Bucket (α)	$C_m _{\alpha=0}$
NACA 4412	0.4833	0.00678	71.28	1.6706	16.25°	Smooth Drop	-0.3 to 1.2	-5° to 7°	-0.1032
AG35	0.4211	0.00546	77.13	1.4513	12.00°	Smooth Drop	-0.4 to 1.4	-5° to 5°	-0.0461
MH113	0.8941	0.00740	120.82	1.9035	16.75°	Smooth Drop	0.3 to 1.4	-5° to 5°	-0.1944
Clark-Y	0.3918	0.00645	60.74	1.5240	14.75°	Smooth Drop	0 to 1.2	-3° to 7°	-0.0826
Eppler 193	0.3659	0.00547	66.89	1.3479	14.5°	Smooth Drop	-0.2 to 1.00	0° to 5°	-0.0750
Eppler 387	0.3915	0.00547	71.57	1.3867	13.25°	Smooth Drop	0.0 to 1.0	-2° to 5°	-0.0794
Eppler 397	0.7149	0.00691	103.46	1.6407	16.75°	Smooth Drop	0.0 to 1.4	-5° to 5°	-0.1704
S1223	1.2030	0.01188	101.26	2.4036	14.5°	Smooth Drop	0.7 to 1.5	-3° to 5°	-0.2740
SD7032	0.4530	0.00538	84.20	1.6333	14.25°	Smooth Stall	-0.2 to 1.2	-5° to 7°	-0.0943
GOE 549	0.4918	0.00838	58.69	1.4786	15°	Smooth Drop	0.0 to 1.5	0° to 5°	-0.0935

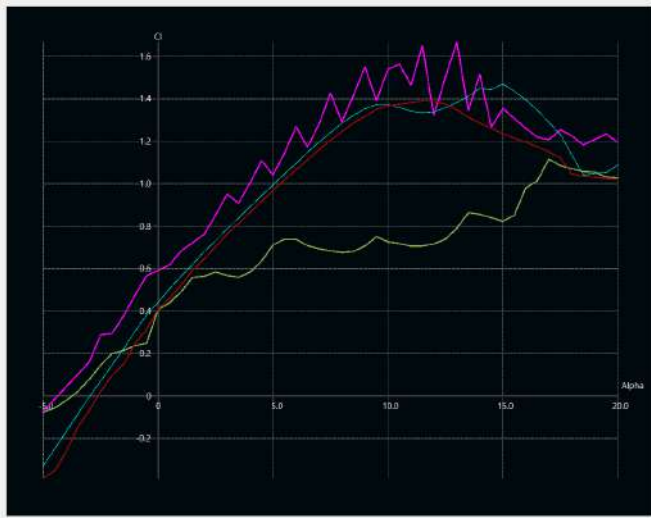
Table 5.2: High-lift airfoils data at $Re = 1,000,000$

We are left with NACA 4412, MH113, SD7032 and GOE549. These airfoils satisfy majority of the required criteria and show Re-independence which is crucial as we are operating across a wide range of Reynolds number. These airfoils also have reasonable thickness, ensuring structural integrity of our high aspect ratio wings. For further analysis of these airfoil shapes, two-dimensional analysis was performed on XFLR5.

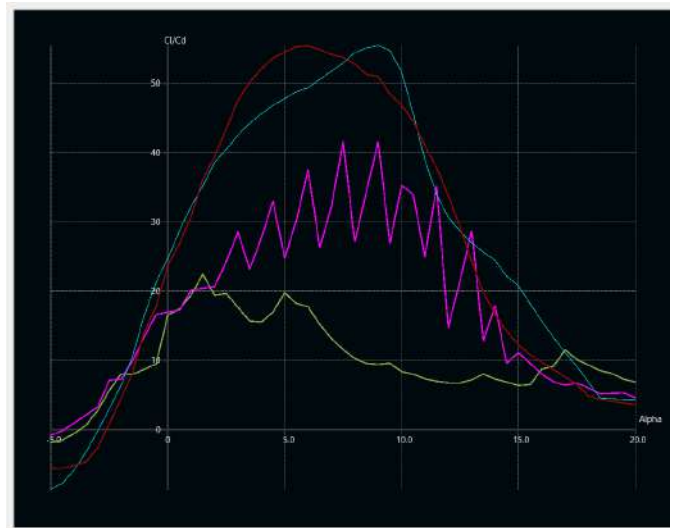
5.1 XFLR5 Analysis of Airfoils

Multiple XFLR5 simulations were performed on different airfoils to achieve the desired lift coefficient (C_l) and lift-to-drag ratio (C_l/C_d). These simulations were conducted over a range of Reynolds numbers and angles of attack to thoroughly evaluate aerodynamic performance. Each airfoil produced distinct graphs over a range of angles of attack from -5 to 20 degrees at a Reynolds number of 100k, 500k, 900k and 1.2M.

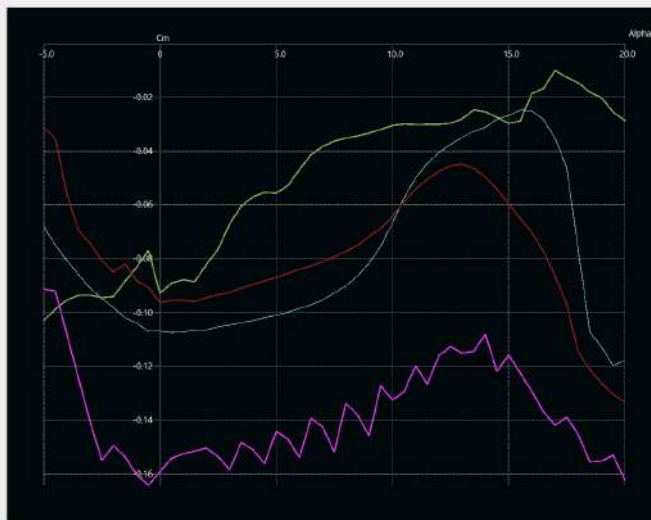
5.1.1 XFLR5 Airfoil Analysis for $Re = 100,000$



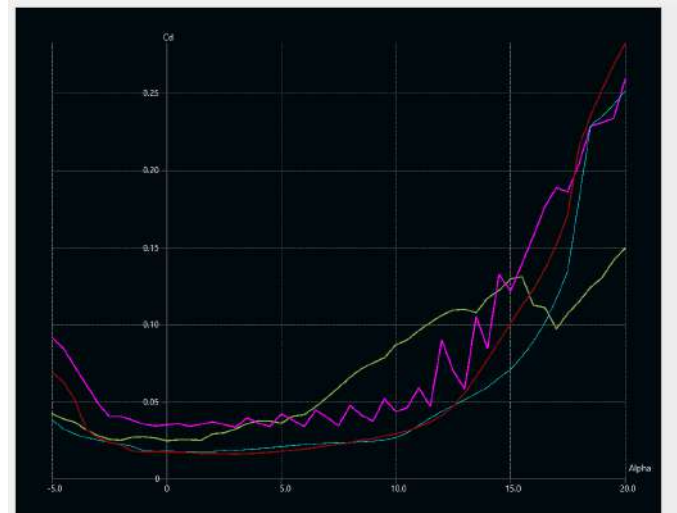
(a) C_l vs α for selected airfoils at $Re = 100,000$



(b) C_l/C_d vs α for selected airfoils at $Re = 100,000$



(c) C_m vs α for selected airfoils at $Re = 100,000$



(d) C_d vs α for selected airfoils at $Re = 100,000$

— MH113 — NACA 4412 — GOE549 — SD7032

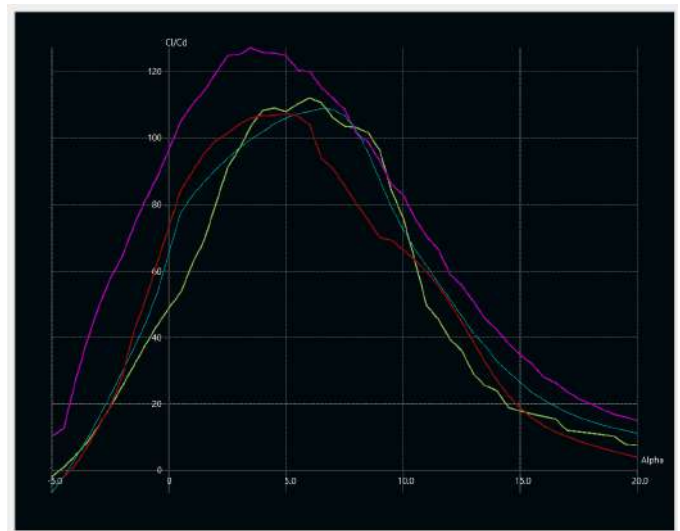
Figure 5.1: XFLR5 Airfoil Analysis for $Re = 100,000$

At $Re = 100k$, all four airfoils show more pronounced low-Reynolds-number effects, including earlier flow separation and stronger sensitivity to angle of attack. The MH113 airfoil continues to demonstrate the highest lift coefficient and a prominent peak in C_l/C_d , indicating that it remains the strongest performer in terms of lift generation and aerodynamic efficiency. However, the presence of noticeable oscillations in both C_l and C_l/C_d curves suggests the influence of laminar separation bubbles, meaning this airfoil becomes more sensitive to small variations in angle of attack or surface conditions at this lower Reynolds number. The NACA4412 airfoil shows moderate but steady lift growth with a smooth pitching moment curve, which indicates more predictable longitudinal stability and easier trim control, making it a reliable choice where handling quality and control stability are important. The GOE549 airfoil achieves reasonable lift at medium angles of attack but shows an earlier efficiency drop and a more noticeable flattening of the lift curve, indicating earlier flow separation and more limited high-lift capability. The SD7032 airfoil produces the lowest lift and exhibits a higher drag rise with increasing angle of attack, making it the least favorable for applications requiring efficient lift or extended low-speed performance. Additionally, stall tendencies at this Reynolds number are sharper, and drag penalties become more significant across the airfoils, especially at high angles of attack.

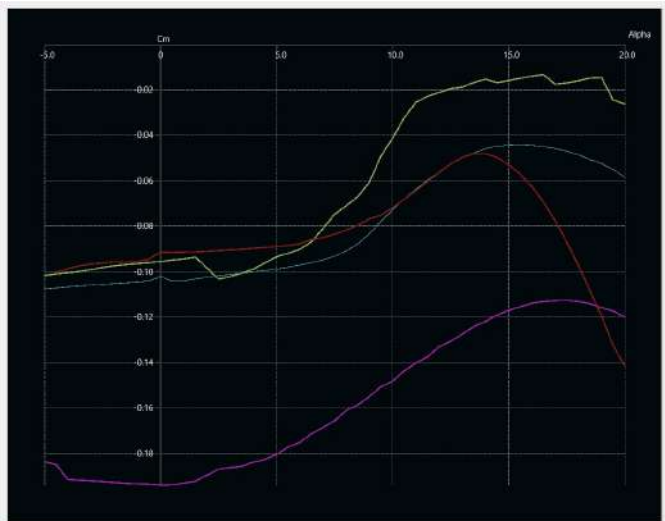
5.1.2 XFLR5 Analysis for $Re = 500,000$



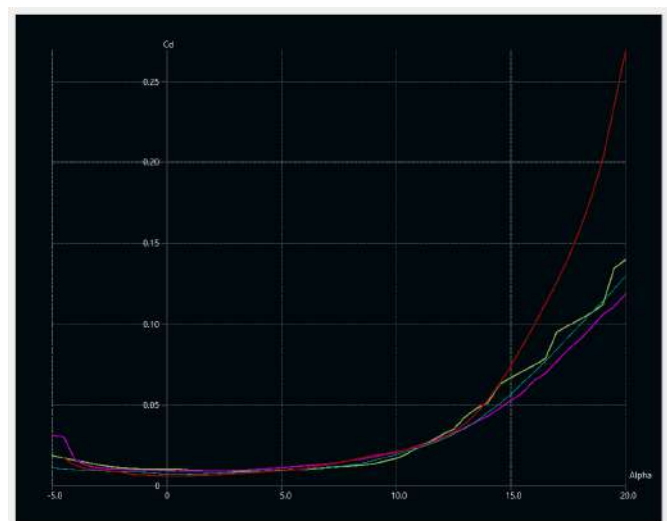
(a) C_l vs α for selected airfoils at $Re = 500,000$



(b) C_l/C_d vs α for selected airfoils at $Re = 500,000$



(c) C_m vs α for selected airfoils at $Re = 500,000$



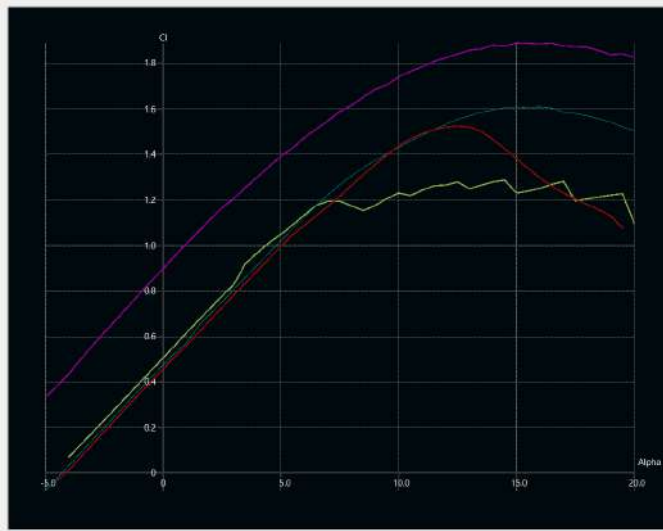
(d) C_d vs α for selected airfoils at $Re = 500,000$

— MH113 — NACA 4412 — GOE549 — SD7032

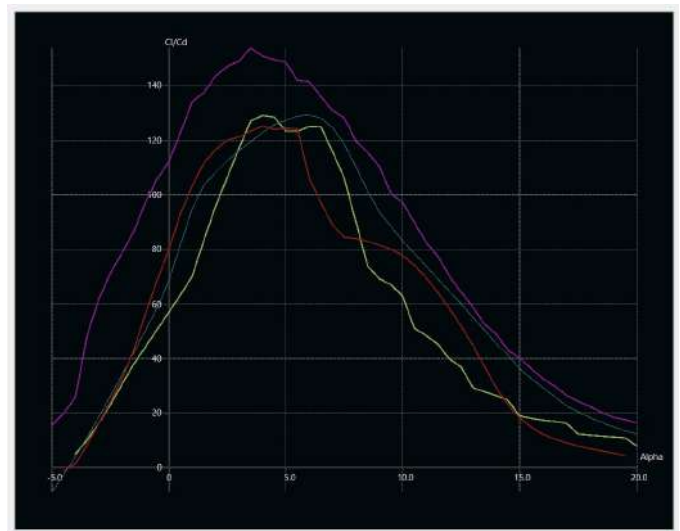
Figure 5.2: XFLR5 Airfoil Analysis for $Re = 500,000$

At $Re = 500k$, the aerodynamic trends of the four airfoils become more distinct, as viscous effects are reduced compared to low Reynolds number, and the flow behavior begins to stabilize. The MH113 airfoil continues to show the highest lift coefficient among the four, with a strong and steady rise in C_l followed by a delayed stall, and it also retains the highest C_l/C_d peak, indicating strong aerodynamic efficiency over a wide operating range. This makes MH113 particularly suitable for high-lift, low-speed performance where maintaining lift at moderate to high angles of attack is desirable. The NACA4412 airfoil again demonstrates balanced and predictable behavior, providing moderate lift with a smooth pitching moment curve and a stable slope, indicating good inherent longitudinal stability, which contributes to ease of trim and control handling. The GOE549 airfoil (yellow) reaches reasonable lift values but shows a more gradual and earlier reduction in C_l/C_d , indicating that its efficiency begins to drop sooner as the angle of attack increases. The stall in GOE549 is also somewhat earlier and less forgiving, limiting its operational envelope in conditions requiring sustained higher lift. The SD7032 airfoil shows lower lift capability overall and a significant rise in drag beyond moderate angles of attack, placing it at a performance disadvantage relative to the others. Its pitching moment variation near stall is more pronounced, which may require additional tail compensation to maintain stability.

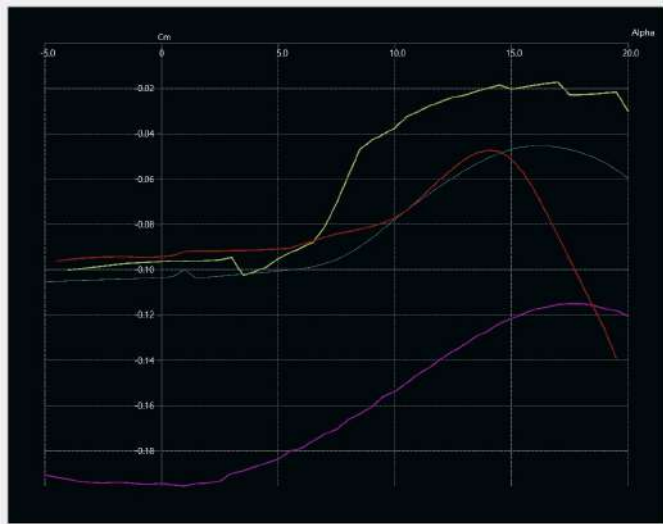
5.1.3 XFLR5 Analysis for $Re = 900,000$



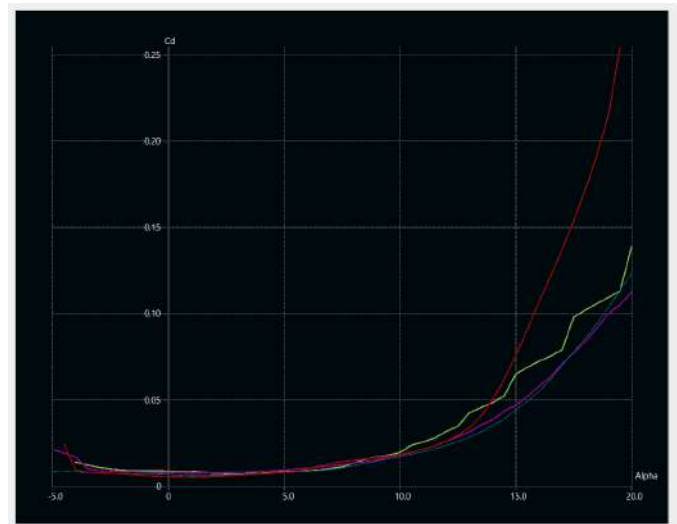
(a) C_l vs α for selected airfoils at $Re = 900,000$



(b) C_l/C_D vs α for selected airfoils at $Re = 900,000$



(c) C_m vs α for selected airfoils at $Re = 900,000$



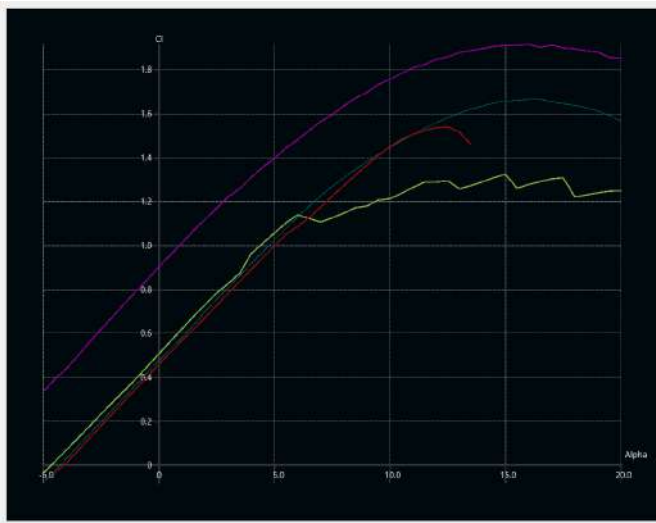
(d) C_D vs α for selected airfoils at $Re = 900,000$

— MH113 — NACA 4412 — GOE549 — SD7032

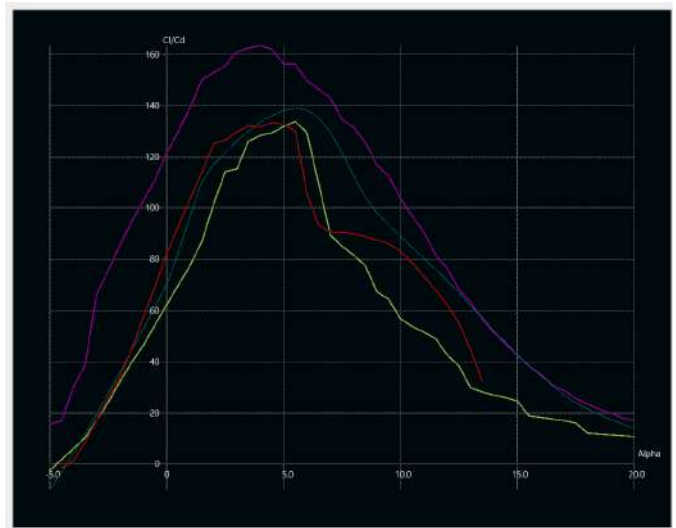
Figure 5.3: XFLR5 Airfoil Analysis for $Re = 900,000$

The plots at $Re = 900k$ show distinct aerodynamic characteristics for each airfoil. MH113 consistently provides the highest lift coefficient across the operating range, and it also exhibits the highest lift-to-drag ratio, indicating that it can generate lift efficiently with relatively low drag. This makes it especially suitable for configurations that require high lift at lower speeds, such as STOL aircraft or UAVs during takeoff, landing, or loitering. However, MH113 has a less negative pitching moment, meaning it contributes less inherent longitudinal stability and may require a larger or more responsive tail surface to maintain trim and control during flight. In comparison, NACA4412 shows a moderate maximum lift but has a more negative and stable pitching moment, which gives better inherent longitudinal stability and smoother stall behavior, making it easier and more predictable to fly, especially when precise attitude control is necessary. GOE549, although it provides reasonable lift at low angles of attack, experiences a drop in efficiency and an earlier stall, reducing its usefulness in high-lift or slow-flight conditions. SD7032 performs the weakest among the four, with lower lift, higher drag growth, and sharper stall, making it less practical where sustained aerodynamic performance is important. Additionally, MH113 performs better in delay of stall, maintaining attached flow at higher angles compared to the others. Similarly, NACA4412 offers a more gradual change in aerodynamic coefficients near stall, which improves controllability and pilot or controller feedback in critical maneuvers.

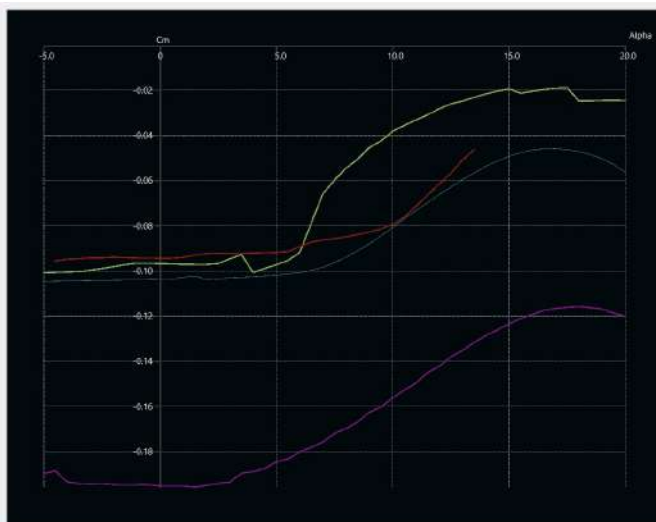
5.1.4 XFLR5 Analysis for $Re = 1,200,000$



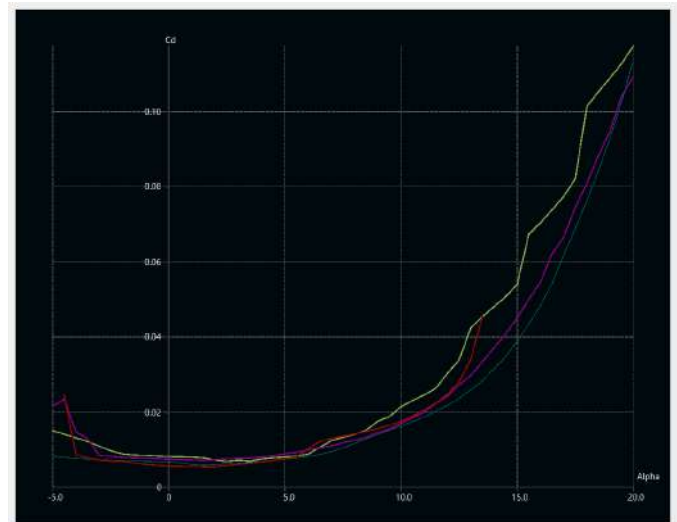
(a) C_l vs α for selected airfoils at $Re = 1,200,000$



(b) C_l/C_d vs α for selected airfoils at $Re = 1,200,000$



(c) C_m vs α for selected airfoils at $Re = 1,200,000$



(d) C_d vs α for selected airfoils at $Re = 1,200,000$

— MH113 — NACA 4412 — GOE549 — SD7032

Figure 5.4: XFLR5 Airfoil Analysis for $Re = 1,200,000$

5.1.5 Shortlisted Airfoils

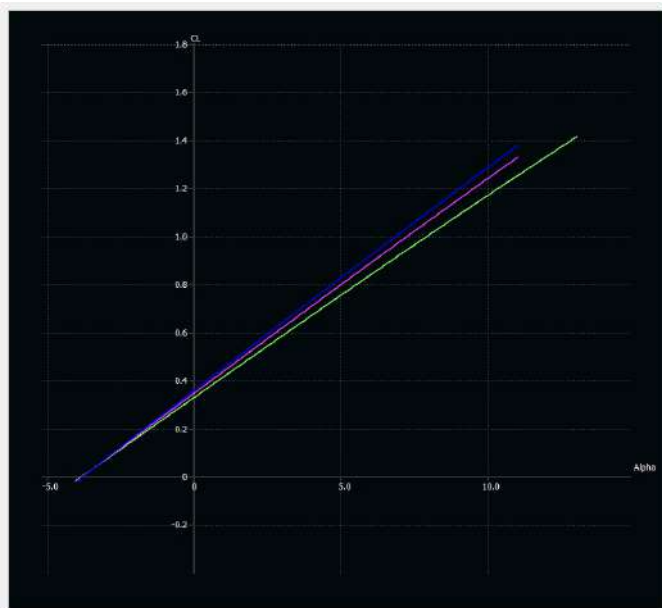
Based on theoretical evaluation, four airfoils—NACA 4412, MH113, Selig/Donovan SD7032, and GOE549—were selected for detailed analysis. XFLR5 simulations were conducted for these airfoils at Reynolds numbers of 100,000, 500,000, 900,000, and 1.2 million to study their aerodynamic characteristics, including lift, drag, and moment variations.

- GOE549: Displayed irregular and unstable aerodynamic characteristics across all Reynolds numbers, making it unsuitable for further consideration.
- MH113: Showed smooth aerodynamic behavior and good lift performance at higher Reynolds numbers, with a favorable stall angle except at $Re=100k$.
- SD7032: Exhibited smooth aerodynamic trends across all Reynolds numbers but had a low stall angle of about 11 degrees.
- NACA 4412: Demonstrated consistent and stable aerodynamic performance with good lift and a desirable stall angle at all Reynolds numbers.

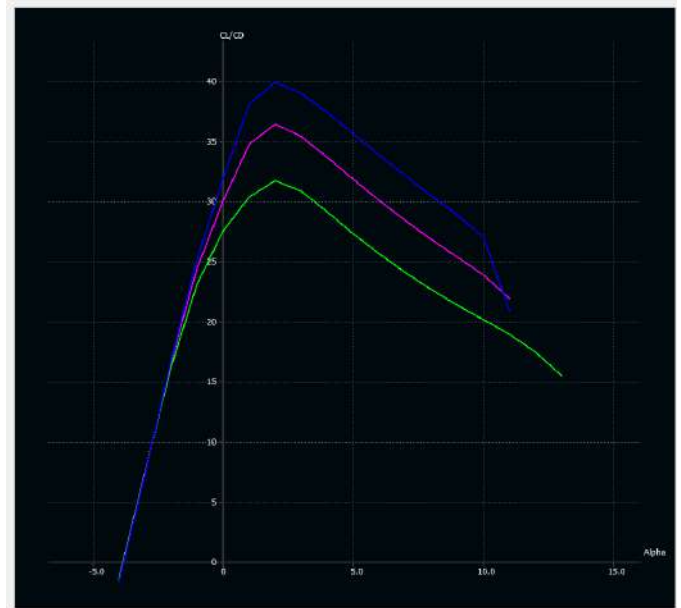
Based on these results, **NACA 4412** and **MH113** are shortlisted for further evaluation.

5.2 Finite Wing Analysis on XFLR5

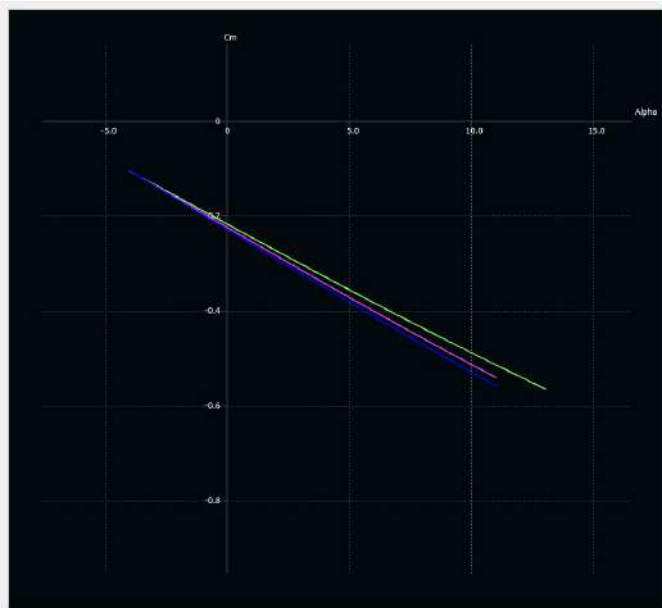
For selection of the optimum airfoil, its performance needs to be examined for the finite wing. The finite wing analysis has been performed for NACA 4412 and MH113 on a clean wing using wing geometry parameters finalized in Section 4 for three aspect ratios: 9, 12, and 15. The analysis has been performed for the cruise conditions since the design is for a clean wing.



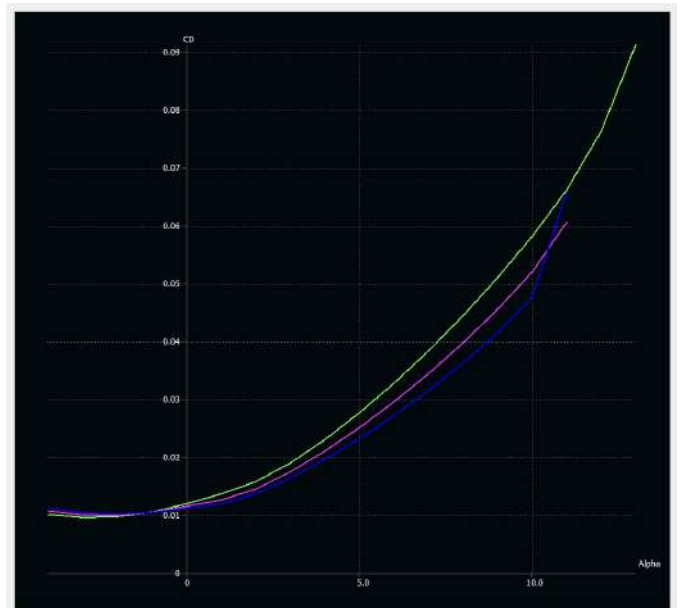
(a) C_l vs α for NACA 4412 for Aspect Ratios 9, 12, 15



(b) C_l/C_D vs α for NACA 4412 for Aspect Ratios 9, 12, 15



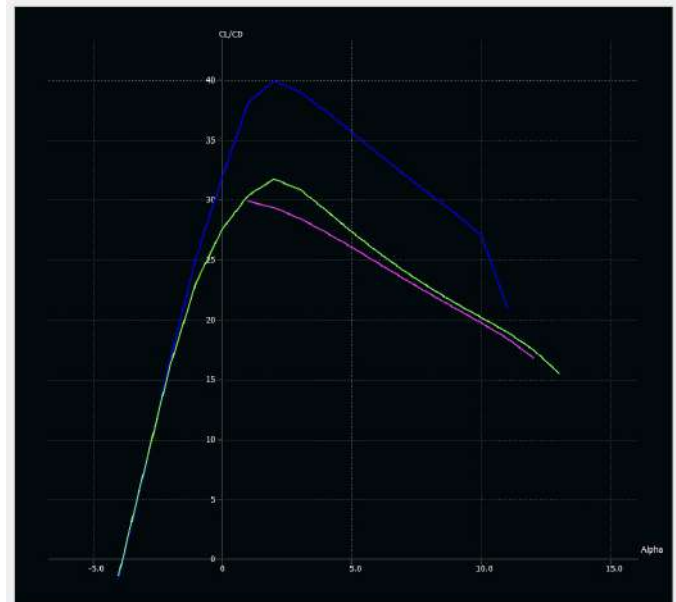
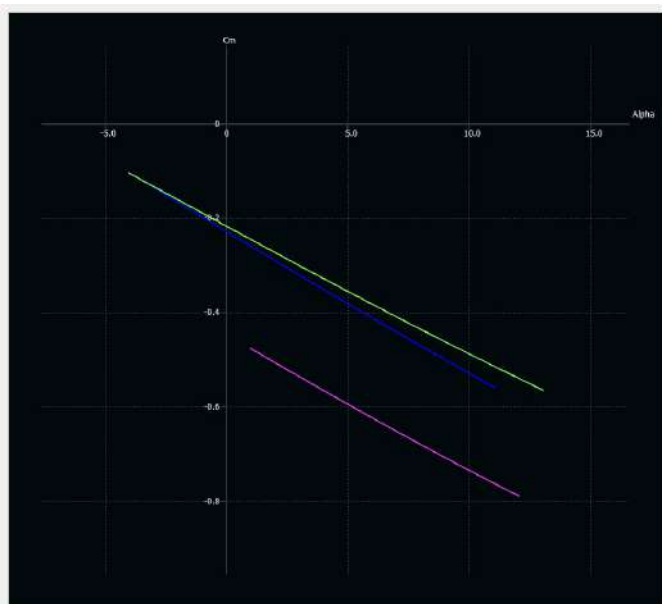
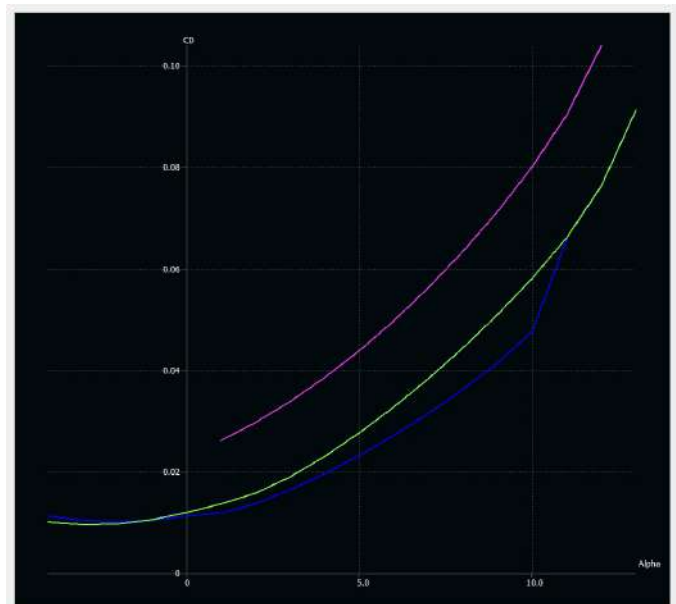
(c) C_m vs α for NACA 4412 for Aspect Ratios 9, 12, 15



(d) C_D vs α for NACA 4412 for Aspect Ratios 9, 12, 15

— AR = 9 — AR = 12 — AR = 15

Figure 5.5: Finite Wing Analysis for NACA 4412


(a) C_L vs α for MH113 for Aspect Ratios 9, 12, 15

(b) C_L/C_D vs α for MH113 for Aspect Ratios 9, 12, 15

(c) C_m vs α for MH113 for Aspect Ratios 9, 12, 15

(d) C_D vs α for MH113 for Aspect Ratios 9, 12, 15

— AR = 9 — AR = 12 — AR = 15

Figure 5.6: Finite Wing Analysis for MH113

5.3 Finalizing Airfoil

At cruise, the trim condition of $L = W$ must be satisfied. The required C_L at cruise was determined to be 0.3058 in Section 4.2. It is required by the chosen airfoil to satisfy this criterion. According to the finite wing analysis on XFLR5, NACA 4412 has C_L near the required value of 0.3058 across a range of aspect ratios for zero angle of attack at cruise. NACA 4412 was selected as the choice for our wing considering its Re-independence and the fact that it performs similarly across a range of aspect ratios enabling us to freely compare and choose the optimum aspect ratio for our wing.

6 Aspect Ratio

A high aspect ratio wing is aerodynamically efficient as it provides a high lift and lower induced drag. But, its long slender wings are susceptible to high deflections under aerodynamic loads and the high bending moments can cause the warping.

Since the lift-to-drag requirement of 20 at cruise, aspect ratios of the range 9 to 15 have been explored.

6.1 Mathematical Modelling for Aspect Ratio

Mathematical modelling using Prandtl's lifting line theory and the thin airfoil theory has been done to explore how the lift coefficient and lift-to-drag ratio vary with the aspect ratio.

During cruise,

$$\frac{C_L}{C_D} \geq 20 \quad (6.1)$$

We know from Section 4.1 the lift coefficient required at cruise.

$$C_D \leq \frac{0.3058}{20} = 0.01529 \quad (6.2)$$

$$C_{D_0} \leq 0.01529 - \frac{(0.3058)^2}{\pi(0.85)AR} \quad (6.3)$$

Using Eq. 17.2, the maximum zero-lift drag coefficient can be used for further calculations depending upon the aspect ratio. Using Eq. 6.2 in Eq. 6.3, we can obtain the theoretical estimation of the lift-to-drag ratio at cruise as shown in Fig. 6.1.

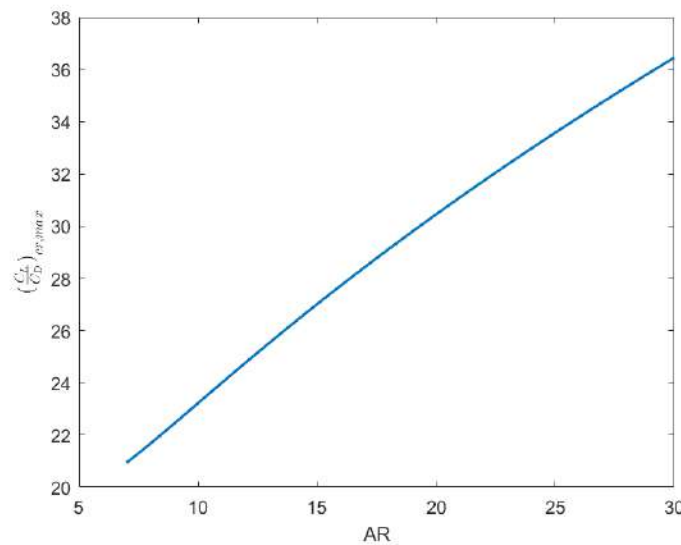


Figure 6.1: Variation of maximum lift-to-drag ratio with aspect ratio

Therefore, theoretically, we are able to achieve a lift-to-drag ratio of greater than 20 for clean wing configuration at cruise for aspect ratios in our desired range.

It is known that the performance of the wing is closely related to the performance of the airfoil. Given the airfoil, an estimate can be made about the wing performance after using the following expressions to account for three-dimensional effects in

finite wings.

$$C_L = C_{L\alpha}(\alpha - \alpha_{C_L=0})$$

$$C_{L\alpha} = \frac{C_{l\alpha}}{1 + \frac{C_{l\alpha}}{\pi e AR}}$$

$$\alpha_{C_L=0} = \alpha_{C_l=0} + \alpha_i$$

$$\alpha_i = \frac{C_L}{\pi e AR}$$

These expressions give the wing lift and drag coefficients completely as a function of aspect ratio, efficiency factor and the zero-lift drag coefficient.

$$C_L = \frac{(\alpha - \alpha_{C_L=0})}{\left(\frac{1}{C_{l\alpha}} + \frac{1}{\pi e AR}\right)\left(1 + \frac{1}{\pi e AR}\right)} \quad (6.4)$$

$$C_D = C_{D_0} + \frac{C_L^2}{\pi e AR} \quad (6.5)$$

Applying the above for NACA 4412 airfoil, and using the efficiency factor and zero-lift drag coefficient calculated previously, we obtain the variation of C_L , C_D and C_L/C_D at cruise as shown in Figure 6.2. It shows that the theoretical approximation of cruise C_L and C_L/C_D satisfy our requirement.

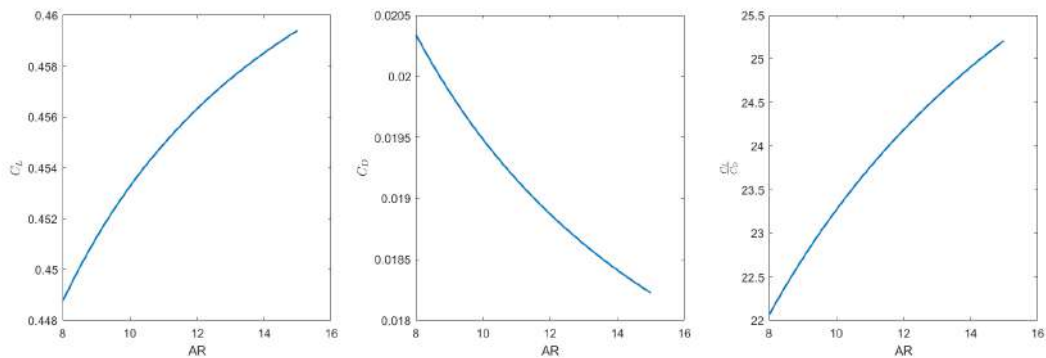


Figure 6.2: Theoretical Variation of C_L , C_D and C_L/C_D for NACA 4412 finite wing

6.2 Preliminary Clean Wing Design

Using the above parameters for the wing geometry, the CAD for the clean wing was designed for different aspect ratios.

Aspect Ratio	9	12	15
C_{root} (in m)	0.18749	0.16237	0.14523
C_{tip} (in m)	0.08437	0.07306	0.06535
$\bar{c}$ (in m)	0.14245	0.12336	0.11034
Span (in m)	1.22339	1.41265	1.57939
Λ_{LE}	2.41333 °	1.81047 °	1.44855 °

Table 6.1: Wing Geometry Parameters for Different Aspect Ratios

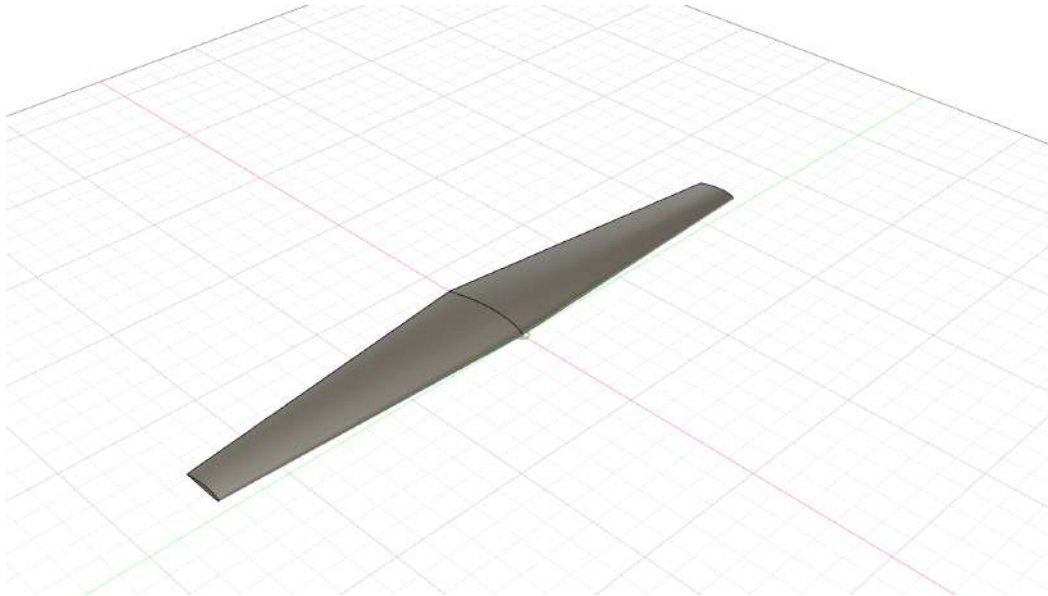


Figure 6.3: Preliminary Clean Wing Design

6.3 CFD of Clean Wing for Different Aspect Ratios

To determine the aspect ratio of the wing, thorough CFD is performed and results are compared for clean wing configuration at different aspect ratios.

6.3.1 Geometry and Case Setup

The geometries considered in this study are illustrated in the Table 6.1 above. The model is three dimensional. As the flow is symmetric about the root chord plane, the wing is divided along this plane, and a symmetry boundary condition is applied on the resulting surface. This approach effectively reduces the computational domain by half, thereby decreasing both computational cost and simulation time.

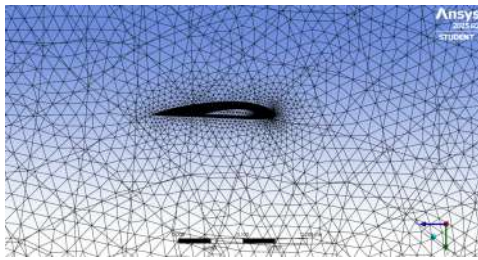
The cruise conditions for which CFD for different aspect ratios has been done are described below:

- Fluid density*: $\rho = 0.902 \text{ kg/m}^3$
- Fluid viscosity*: $\mu = 1.69 \times 10^{-5} \text{ kg/(m}\cdot\text{s)}$
- Freestream velocity: $U_\infty = 80 \text{ m/s}$
- Characteristic length: $L = 0.144 \text{ m}$
- Reynolds number: $Re = \frac{\rho U_\infty L}{\mu} \approx 6 \times 10^5$
- Turbulence model: $k-\omega$ Shear Stress Transport (SST)

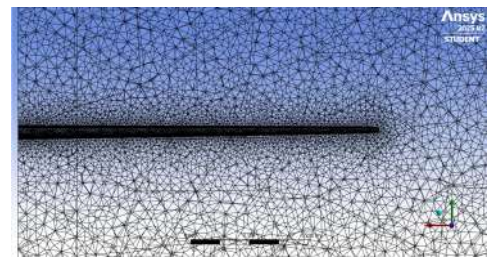
* Fluid properties obtained from the US Standard Atmosphere at 10,000 ft altitude.

6.3.2 Mesh Details

A tetrahedron-dominated unstructured mesh was created, with structured prism-layer inflation around the wing surface. The y^+ value was set to 30, to capture the log layer using the $k-\omega$ SST RANS model, resulting in a first cell height of approximately 0.02 mm, determined using the same calculator. The mesh for AR=9,12,15 with the number of elements and nodes.

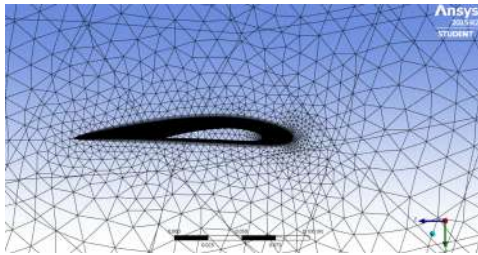


(a) Airfoil cross section

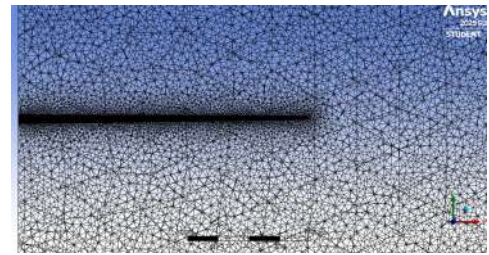


(b) Wing cross section

Figure 6.4: AR=9(Number of Nodes=1022686, Number of Elements=5709949)

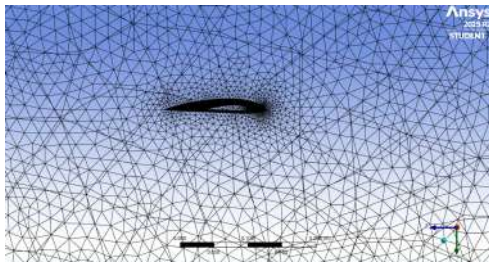


(a) Airfoil cross section

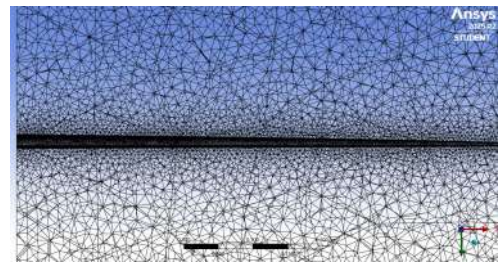


(b) Wing cross section

Figure 6.5: AR=12(Number of Nodes=1094162, Number of Elements=6106980)



(a) Airfoil cross section



(b) Wing cross section

Figure 6.6: AR=15(Number of Nodes=1062568, Number of Elements=5936277)

6.3.3 Results

The table presents the variation of aerodynamic parameters with aspect ratio (AR). As the aspect ratio increases from 9 to 15, the lift coefficient (C_L) shows a gradual increase, indicating improved lift generation efficiency. The drag coefficient (C_D) remains nearly constant, leading to a steady rise in the lift-to-drag ratio (C_L/C_D), which reflects enhanced aerodynamic performance. The Moment Coefficient (C_M) also increases with AR, signifying reduced induced drag and improved overall wing efficiency for higher aspect ratios.

AR	C_L	C_D	C_L/C_D	C_M
9	0.31417	0.020885	15.04285	0.1453
12	0.32965	0.021218	15.53634	0.1767
15	0.33739	0.021566	15.64453	0.2026

Table 6.2: Variation of aerodynamic parameters with aspect ratio (AR) for cruise conditions

6.4 Selection of Aspect Ratio

An aspect ratio of **15** is finalized based on the above CFD results since the lift coefficient as well as the lift-to-drag ratio are increasing. A higher aspect ratio than 15 can be dangerous as the moments on the wing are high and can cause structural damage due to warping and large flexion on the wing.

7 Flaps

Flaps are high-lift devices located on the trailing edges of wings, deployed to significantly increase the airfoil's camber (curvature) and lift coefficient. This mechanism allows aircraft to generate necessary lift at lower airspeeds which is crucial for reducing the required takeoff and landing speeds and distances.

7.1 Math Modelling of Single Slotted Flaps

In this mathematical modeling, the flap airfoil is considered to have the same geometry as the main airfoil, namely the profile NACA 4412. This was important because it maintains the same aerodynamic characteristics between main and flap elements. The formulations used for the mathematical modeling are based on semi-empirical methods presented by Olson [19], who developed predictive approaches for low-speed aerodynamic characteristics. Furthermore, some parameters have been adopted from experimental data given by Abbott and Von Doenhoff [20], particularly from the section on *High-Lift Devices* in their work *Theory of Wing Sections*.

A flap primarily increases the lift coefficient through an increase in effective camber of the airfoil, which delays flow separation. The outcome is a higher circulation around the airfoil and therefore an increase in lift.

When a trailing-edge flap is extended, the following changes occur to the section lift-curve slope, as shown in Fig. 7.1: the lift at zero angle of attack is increased by an increment $\Delta_f c_{l,0}$, the lift-curve slope changes to a modified value of $c'_{l,\alpha}$, and the maximum sectional lift coefficient increases by an increment $\Delta_f c_{l,max}$.

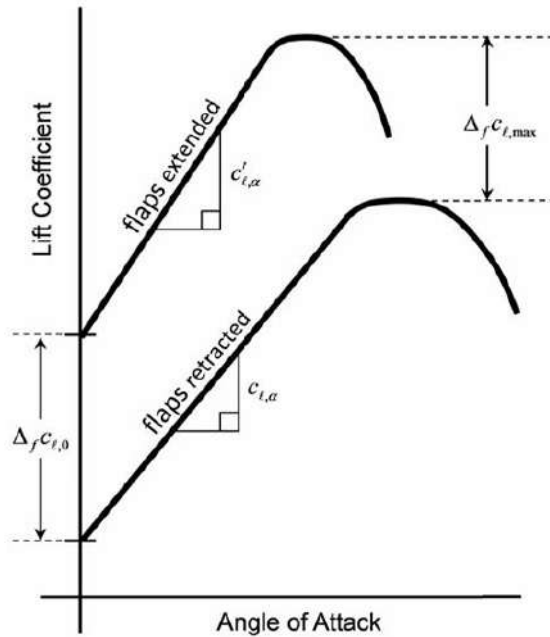


Figure 7.1: Two-dimensional trailing-edge flap effects

7.1.1 Lift Increment at Zero Angle of Attack

In addition to changing the camber and incidence of the base airfoil, slotted and Fowler flaps also have: the benefit of increasing the effective chord of the section through rearward motion of the flap segments. Since the circulation has a larger chord length to operate on, the flap extension results in an additional lift increment. Since the section $c_{l,0}$ is not affected by the chord extension, the lift coefficient increment with flaps extended can be determined for the extended chord and then referred back to the original chord as follows:

$$\Delta_f c_{l,0} = \Delta_f c'_{l,0} \frac{c'}{c} + c_{l,0} \left(\frac{c'}{c} - 1 \right),$$

where $c' \equiv \zeta c + c_f \cos(\delta_f)$ is the extended chord and $\Delta_f c'_{l,0}$ is the lift increment referenced against the extended chord:

$$\Delta_f c'_{l,0} = \eta_\delta \alpha'_\delta c'_{l,\alpha} \delta_f.$$

The parametric expressions for the flap effectiveness factor are

$$\alpha'_\delta = 1 - \frac{\theta'_f - \sin \theta'_f}{\pi}, \quad \theta'_f = \cos^{-1} \left(2 \frac{c_f}{c'} - 1 \right).$$

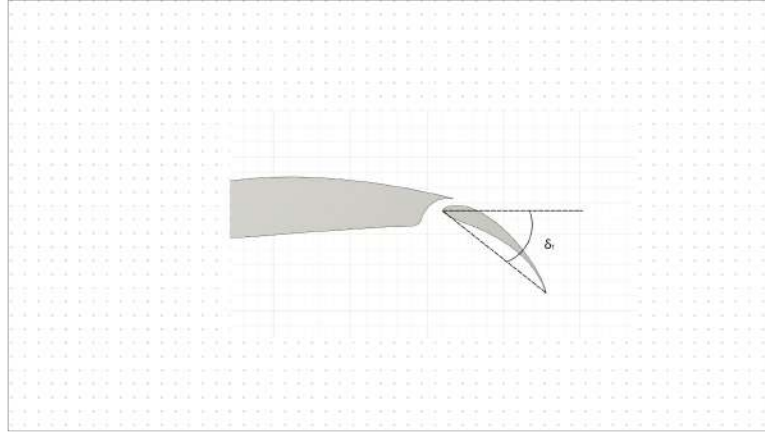


Figure 7.2: Single-slotted flap geometry

Using the chord relation $c' = \zeta c + c_f \cos(\delta_f)$, the zero-angle lift increment becomes

$$\Delta_f c_{l,0} = \Delta_f c'_{l,0} \frac{\zeta c + c_f \cos(\delta_f)}{c} + c_{l,0} \left(\frac{\zeta c + c_f \cos(\delta_f)}{c} - 1 \right),$$

and the geometric parameter in α'_δ evaluates to

$$\theta'_f = \cos^{-1} \left(2 \frac{c_f}{\zeta c + c_f \cos(\delta_f)} - 1 \right).$$

To determine the variation of flap lift-effectiveness η_l with the flap-to-chord ratio (c_f/c) at different flap deflection angles, the data shown in Fig. 7.3 was digitally extracted and curve-fitted using a second-order polynomial of the form

$$\eta_l = a_2 \left(\frac{c_f}{c} \right)^2 + a_1 \left(\frac{c_f}{c} \right) + a_0.$$

It is noted that the original plot corresponds to plain trailing-edge flaps with a finite hinge gap. Since the lift effectiveness behavior of single-slotted flaps follows a similar trend, this relationship is treated as an approximate representation for slotted flaps in the present analysis.

The fitted curves for each deflection angle (0° – 60°) yield the following relations:

$$\text{Deflection } 0^\circ : \eta_l = -2.2500 \left(\frac{c_f}{c} \right)^2 + 1.5350 \left(\frac{c_f}{c} \right) + 0.6525$$

$$\text{Deflection } 10^\circ : \eta_l = -2.5000 \left(\frac{c_f}{c} \right)^2 + 1.7300 \left(\frac{c_f}{c} \right) + 0.6050$$

$$\text{Deflection } 20^\circ : \eta_l = -2.2500 \left(\frac{c_f}{c} \right)^2 + 1.6550 \left(\frac{c_f}{c} \right) + 0.4575$$

$$\text{Deflection } 30^\circ : \eta_l = -2.2500 \left(\frac{c_f}{c} \right)^2 + 1.6550 \left(\frac{c_f}{c} \right) + 0.3575$$

$$\text{Deflection } 40^\circ : \eta_l = -1.7500 \left(\frac{c_f}{c} \right)^2 + 1.3050 \left(\frac{c_f}{c} \right) + 0.3375$$

$$\text{Deflection } 50^\circ : \eta_l = -1.5000 \left(\frac{c_f}{c} \right)^2 + 1.1900 \left(\frac{c_f}{c} \right) + 0.2950$$

$$\text{Deflection } 60^\circ : \eta_l = -0.7500 \left(\frac{c_f}{c} \right)^2 + 0.6850 \left(\frac{c_f}{c} \right) + 0.2975$$

These expressions were obtained by fitting the extracted data points from the plot using `polyfit` in MATLAB, ensuring smooth regression curves for use in the design.

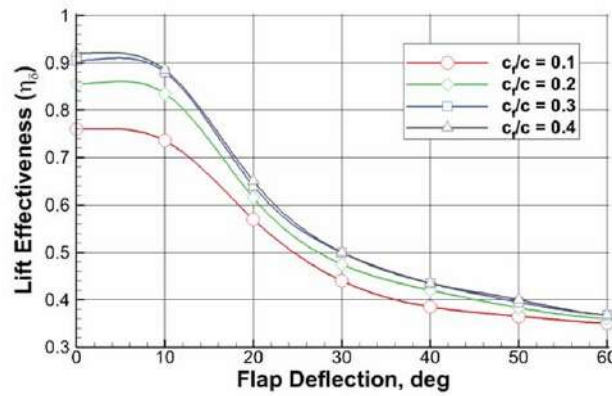


Figure 7.3: Variation of lift effectiveness η_l with flap chord ratio c_f/c at different flap deflections

In the numerical implementation, the lift increment at zero angle of attack is evaluated using

We take $c'_{l,\alpha} \approx C_{l,\alpha} = 2\pi$ (thin-airfoil value) and use the fitted $\eta_l((c_f/c), \delta_f)$. The effectiveness factor is

$$\alpha'_\delta = 1 - \frac{\theta'_f - \sin \theta'_f}{\pi}, \quad \theta'_f = \cos^{-1} \left(2 \frac{(c_f/c)}{\zeta + (c_f/c) \cos \delta_f} - 1 \right).$$

Thus the quantity evaluated in the code is

$$\Delta_f c_{l,0} = \eta_l((c_f/c), \delta_f) \alpha'_\delta((c_f/c), \zeta, \delta_f) C_{l,\alpha} \delta_f \left[\zeta + (c_f/c) \cos \delta_f \right] + c_{l,0} \left(\zeta + (c_f/c) \cos \delta_f - 1 \right)$$

(where angles are in radians). Note that $c_{l,0}$ is retained (nonzero base-section lift at zero AoA). For numerical robustness in computing θ'_f , the argument of $\cos^{-1}(\cdot)$ is clamped to $[-1, 1]$.

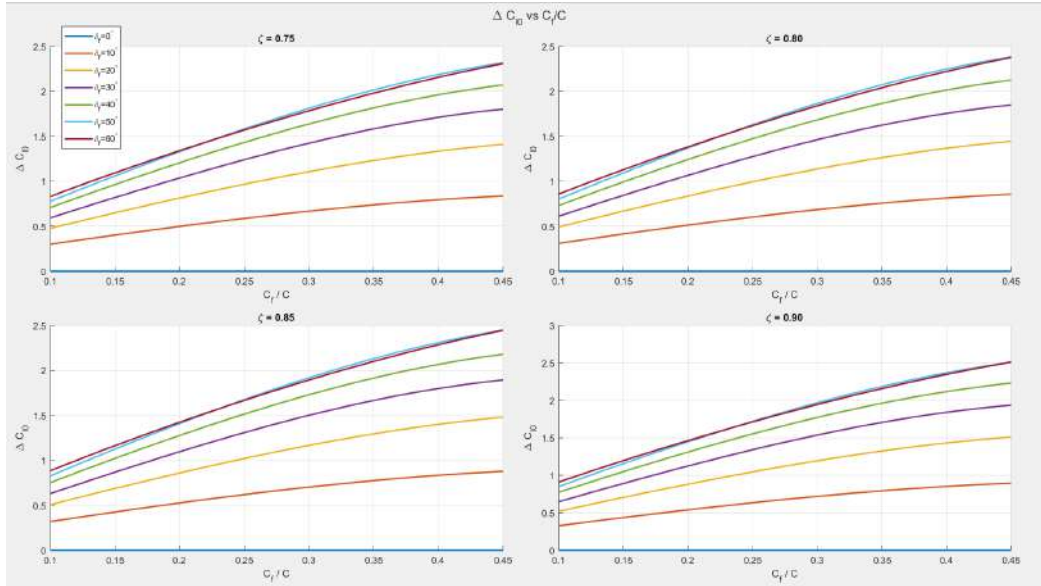


Figure 7.4: Variation of zero-angle lift increment $\Delta_f C_{l,0}$ with flap chord ratio c_f/c for different flap deflections and chord-extension factors ζ .

The above results were generated using the implemented MATLAB routine, which evaluates the analytical expression for $\Delta_f C_{l,0}$ over a range of flap chord ratios, deflection angles, and values of ζ . The plots therefore provide a direct numerical visualization of the closed-form lift increment model.

7.1.2 Drag increment due to flap deflection.

The increment in section profile drag coefficient associated with flap deployment is evaluated in the form

$$\Delta_f C_{dp0} = k_d C_{l,\alpha} \alpha'_\delta \left(\frac{c_f}{c} \right) \delta_f \sin \delta_f + C_{dp0} \left(\frac{c'}{c} - 1 \right),$$

where the first term represents the induced drag contribution associated with the additional circulation generated by the flap, and the second term accounts for the increase in wetted surface and form drag caused by the extended chord. The extended chord ratio is again

$$\frac{c'}{c} = \zeta + \frac{c_f}{c} \cos \delta_f.$$

The same geometric factor used in the lift relation appears here through

$$\alpha'_\delta = 1 - \frac{\theta'_f - \sin \theta'_f}{\pi}, \quad \theta'_f = \cos^{-1} \left(2 \frac{(c_f/c)}{\zeta + (c_f/c) \cos \delta_f} - 1 \right),$$

and $C_{l,\alpha} \approx 2\pi$ is retained from thin-airfoil theory. The parameter k_d is an empirical drag-due-to-lift scaling factor and C_{dp0} is the baseline section profile drag coefficient of the undeflected airfoil. Both terms increase with flap deflection because of the coupled effects of circulation growth and chord extension.

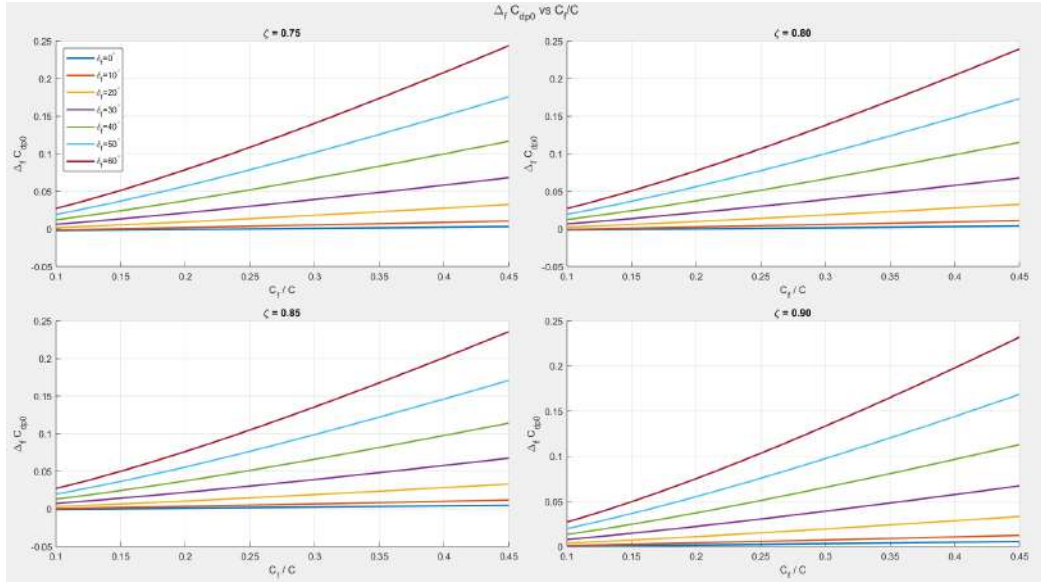


Figure 7.5: Variation of drag increment $\Delta_f C_{dp0}$ with flap chord ratio c_f/c for different flap deflections and chord-extension factors ζ .

7.1.3 Increment in Maximum Lift Coefficient

The effect of trailing-edge flap deflection on wing maximum sectional lift coefficient is strongly influenced by the stalling behavior of the base airfoil section. This stalling behavior is closely correlated with the leading-edge sharpness parameter, Δy , which is the difference between the upper-surface ordinates at 6% chord and 0.15% chord, respectively, normalized by chord. Airfoils with sharp noses (Δy up to 1.5% chord) exhibit leading-edge, or long-bubble, stall behavior, in which a region of separated flow at the leading edge progresses rearward as the angle of attack is increased, leading to stalling of the entire section. For these sections, thin-airfoil theory finds that the increment in maximum lift coefficient due to flap deflection is related to the theoretical lift increment:

$$\Delta_f c_{l,\max} = \Delta_f c_{l,0}^{\text{theory}} \frac{\sin \theta_f}{\pi - (\theta_f - \sin \theta_f)}$$

For each flap configuration, the maximum sectional lift coefficient referenced to the extended chord is computed from

$$c'_{l,\max} = \frac{1}{2} (c_{l,0}^{\text{base}} + \Delta_f c'_{l,0}),$$

where $c_{l,0}^{\text{base}}$ is the maximum lift coefficient of the undeflected airfoil (taken here as $c_{l,0}^{\text{base}} = 0.5$), and $\Delta_f c'_{l,0}$ is the theoretical zero-AoA lift increment referenced to the extended chord. The latter is evaluated using the same expression derived previously,

$$\Delta_f c'_{l,0} = \eta_l \left(\frac{c_f}{c}, \delta_f \right) \alpha'_\delta \left(\frac{c_f}{c}, \zeta, \delta_f \right) C_{l,\alpha} \delta_f \left(\frac{c'}{c} \right) + c_{l,0} \left(\frac{c'}{c} - 1 \right),$$

with the extended-chord relation

$$\frac{c'}{c} = \zeta + \frac{c_f}{c} \cos \delta_f.$$

The dependence of $c'_{l,\max}$ on flap chord ratio c_f/c , flap deflection δ_f , and chord-extension factor ζ therefore arises entirely from the incremental lift term $\Delta_f c'_{l,0}$, which embeds the fitted effectiveness function η_l , the geometric effectiveness factor α'_δ , and the extended-chord ratio. The resulting curves illustrate how the attainable maximum lift coefficient increases with both flap deflection and flap spanwise length, but is moderated by the chosen value of ζ , which controls the effective chord growth of the slotted flap.

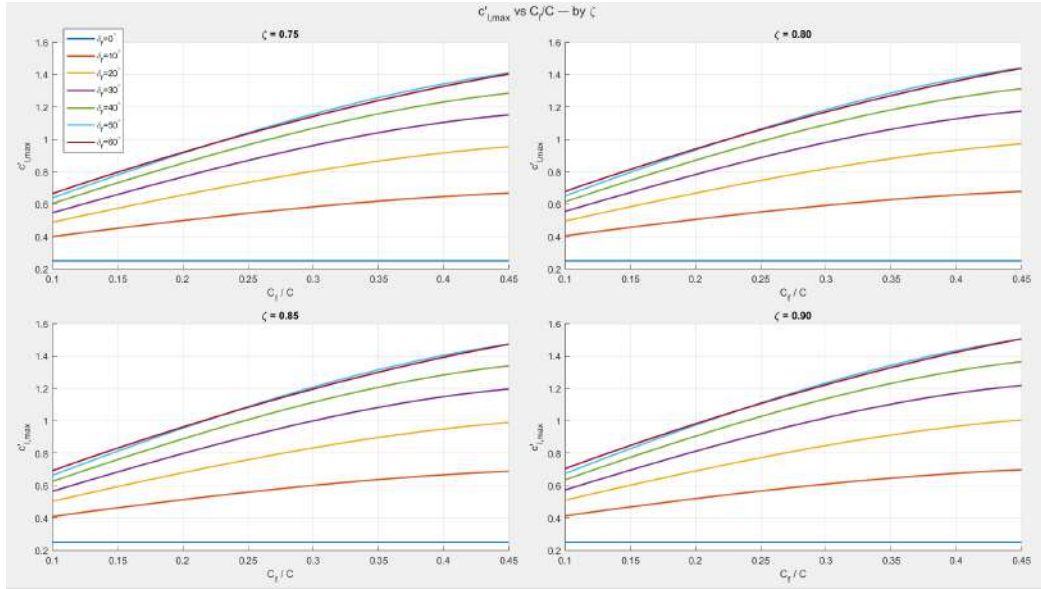


Figure 7.6: Maximum sectional lift coefficient $c'_{l,max}$ as a function of c_f/c for each flap deflection and chord–extension factor ζ .

7.1.4 Lift Curve Variation with Flap Deflection

The effect of flap deflection on the complete sectional lift curve is evaluated by reconstructing the linearised $C_l(\alpha)$ relationship using two anchor points. For each flap configuration, the incremental lift at zero angle of attack is first obtained from the previously derived expression for $\Delta_f c'_{l,0}$, and this value is then added to a reference clean-wing lift coefficient at $\alpha = 0^\circ$. A second anchor point at $\alpha = 10^\circ$ is constructed in a similar manner, with an additional incremental term accounting for the progressive growth of lift with angle of attack.

Once the two anchor values are defined, the full lift curve is generated by assuming a linear variation between them, i.e.

$$C_l(\alpha) = C_l(0^\circ) + \frac{C_l(10^\circ) - C_l(0^\circ)}{10^\circ} \alpha,$$

where the slope is evaluated separately for each combination of flap chord ratio (c_f/c), flap deflection δ_f , and chord–extension factor ζ . In this formulation, the influence of flap geometry enters only through the incremental lift term $\Delta_f c'_{l,0}$, while the linear C_l – α trend is preserved for clarity of comparison.

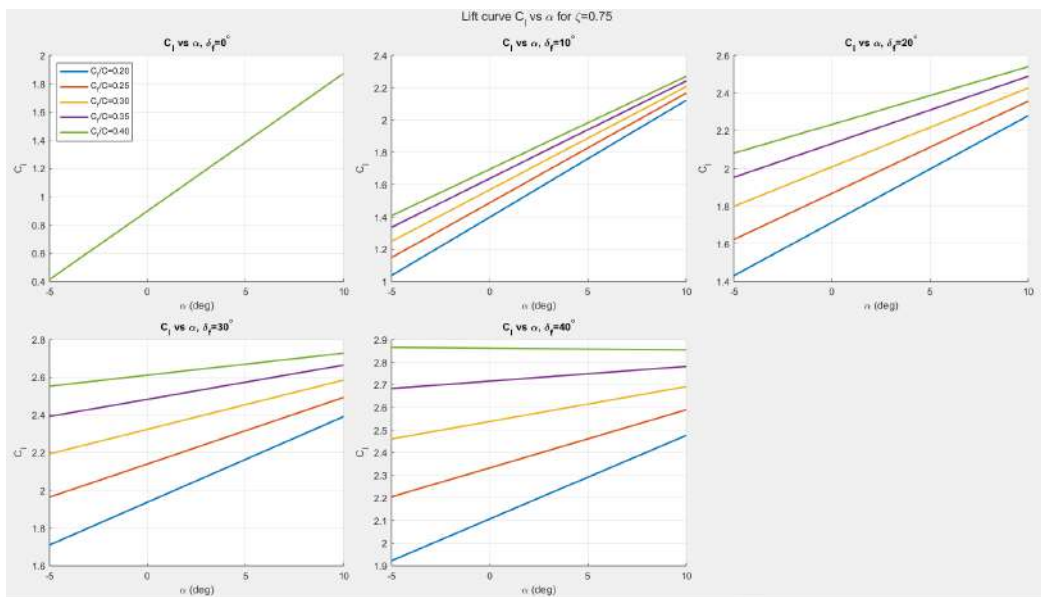


Figure 7.7: $C_l(\alpha)$ variation for $\zeta = 0.75$

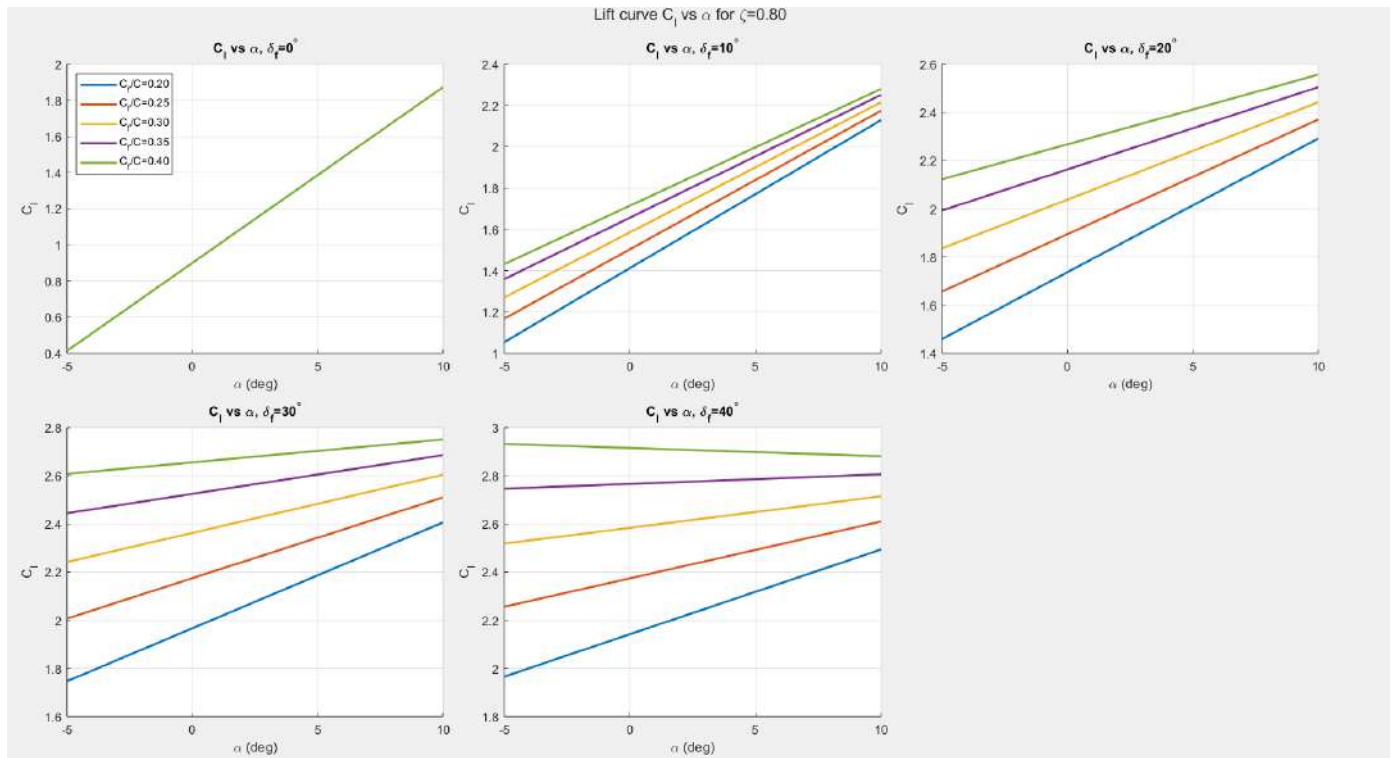


Figure 7.8: $C_l(\alpha)$ variation for $\zeta = 0.80$

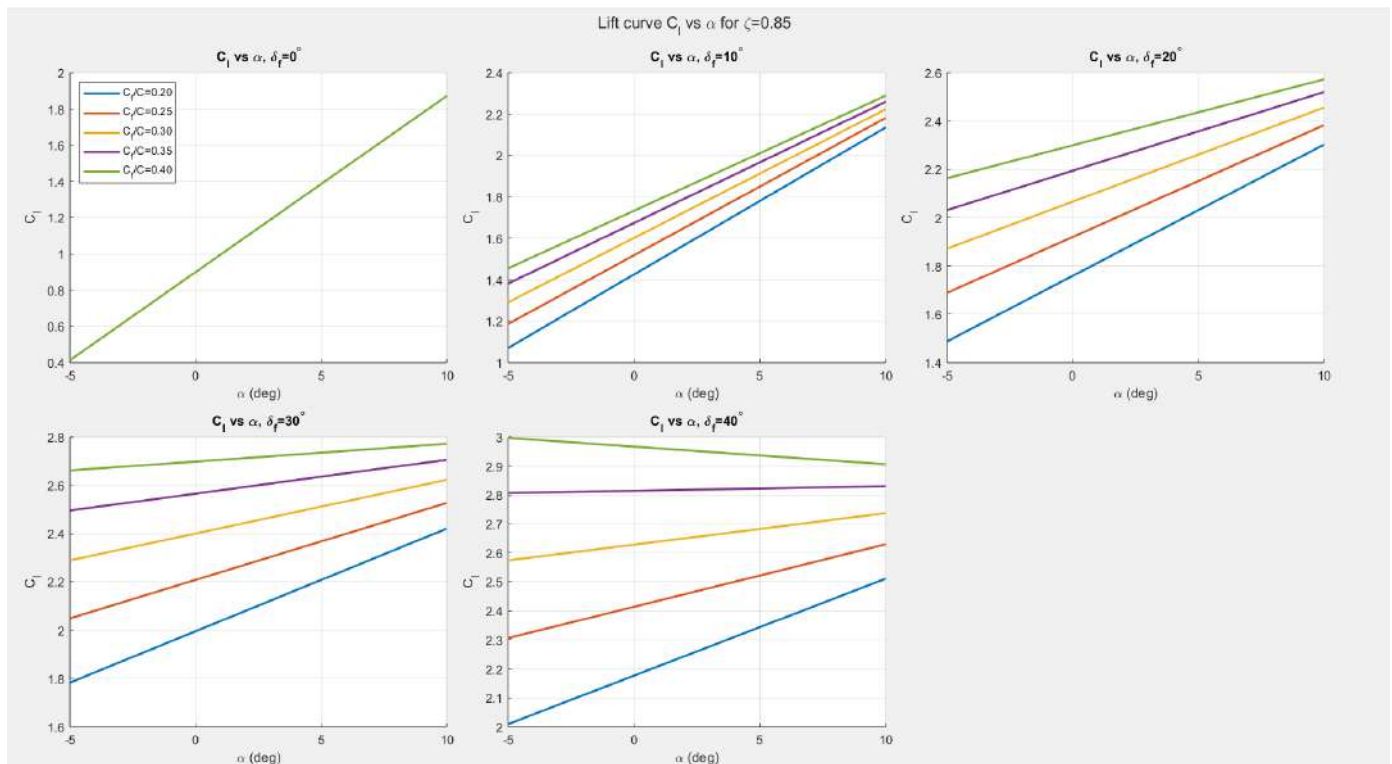


Figure 7.9: $C_l(\alpha)$ variation for $\zeta = 0.85$

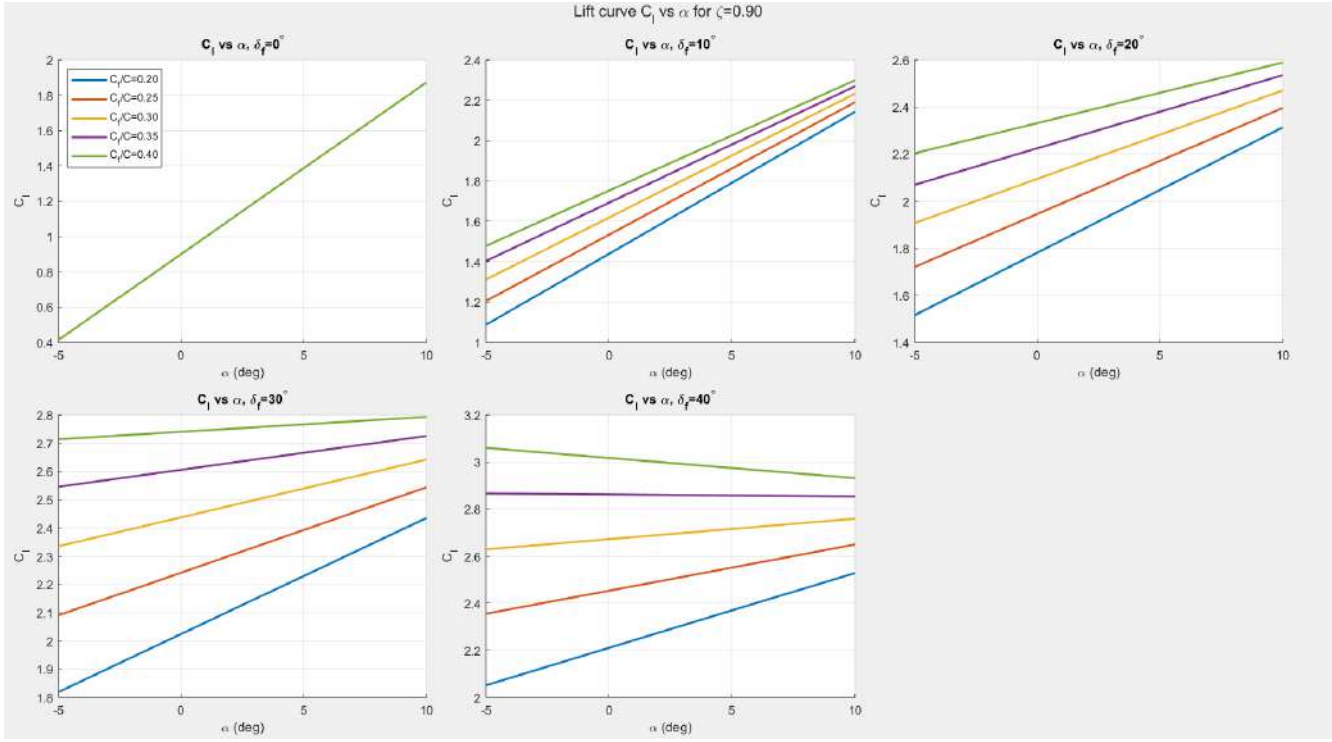


Figure 7.10: $C_l(\alpha)$ variation for $\zeta = 0.90$

The MATLAB-generated subplots clearly show how both the lift-curve offset at $\alpha = 0^\circ$ and the slope of the $C_l(\alpha)$ relation are progressively altered by increasing flap deflection, larger flap chord ratios, and higher values of ζ . This provides a direct visual comparison of how flap geometry and deflection influence the usable lift range of the section.

7.1.5 Drag Polar Variation with Flap Deflection

The sectional drag coefficient in the presence of flap deflection is modelled as the sum of a modified zero-lift drag term and an induced drag term of the classical lifting-line form,

$$C_D(\alpha) = C_{D0} + \frac{C_l^2(\alpha)}{\pi e AR},$$

where the induced component is evaluated using an Oswald efficiency factor $e = 0.85$ and an aspect ratio $AR = 9$. Both e and AR values are adopted from the estimates obtained in the preceding sections, where the efficiency factor was determined from preliminary aerodynamic analyses and the aspect ratio was selected from the range of considered values $AR = 9, 12, \text{ and } 15$ as being representative of the baseline configuration.

The modified profile drag term is written as

$$C_{D0} = C_{D0}^{\text{clean}} + \Delta_f C_{dp0}, \quad C_{D0}^{\text{clean}} = 0.018,$$

so that the contribution of flap deployment enters entirely through the previously evaluated drag increment $\Delta_f C_{dp0}$. The lift term $C_l(\alpha)$ is the same linear reconstruction introduced in the preceding subsection and therefore embeds the dependence on flap deflection δ_f , flap chord ratio c_f/c , and chord-extension factor ζ .

The resulting drag polar is generated over the range $-15^\circ \leq \alpha \leq 20^\circ$, and is plotted for multiple values of c_f/c within each flap deflection case. As in the lift analysis, four values of the chord-extension factor are examined:

$$\zeta = 0.75, 0.80, 0.85, 0.90.$$

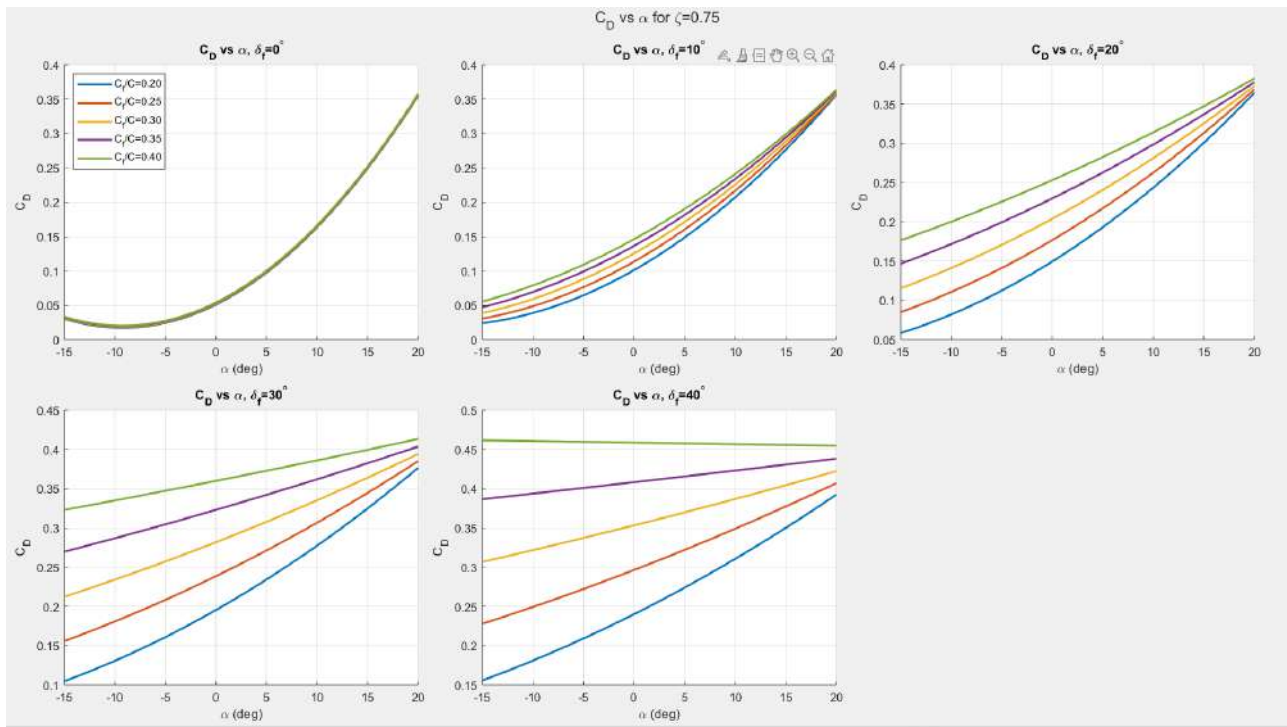


Figure 7.11: Drag polar $C_D(\alpha)$ for $\zeta = 0.75$ at multiple flap chord ratios c_f/c and flap deflections δ_f .

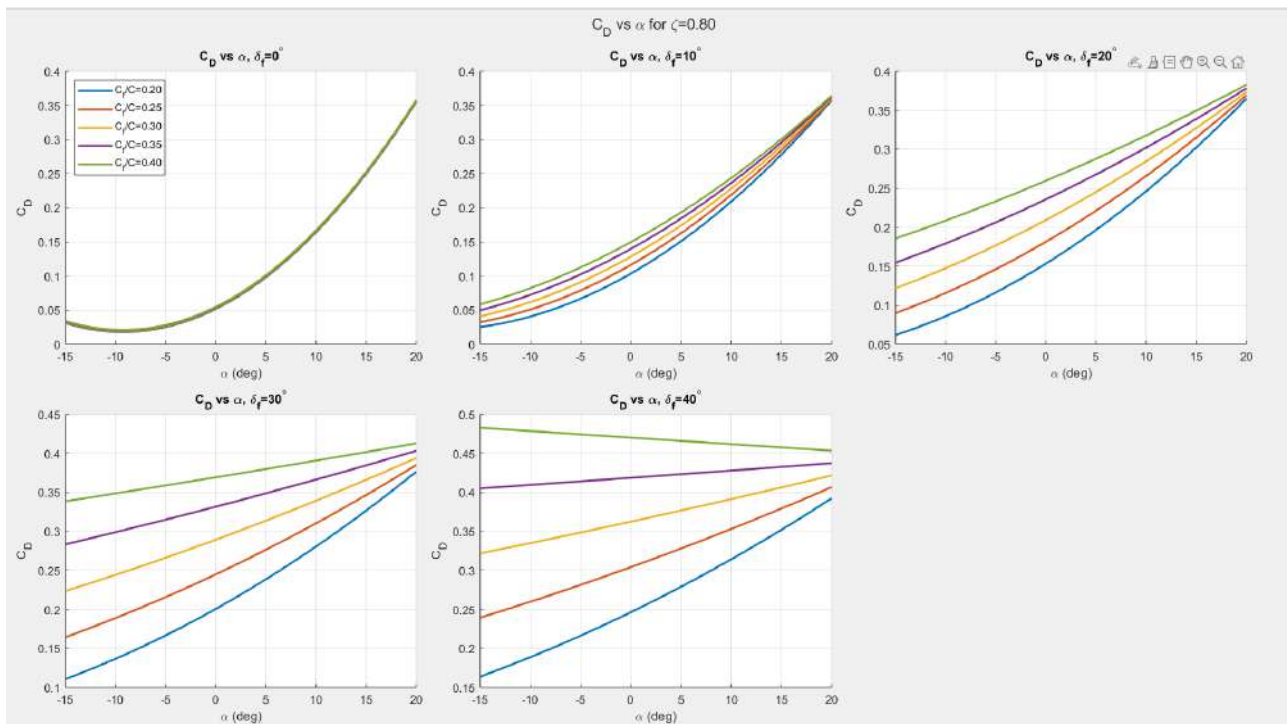


Figure 7.12: Drag polar $C_D(\alpha)$ for $\zeta = 0.80$ at multiple flap chord ratios c_f/c and flap deflections δ_f .

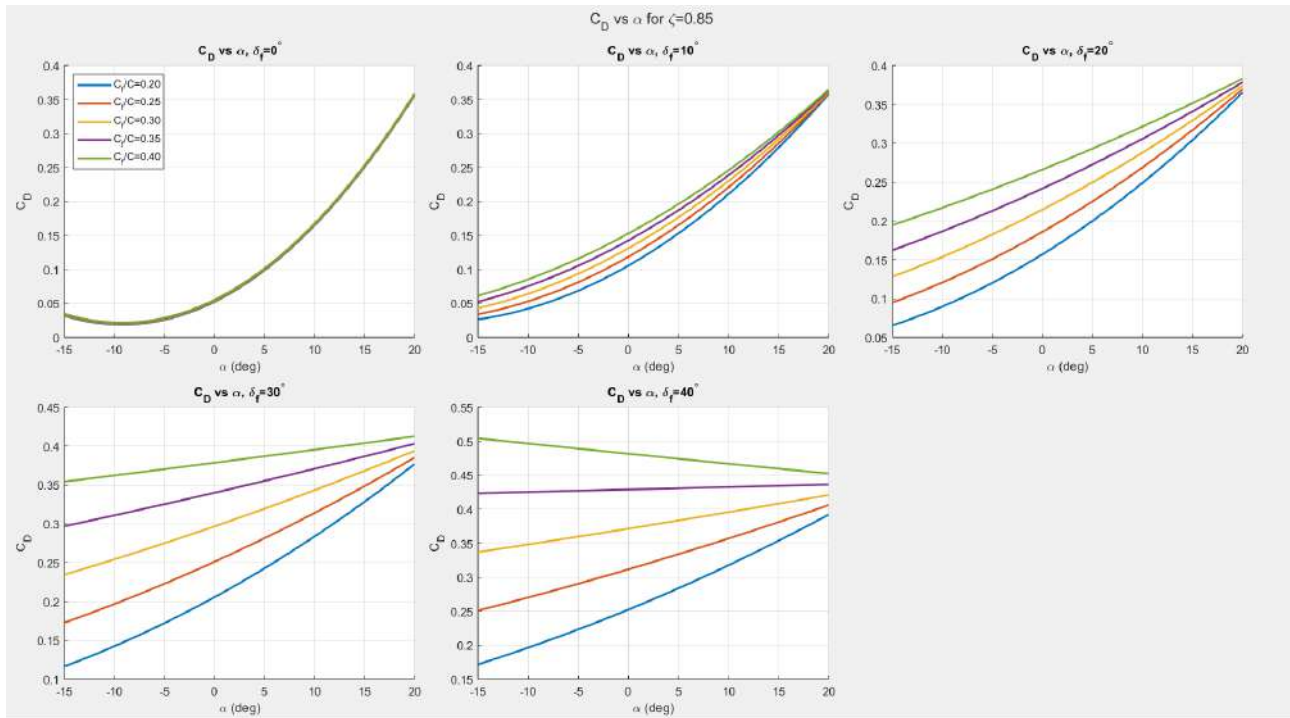


Figure 7.13: Drag polar $C_D(\alpha)$ for $\zeta = 0.85$ at multiple flap chord ratios c_f/c and flap deflections δ_f .

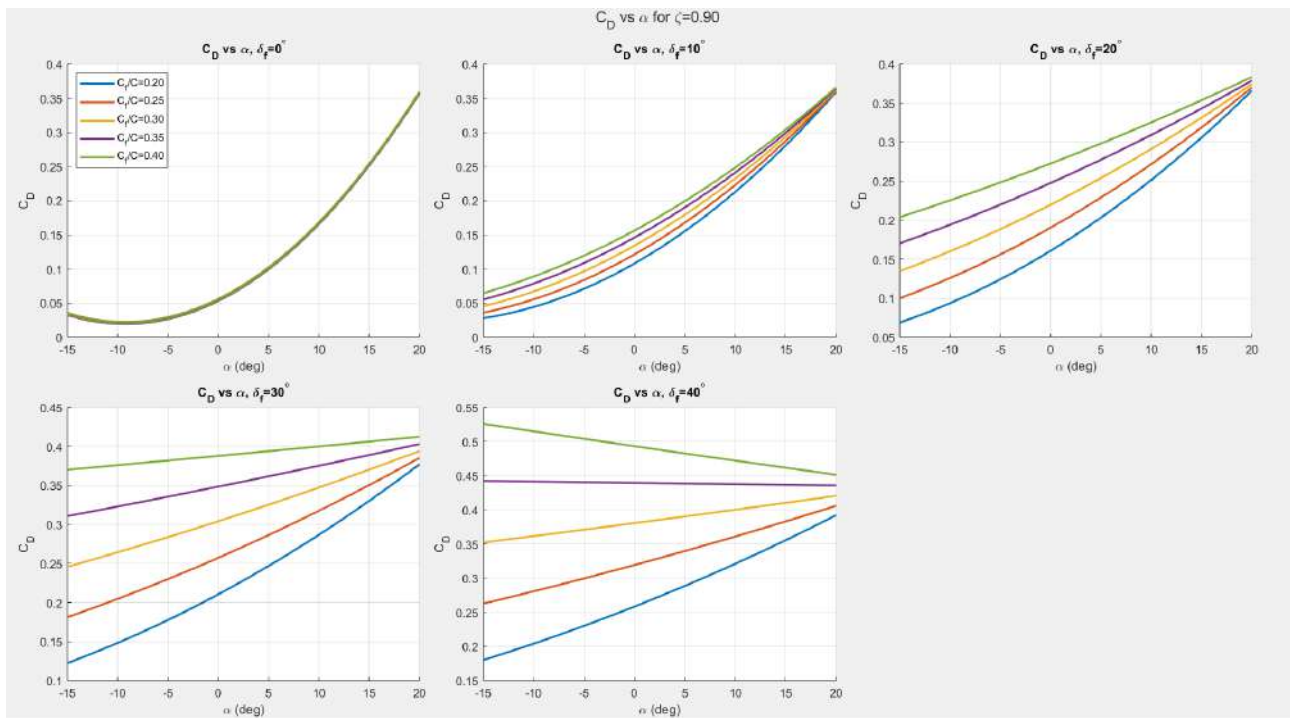


Figure 7.14: Drag polar $C_D(\alpha)$ for $\zeta = 0.90$.

These plots show the overall impact from flap deflection along with flap chord ratio and chord extension. They affect how drag builds up as angles of attack climb higher. Bigger flap chords tend to raise the starting drag level quite a bit. Higher deflection angles do the same for that baseline and also speed up the rise in induced drag. Then higher ζ values make things worse still. They amplify the extra drag through more chord extension.

7.1.6 Aerodynamic Efficiency Variation: C_l/C_D vs α

The aerodynamic efficiency of the airfoil–flap system is assessed using the ratio

$$\frac{C_l}{C_D},$$

which provides a direct measure of lift produced per unit drag. Since both $C_l(\alpha)$ and $C_D(\alpha)$ have already been constructed as functions of flap chord ratio (c_f/c), flap deflection δ_f , and chord–extension factor ζ , the efficiency ratio follows directly from

$$\frac{C_l}{C_D}(\alpha) = \frac{C_l(\alpha)}{C_{D0} + \frac{C_l^2(\alpha)}{\pi e AR}},$$

where the zero–lift drag term is

$$C_{D0} = C_{D0}^{\text{clean}} + \Delta_f C_{dp0}, \quad C_{D0}^{\text{clean}} = 0.018,$$

and the induced drag contribution uses the same aspect ratio and efficiency factor as before,

$$e = 0.85, \quad AR = 9.$$

Because both $C_l(\alpha)$ and $\Delta_f C_{dp0}$ grow with flap deflection, the resulting efficiency curve reflects the trade–off between the beneficial lift increment and the associated drag penalty. The plots therefore reveal how the usable lift–to–drag ratio shifts with increasing flap size, flap angle, and effective chord growth through ζ .

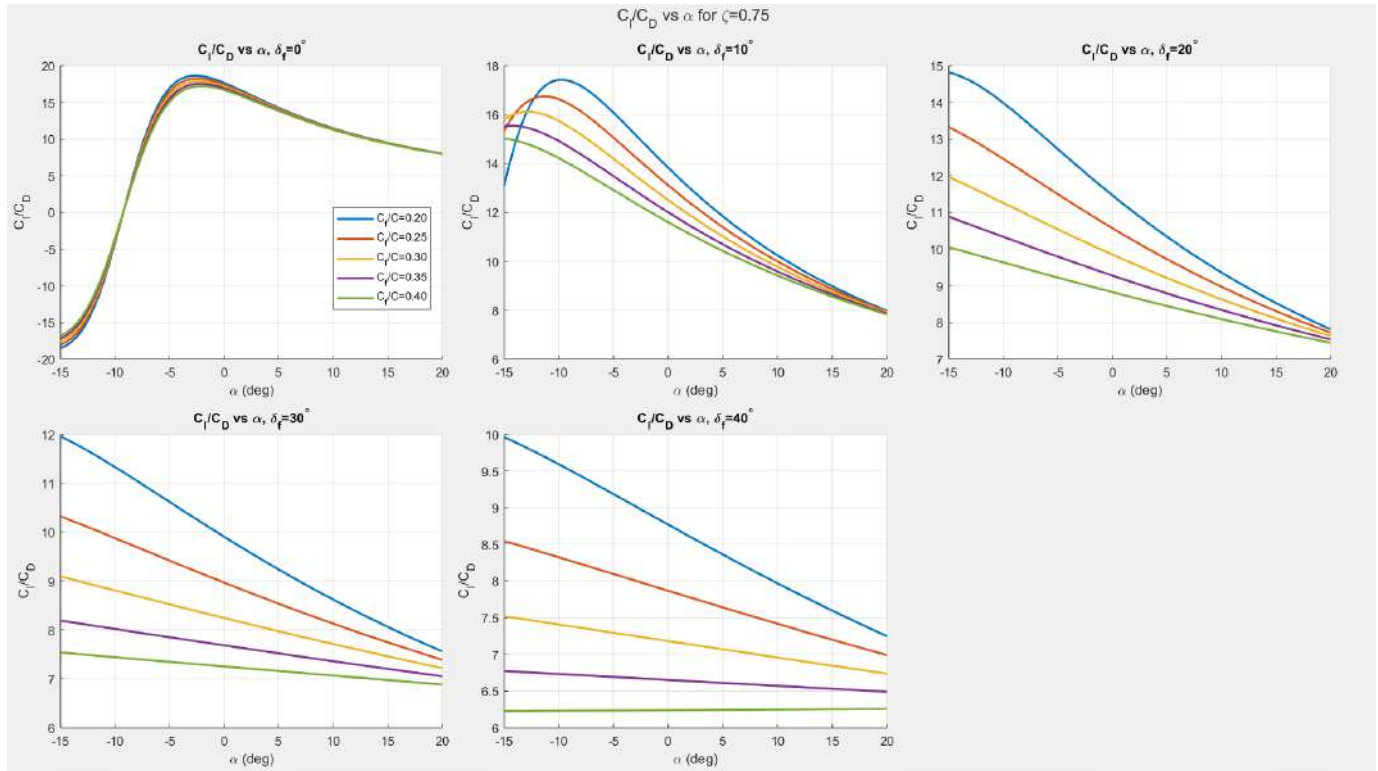


Figure 7.15: Variation of C_l/C_D with angle of attack for $\zeta = 0.75$ at different flap chord ratios c_f/c .

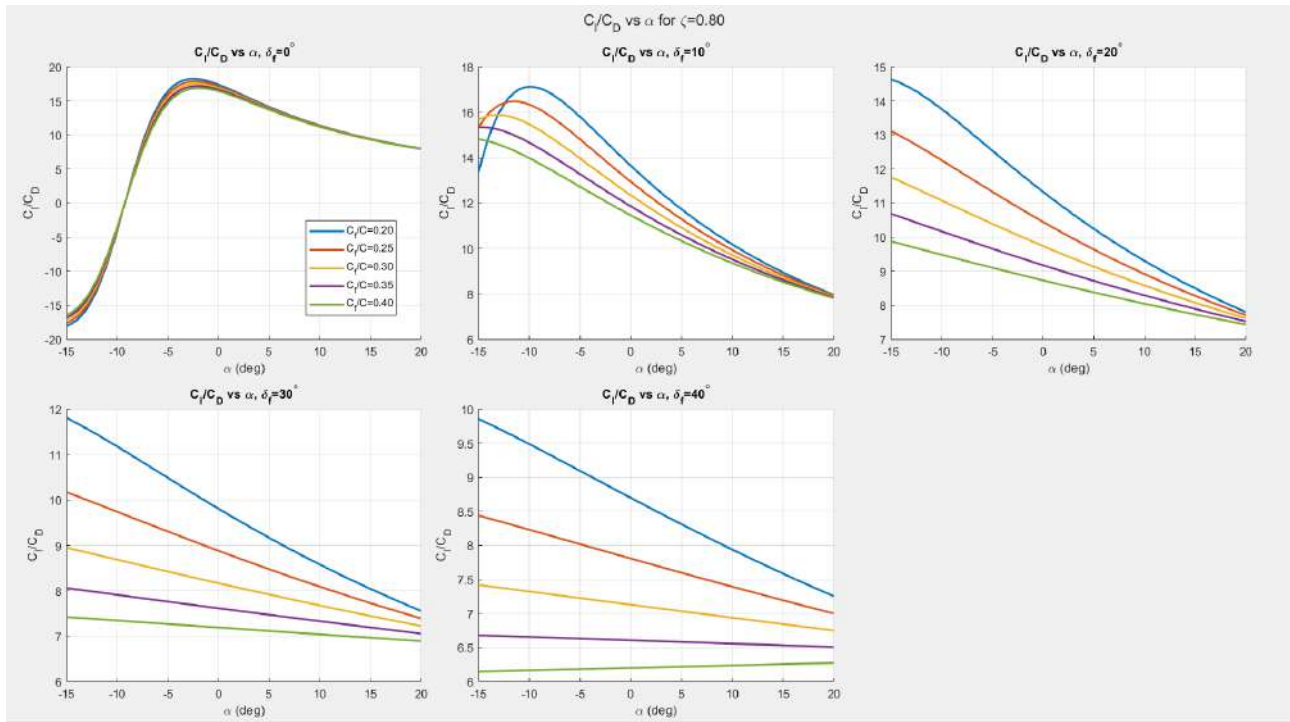


Figure 7.16: C_L/C_D variation for $\zeta = 0.80$.

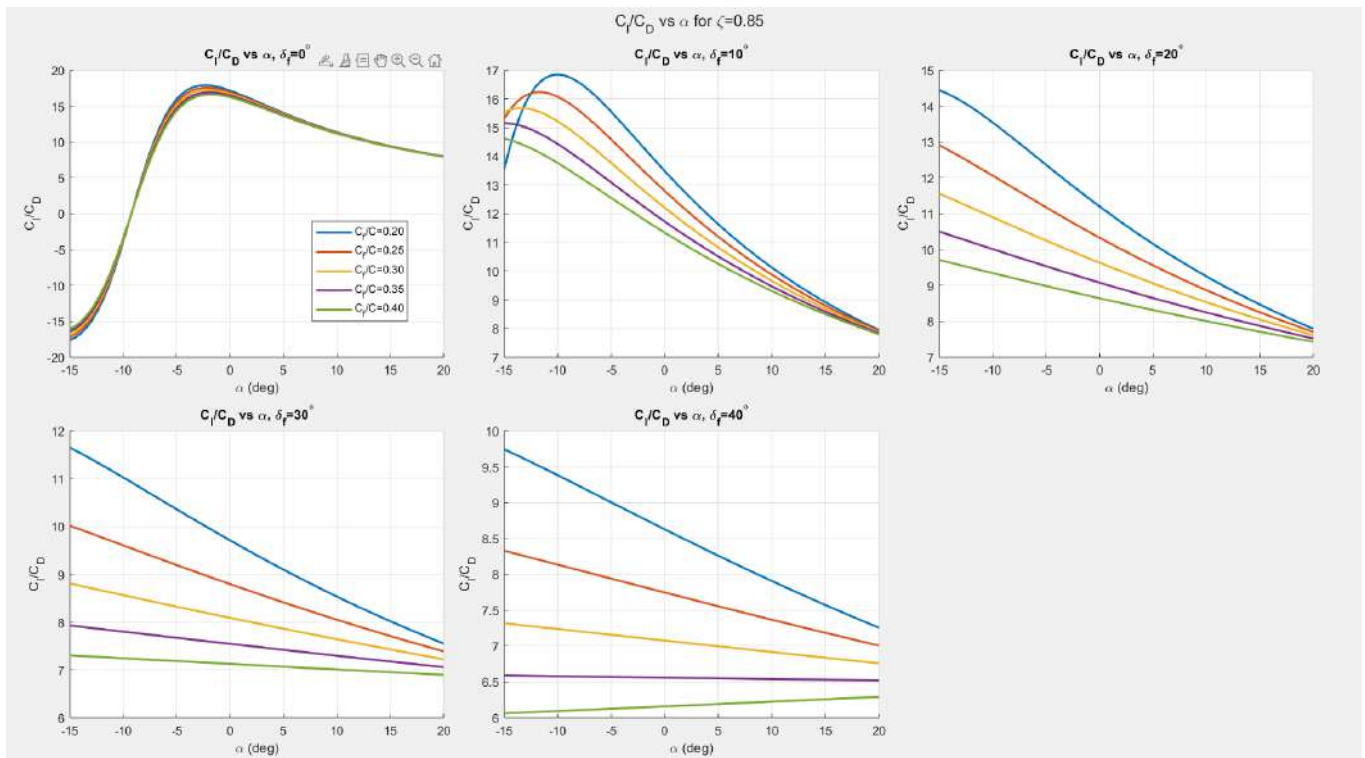


Figure 7.17: C_L/C_D variation for $\zeta = 0.85$.

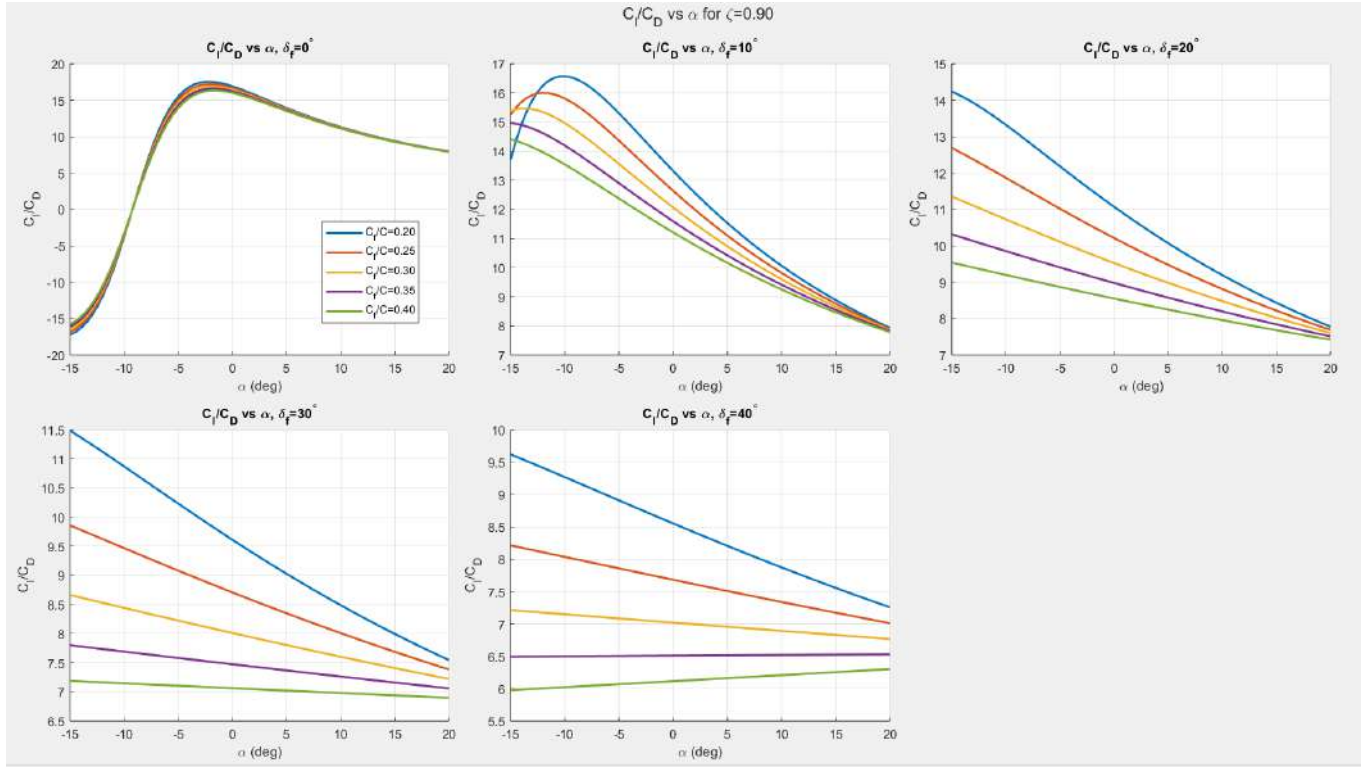


Figure 7.18: C_l/C_D variation for $\zeta = 0.90$.

Across all cases, the peak value of C_l/C_D typically shifts to lower angles of attack as flap deflection increases, while the maximum efficiency level decreases due to the drag growth. Increasing ζ further reduces the efficiency ceiling because of the additional chord extension drag, despite the accompanying lift benefit.

7.2 Selection of Geometry for Flaps

7.2.1 Effect of ζ

An increase in the chord–extension factor ζ from 0.75 to 0.90 produces a noticeable rise in the lift coefficient at zero angle of attack, $\Delta_f C_{l,0}$, whereas the slope of the $\Delta_f C_{l,0}$ versus (c_f/c) curve remains nearly unchanged. The increment in zero–lift drag coefficient, $\Delta_f C_{dp0}$, decreases with increasing ζ , while the increment in maximum lift coefficient, $\Delta_f C_{l,max}$, shows a clear upward trend as ζ increases.

For the subsequent plots of C_l vs. α , C_D vs. α , and C_l/C_D vs. α , the variations are predominantly governed by the flap chord ratio (c_f/c) and the flap deflection angle δ_f , with ζ having only a secondary influence.

From these observations—particularly those derived from the increment plots ($\Delta_f C_{l,0}$ vs. c_f/c , $\Delta_f C_{l,max}$ vs. c_f/c , and $\Delta_f C_{dp0}$ vs. c_f/c)—a value of $\zeta = 0.90$ provides the most favourable aerodynamic characteristics and is therefore selected for further analysis.

$$\zeta = 0.90$$

(7.1)

7.2.2 Effect of Flap Chord Ratio (c_f/c)

Considering $\zeta = 0.90$, the first three increment plots show that $\Delta_f C_{l,0}$, $\Delta_f C_{dp0}$, and $\Delta_f C_{l,max}$ all increase with increasing flap chord ratio (c_f/c) . However, examination of the lift curves C_l vs. α for $\zeta = 0.90$ indicates that, at higher flap deflections, configurations with $(c_f/c) > 0.30$ tend to exhibit either reduced positive slope, nearly flat characteristics, or even a slightly negative slope.

Similarly, for the C_l/C_D vs. α plots at $\zeta = 0.90$, the slope approaches zero for $(c_f/c) > 0.30$ at large deflection angles, implying diminished aerodynamic efficiency.

Taking all these effects into account, a value slightly greater than $(c_f/c) = 0.25$ but below 0.30 is deemed optimal. The absolute slope of both the C_l vs. α and C_l/C_D vs. α curves should remain distinctly positive, leading to an ideal design range of

$$0.25 < \frac{c_f}{c} < 0.30,$$

with the chosen representative value

$$\frac{c_f}{c} = 0.256.$$

7.2.3 Estimation of Gap and Overlap for a Single-Slotted Flap

The geometric parameters of a single-slotted flap — namely the gap g and overlap O — are known to vary approximately linearly with flap deflection angle δ_f . At zero deflection ($\delta_f = 0^\circ$), the flap has a small finite gap, while at the maximum deflection ($\delta_f = 45^\circ$) the gap and overlap both increase slightly due to flap rotation and hinge offset geometry.

Based on available reference data and geometric approximations:

$$g = 0.008c + \frac{0.017c}{45} \delta_f, \quad O = 0.020c + \frac{0.015c}{45} \delta_f,$$

where δ_f is the flap deflection angle in degrees and c is the main airfoil chord. Both expressions are written in the standard straight-line form,

$$y = mx + c,$$

where y represents either the non-dimensional gap or overlap, $x = \delta_f$, the slope m denotes the rate of change per degree of flap deflection, and the constant term c corresponds to the initial value at zero deflection.

Using these linear relations:

$$\frac{dg}{d\delta_f} = \frac{0.017c}{45} = 3.78 \times 10^{-4} c/^\circ, \quad \frac{dO}{d\delta_f} = \frac{0.015c}{45} = 3.33 \times 10^{-4} c/^\circ.$$

Thus, the gap and overlap at any intermediate flap deflection can be directly estimated using the linear equations above. This linear assumption provides a simple and sufficiently accurate geometric model for preliminary aerodynamic analysis of the slotted-flap system.

7.2.4 Curvature of the Slot

Experimental observations in the classic *Theory of Wing Sections*, from NACA Report 824, point out something key about slotted flaps. Those with a longer lip or more extended slot curvature pretty much deliver higher lift gains. This stands in contrast to setups with shorter or sharper slot shapes. The reason behind this boost comes down to how a longer and gently curved slot handles things. It guides the high energy flow from the lower surface more effectively right over the flap's upper side. That action delays flow separation quite a bit. It also improves pressure recovery in a solid way.

For the present configuration, the flap chord is taken as

$$\frac{c_f}{c} = 0.30$$

where c is the main airfoil chord. In the fully retracted position, approximately half of the flap length is located beneath the main airfoil, ensuring smooth surface continuity and minimizing aerodynamic drag. Accordingly, the slot opening on the lower surface begins at approximately

$$x_s = 0.73 c$$

measured from the leading edge of the main airfoil.

Based on the geometric trends and empirical data reported in the literature, the curvature radius of the slot centerline,

denoted as R_s , should typically lie within the range

$$0.04c \leq R_s \leq 0.08c.$$

Larger curvatures (smaller R_s) cause the slot jet to turn too sharply, resulting in flow separation and loss of lift, while excessively small curvatures (larger R_s) lead to an ineffective slot that does not provide sufficient re-energization of the boundary layer.

Hence, an intermediate value provides the best balance between flow guidance and lift augmentation.

For the present slotted-flap design, the curvature radius of the slot is therefore selected as

$$R_s = 0.07c.$$

This value ensures smooth flow redirection through the slot while maintaining strong lift enhancement due to the favorable pressure distribution over the flap surface.

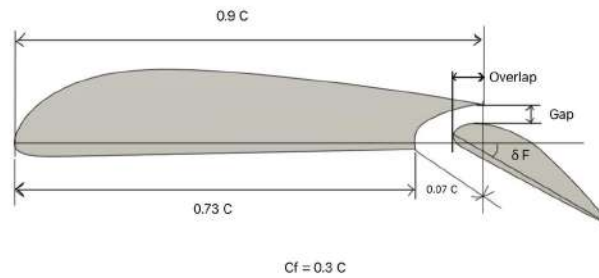


Figure 7.19: Representative slot curvature geometry for a single-slotted flap. The chosen curvature radius $R_s = 0.07c$ ensures smooth flow transition and effective boundary-layer re-energization.

7.3 Flap Span

The total wingspan of the aircraft is represented by b . For a conventional configuration, the aileron chord length is typically about $0.25c$, where c denotes the mean aerodynamic chord. Based on standard aileron design guidelines and trends obtained from the corresponding reference graph, the total spanwise extent of the ailerons corresponds to approximately $0.38b$ of the full wing span.

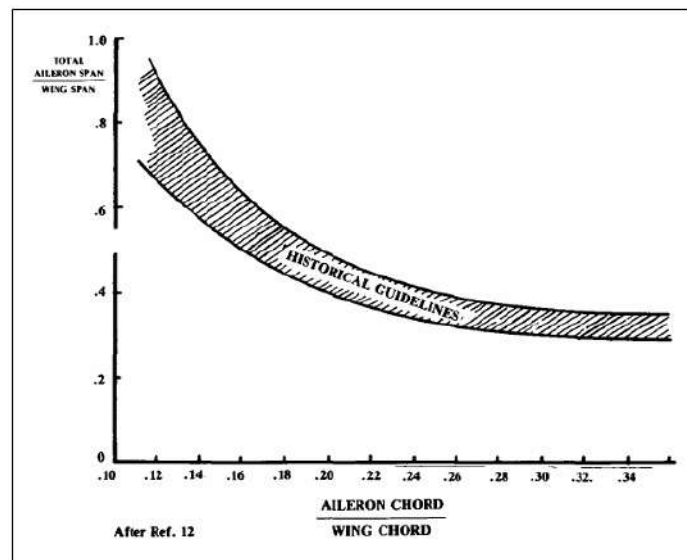


Figure 7.20: Aileron Chord and Coverage of Span

The remaining portion of the span is allocated for flap installation. However, to ensure smooth aerodynamic transitions and avoid structural interference, clearance regions are maintained near both the fuselage and the ailerons. These clearances, each equivalent to $0.015b$ of the span, are filled with elastomeric materials to allow limited flexibility and smooth deflection at the junctions. This design approach helps minimize torsional stresses and prevents undesirable twisting of the wing structure during control surface deflection.

After accounting for the designated clearances, the effective span available for flap placement is $b(f) = 0.59b$. Below is the graph for aileron chord and span distribution and the position and span of the flaps on the half wing.

7.4 2D Flap Deflection CFD Study for Takeoff

This study examines the variation of the aerodynamic coefficients with flap deflection angles over a range of angles of attack for a given airfoil and flap configuration in 2D. The analysis is performed for the NACA 4412 airfoil with NACA 4412 flaps employing a single slotted flap configuration. The flap deflections are set in accordance with the relations described earlier in Section 7. The optimal deflection angle and angle of attack are determined based on the maximum lift coefficient while maintaining reasonable lift to drag ratio under the takeoff conditions.

7.4.1 Geometry and Case Setup

The figure below illustrates the geometries analyzed in this study, indicating both the angle of attack and the flap deflection angles. The geometry is enclosed within a rectangular computational domain, extending $5c$ upstream, $10c$ downstream, and $5c$ laterally. Additionally, two refinement regions are included, with their respective extents illustrated in the figures below.

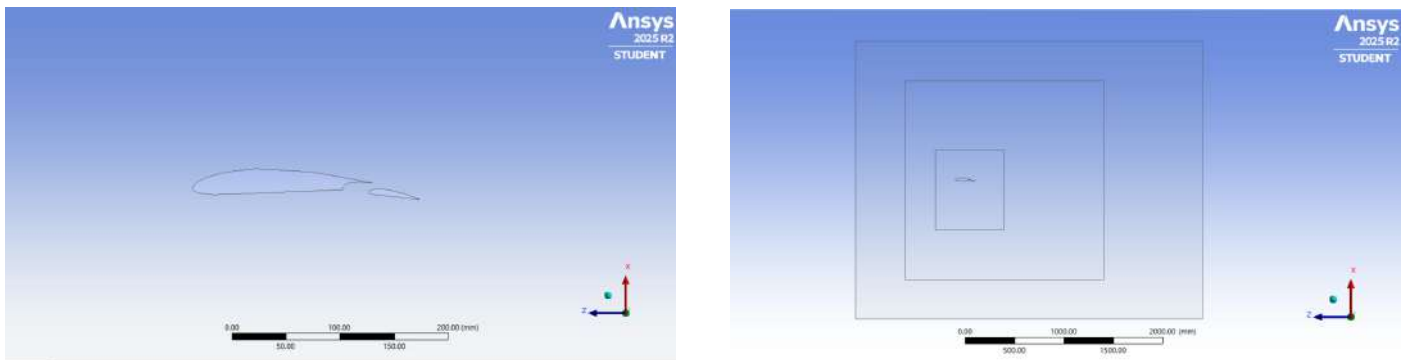


Figure 7.21: 2D Geometry for Deployed Flaps

Boundary conditions: A uniform freestream velocity (U_∞) is applied at the upstream boundary whereas at the downstream boundary, a Neumann-type boundary condition is imposed for velocity, corresponding to a zero viscous stress vector. Along the upper and lower boundaries, the velocity component normal to the stress vector on these surfaces is set to zero. The boundary conditions, domain dimensions, and refinement regions are kept nearly identical for all subsequent simulations.

7.4.2 Mesh Details

A quadrilateral-dominated unstructured mesh has been generated featuring structured inflation layers near the airfoil vicinity. The dimensionless wall distance (y^+) was set to 1, as required by the $k-\omega$ SST model for 2D cases. This corresponds to a first cell height of 0.02 mm, as estimated using an online y^+ calculator. Based on the Blasius flat-plate solution, the boundary layer thickness was estimated to be approximately 2 mm. Using this information and a growth rate of 1.2, a total of two inflation layers were applied to accurately resolve the boundary layer region. The resulting mesh is shown below along with the boundary layer refinement around the airfoil. It contains approximately 34397 nodes and 33805 elements with similar mesh characteristics applied to remaining cases.

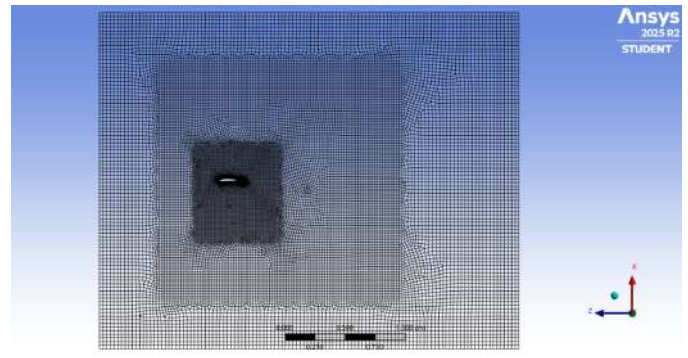
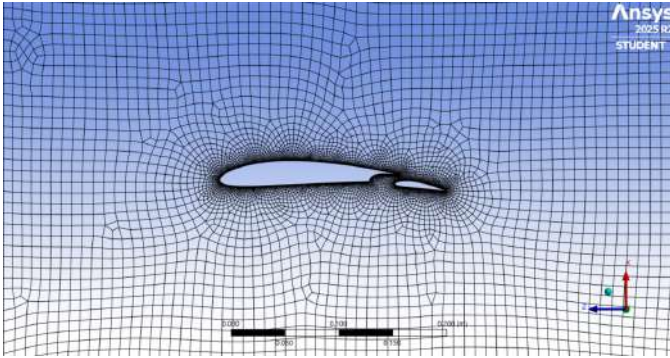


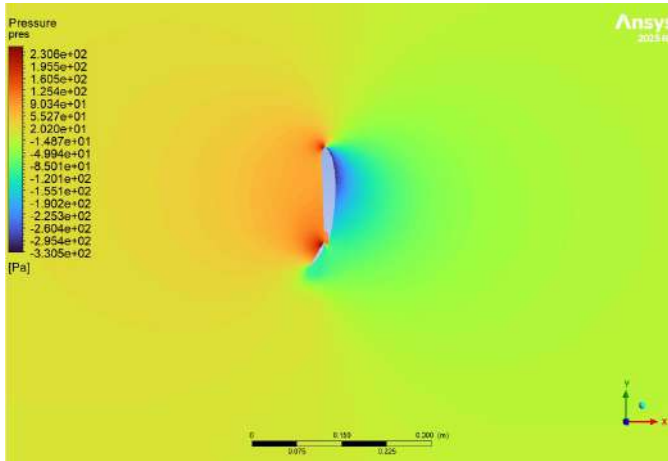
Figure 7.22: Meshing for Deployed Flaps

7.4.3 Results

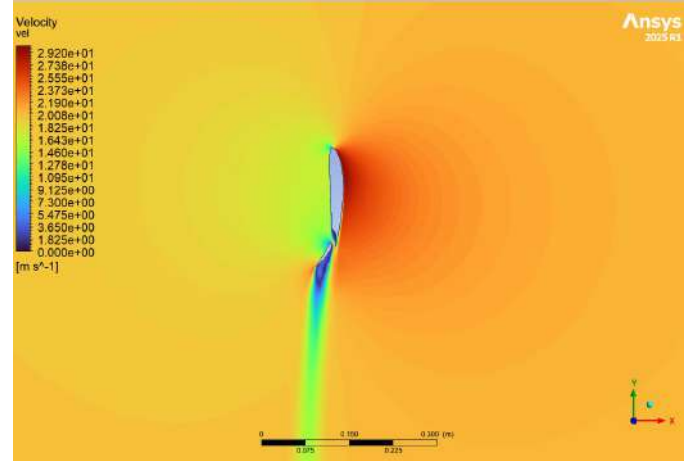
AoA (°)	Defl. (°)	C_L	C_D	C_L/C_D
0	5	0.7309	0.02197	33.26
	10	0.9674	0.02550	37.93
	15	1.1657	0.03094	37.67
	20	1.2621	0.04094	30.83
	25	1.2124	0.06262	19.36
	30	1.3190	0.07662	17.21
	35	1.4005	0.08831	15.86
	40	1.4794	0.10208	14.49
	45	1.5421	0.11320	13.62
2	5	0.9634	0.02464	39.09
	10	1.1925	0.02862	41.67
	15	1.3645	0.03508	38.89
	20	1.4767	0.04615	32.00
	25	1.4316	0.06612	21.65
	30	1.5154	0.08001	18.94
	35	1.6003	0.09382	17.06
	40	1.6633	0.10491	15.85
	45	1.7247	0.11643	14.81
4	5	1.1851	0.02861	41.42
	10	1.4259	0.03366	42.36
	15	1.6125	0.04077	39.55
	20	1.7161	0.05252	32.68
	25	1.6101	0.07206	22.35
	30	1.6895	0.08511	19.85
	35	1.7740	0.09794	18.11
	40	1.8382	0.11009	16.70
	45	1.9012	0.12022	15.81

AoA (°)	Defl. (°)	C_L	C_D	C_L/C_D
6	5	1.4157	0.03348	42.28
	10	1.6544	0.03875	42.69
	15	1.8474	0.04686	39.43
	20	1.8820	0.05951	31.62
	25	1.8069	0.07939	22.76
	30	1.8699	0.09229	20.26
	35	1.9447	0.10414	18.67
	40	2.0157	0.11677	17.26
	45	2.0684	0.12934	15.99
8	5	1.6488	0.04101	40.21
	10	1.8611	0.04706	39.54
	15	1.9825	0.05694	34.82
	20	2.0461	0.07047	29.04
	25	1.9867	0.08849	22.45
	30	2.0378	0.10174	20.03
	35	2.1042	0.11626	18.10
	40	2.1634	0.12689	17.05
	45	2.2117	0.13951	15.85
10	5	1.8278	0.05164	35.39
	10	2.0258	0.06072	33.36
	15	2.1523	0.06843	31.45
	20	2.1365	0.08313	25.70
	25	2.0996	0.10230	20.52
	30	2.1713	0.11669	18.61
	35	2.2226	0.12953	17.16
	40	2.2620	0.14501	15.60
	45	2.3089	0.15379	15.01

Table 7.1: Lift and Drag Data for Various Angles of Attack and Flap Deflections



(a) Pressure Contour



(b) Velocity Contour

Figure 7.23: Pressure and velocity contours at $\alpha = 0^\circ$ with flap deflection 30° .

Till 10° angle of attack, only flaps are deployed. For angles of attack 0 - 10° , based on the above flap deflection study, the optimal flap deflection is 30° because it gives high lift-coefficient while maintaining high lift-to-drag ratio at various angles of attack.

8 Slats

Leading-edge slats are deployed to delay stall by controlling the flow over the nose of the airfoil. Unlike trailing-edge flaps, which primarily increase camber, a slat acts by re-energizing the boundary layer, allowing the main airfoil to sustain higher angles of attack before flow separation occurs. This results in both an increase in lift at zero angle of attack and a significant rise in the maximum sectional lift coefficient, making slats highly effective in high-lift systems for take-off and landing.

8.1 Mathematical Modeling of Leading-Edge Slats

Following subsections provide the insights of the mathematical and analytical approach in estimating in chord length of the slat.

8.1.1 Increment in Lift Coefficient at Zero Angle of Attack

For a leading-edge slat of chord ratio (c_s/c) deflected by an angle δ_s , the change in section lift coefficient at zero angle of attack is given by

$$\Delta_s C_{l0} = -\frac{(\theta_s - \sin \theta_s)}{\pi} \delta_s C_{l,\alpha}, \quad \theta_s = \cos^{-1} \left(1 - 2 \frac{c_s}{c} \right),$$

with $C_{l,\alpha} \approx 2\pi$. Since $\theta_s - \sin \theta_s > 0$ for $0 < c_s/c < 1$ and $\delta_s > 0$, it follows that $\Delta_s C_{l0} \leq 0$: A slat reduces the lift at zero AoA. Physically, the slat relieves the leading-edge suction at small angles, trading a slight low-lift penalty for a much larger gain in maximum lift and stall margin. The magnitude $|\Delta_s C_{l0}|$ increases with both c_s/c and δ_s .

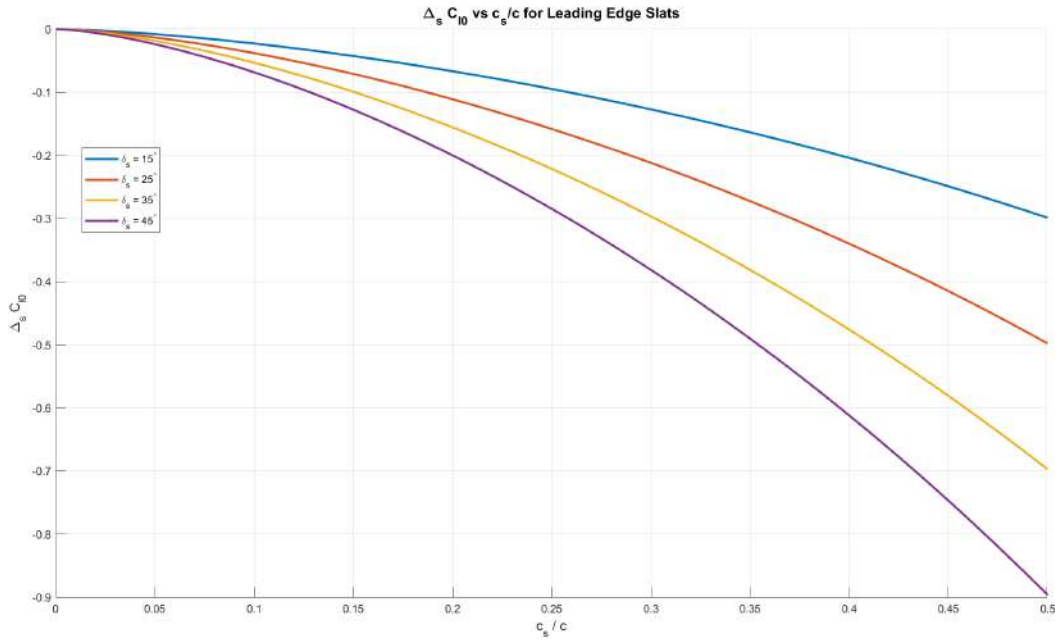


Figure 8.1: Change in zero-angle lift $\Delta_s C_{l0}$ (negative) versus slat chord ratio c_s/c for different slat deflections δ_s .

8.1.2 Increment in Maximum Lift Coefficient

The improvement in maximum sectional lift coefficient produced by the slat is modelled using the empirical relation

$$\Delta_s C_{l,\max} = 0.58 \sqrt{\frac{c_s}{c}} \delta_s C_{l,\alpha},$$

which reflects the ability of the slat to delay leading-edge stall and maintain attached flow over a wider range of angles of attack. Here the dependence on $\sqrt{c_s/c}$ indicates a diminishing return for very large slat chords, while the linear dependence on δ_s shows the continued benefit of increasing slat deflection.

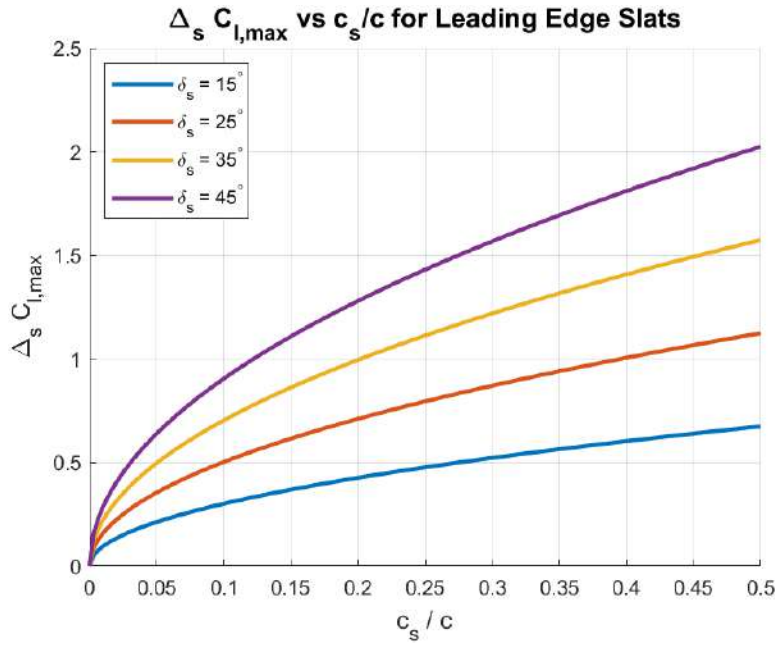


Figure 8.2: Increment in maximum sectional lift coefficient $\Delta_s C_{l,\max}$ as a function of slat chord ratio c_s/c for multiple slat deflections δ_s .

8.2 Effect of Slats on the Section Lift Curve

The influence of leading-edge slats on the complete sectional lift curve is examined by reconstructing $C_l(\alpha)$ for a range of slat chord ratios (c_s/c) and slat deflections δ_s . Two anchor points are defined for each case:

$$C_l(0^\circ) = C_{l,0}^{\text{base}} + \Delta_s C_{l,0}, \quad C_l(10^\circ) = C_{l,10}^{\text{base}} + \Delta_s C_{l,\max},$$

where the baseline (no-slat) values are

$$C_{l,0}^{\text{base}} = 0.5, \quad C_{l,10}^{\text{base}} = 1.4,$$

and the slat-induced increments are given by the relations

$$\Delta_s C_{l,0} = -\frac{(\theta_s - \sin \theta_s)}{\pi} \delta_s C_{l,\alpha}, \quad \Delta_s C_{l,\max} = 0.58 \sqrt{\frac{c_s}{c}} \delta_s C_{l,\alpha}.$$

A linear lift curve is then constructed between the two anchor points:

$$C_l(\alpha) = C_l(0^\circ) + \frac{C_l(10^\circ) - C_l(0^\circ)}{10^\circ} \alpha,$$

so that the effect of the slat appears only through the shifted endpoints. Because $\Delta_s C_{l,0} < 0$ and $\Delta_s C_{l,\max} > 0$, the result is a rotation of the lift curve: the zero-lift offset decreases, while the maximum attainable lift increases.

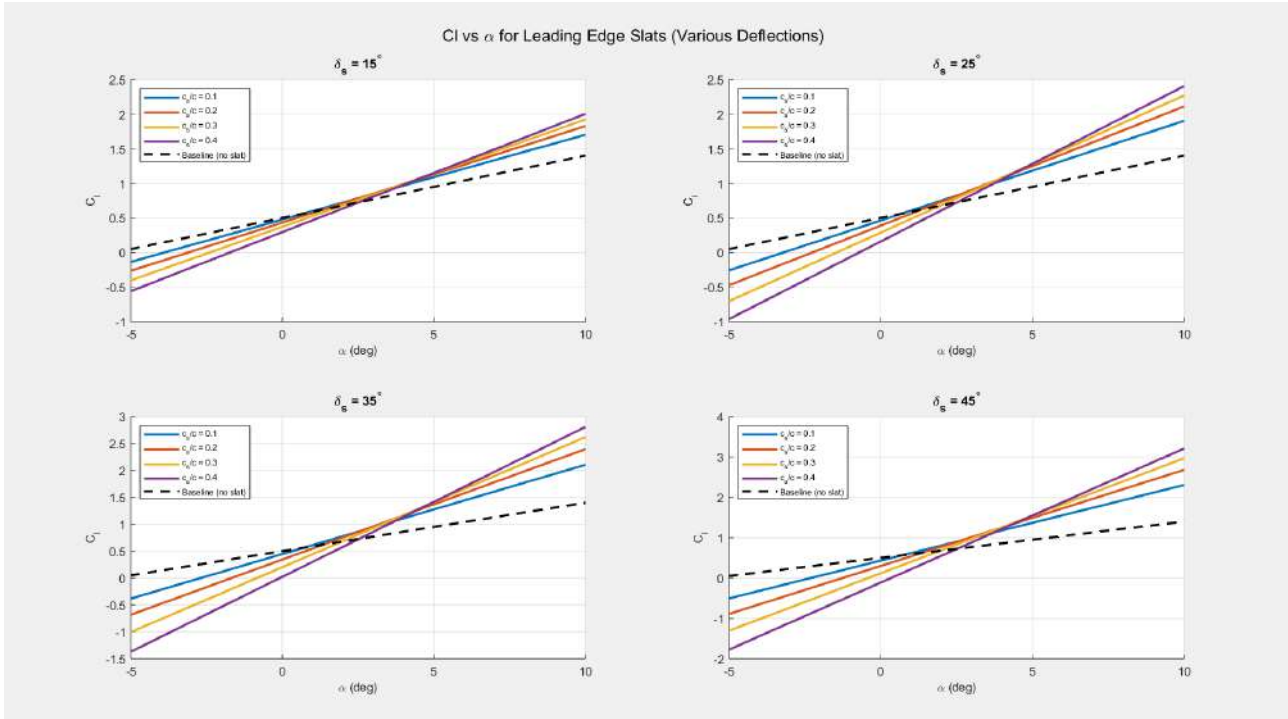


Figure 8.3: Modified lift curves $C_l(\alpha)$ for leading-edge slats at different deflections δ_s . Each subplot corresponds to one slat deflection, and each coloured line represents a different slat chord ratio c_s/c . The dashed black line shows the baseline (no-slat) lift curve.

The plots show that increasing the slat chord or deflection progressively lowers the lift at small angles of attack while significantly increasing the lift at higher angles, illustrating the primary purpose of the slat: a controlled loss of low-AoA lift in exchange for a much higher stall angle and maximum lift coefficient.

8.3 Slat Chord Selection

Leading-edge slats are primarily employed to delay stall by re-energising the boundary layer; however, they also reduce the lift coefficient at zero angle of attack.

Therefore, the lift increment produced by the trailing-edge flap at zero AoA must exceed the lift decrement caused by the slat to ensure a net positive overall lift.

From the analysis, as the slat chord ratio (c_s/c) increases, the decrease in lift at zero angle of attack ($\Delta_s C_{l,0}$, negative) becomes larger in magnitude. Conversely, the increment in maximum lift coefficient ($\Delta_s C_{l,max}$) increases with (c_s/c). At the maximum slat deflection, however, the reduction in zero-AoA lift exceeds the gain in maximum lift when (c_s/c) becomes too large.

Moreover, all the $\Delta_s C_{l,max}$ curves tend to converge at higher lift values, while they remain widely spaced near $\alpha = 0^\circ$. This indicates that the effect of (c_s/c) is more pronounced at small angles of attack than near the stall region. Hence, the loss in lift at zero AoA due to slats must be compensated by the flap-induced lift increment.

Based on these considerations, a suitable range for the slat chord ratio is

$$0.10 \leq \frac{c_s}{c} \leq 0.20,$$

and the selected design value is

$$\boxed{\frac{c_s}{c} = 0.16} \quad (8.1)$$

8.4 2D Slat Deflection CFD Study for Takeoff

The slat-only configuration with the airfoil was used for CFD analysis to determine the optimal position and deflection of slats when deployed to maximize lift during takeoff. The slat geometry as discussed in the previous sections has been used for the analysis.

8.4.1 Geometry and Case Setup

The NACA 4412 airfoil along with the slat and corresponding variable parameters of this study are shown in the figure below. The slat deflections are varied from 0° to 25° . All simulations are conducted under takeoff conditions using the $k-\omega$ SST turbulence model.

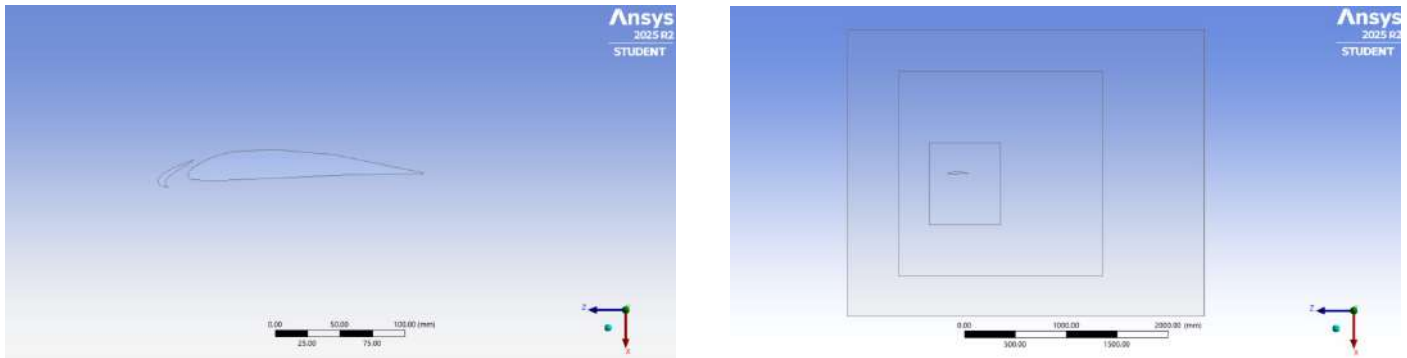


Figure 8.4: 2D Geometry for Deployed Slats

8.4.2 Mesh Details

The mesh is generated in a manner similar to the previous study and consists of approximately 35380 nodes and 34784 elements.

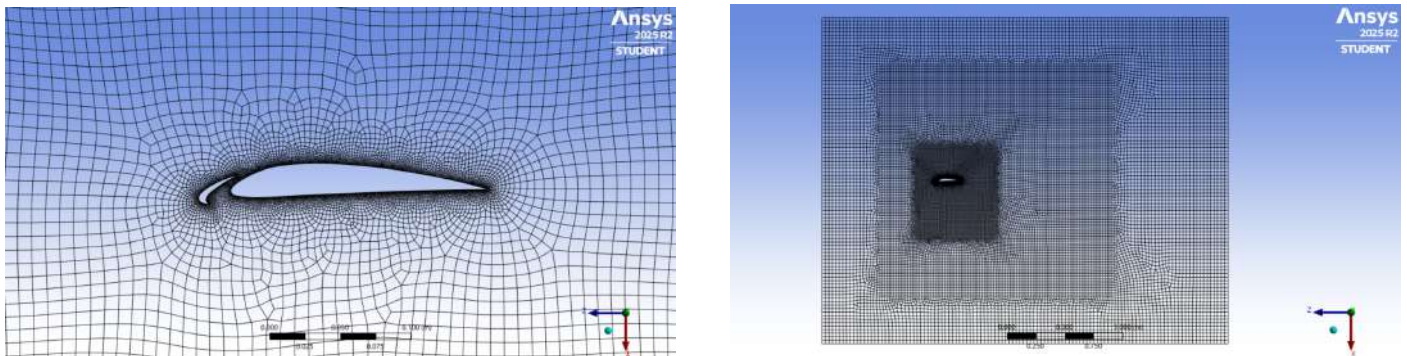


Figure 8.5: Meshing for Deployed Slats

8.4.3 CFD Results for 30P30N Slat Configuration

The 30P30N configuration for the three-element airfoil is widely used in industry and research in aerodynamics and has been used for analysis as a starting point to determine the slat position and deflection. In accordance with the **30P30N** configuration (Fig. 8.6), the overhang, the gap and the depth during slat deployment were determined, and a two-dimensional CFD analysis was conducted on the airfoil-slat system for the different deflection angles.

A slat overhang used was $2.5\%c$ and gap radius used was $2.95\%c$. The slat deflection angle was varied during the 2D deflection study.

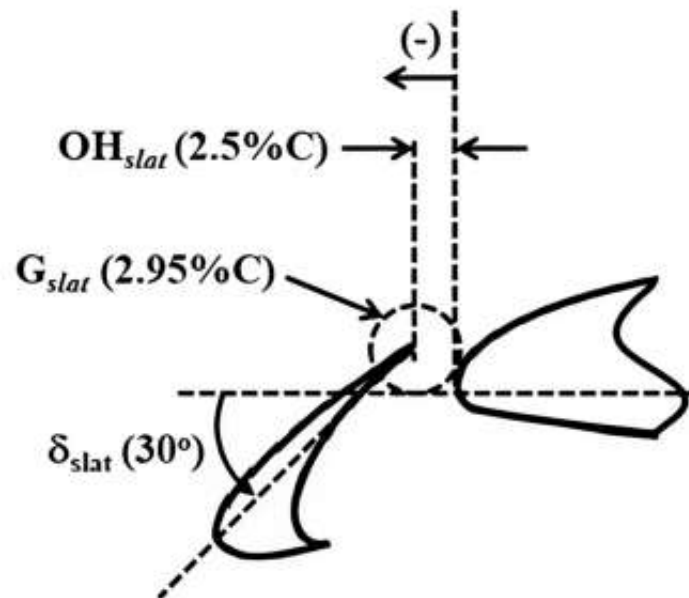
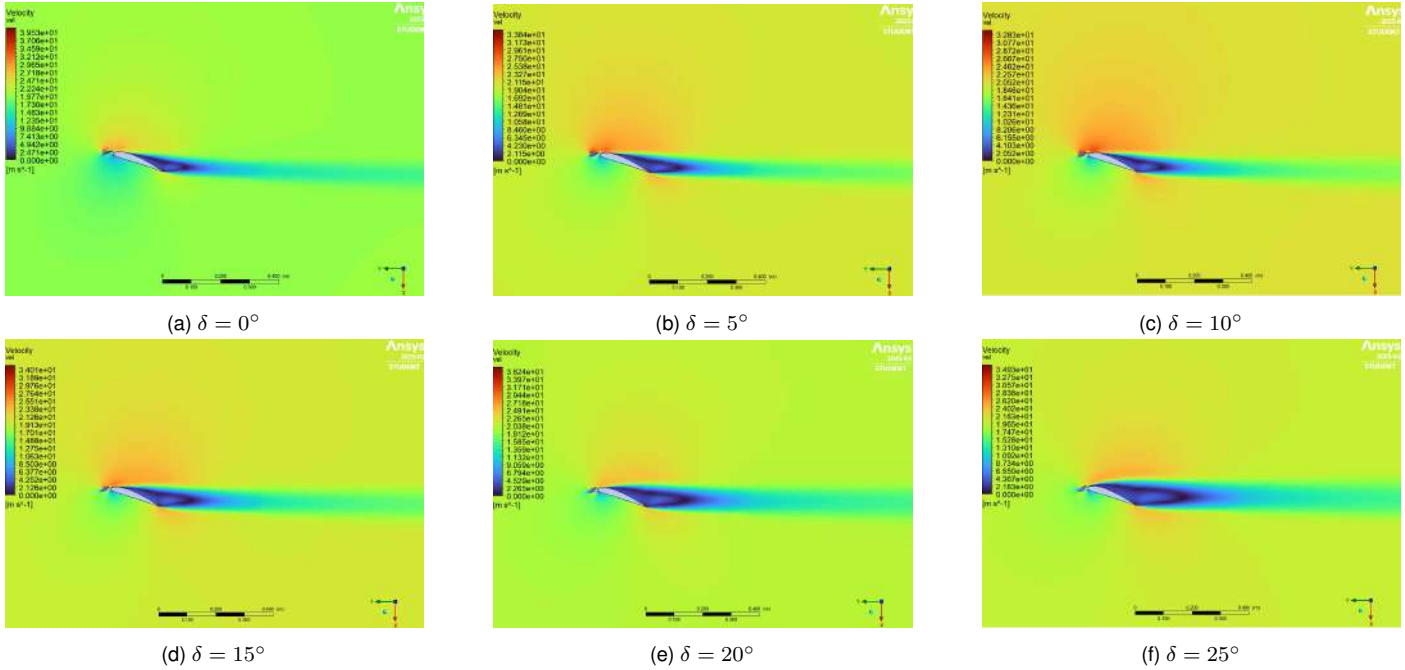


Figure 8.6: 30P30N Slat Configuration

AoA (°)	Defl. (°)	C_L	C_D	C_L/C_D
0	25	0.0142	0.08390	0.17
	20	0.0782	0.07868	0.99
	15	0.1522	0.05628	2.70
	10	0.2177	0.05142	4.23
	5	0.2705	0.04936	5.48
5	25	0.2891	0.07327	3.95
	20	0.5026	0.05153	9.75
	15	0.5145	0.05080	10.13
	10	0.5110	0.05123	9.97
	5	0.5480	0.04769	11.49
10	25	0.4894	0.13444	3.64
	20	0.5568	0.11766	4.73
	15	0.8459	0.07629	11.09
	10	1.0332	0.06027	17.14
	5	1.1708	0.05137	22.79

AoA (°)	Defl. (°)	C_L	C_D	C_L/C_D
15	25	0.6224	0.20944	2.97
	20	0.8234	0.15711	5.24
	15	0.7744	0.15636	4.95
	10	1.0179	0.12503	8.14
	5	1.5838	0.07074	22.39
20	25	0.7381	0.28067	2.63
	20	0.8613	0.24483	3.52
	15	1.0034	0.21571	4.65
	10	1.1686	0.18913	6.18
	5	1.1723	0.18230	6.43
25	25	0.8985	0.34617	2.60
	20	1.0266	0.30115	3.41
	15	1.1253	0.28053	4.01
	10	1.2615	0.25264	4.99
	5	1.3869	0.23715	5.85

Table 8.1: Lift and Drag Data for Various Angles of Attack and Slat Deflections


Figure 8.7: Velocity Contours for Slats for $\alpha = 20^\circ$ Section 9

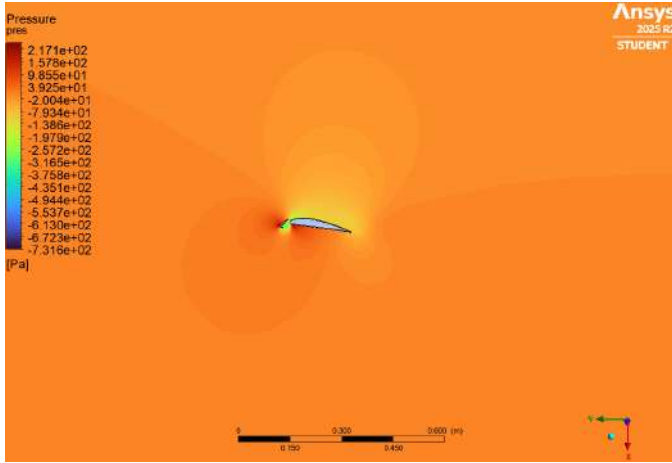
8.4.4 CFD Results of Slats at Finalized Position

The slats designed according to the 30P30N configuration used yield undesirable results. From the velocity contours, it is clearly visible that the slats at 2.95% c gap radius and 2.5% c overhang is ineffective at any angle of slat deflection despite the high C_l values because the airfoil stalls in each case. The role of slats is to delay stall by redirecting flow over the surface of the airfoil. The reason for the high lift values in Section 8.4 because the slat is producing the lift while the airfoil is in stall.

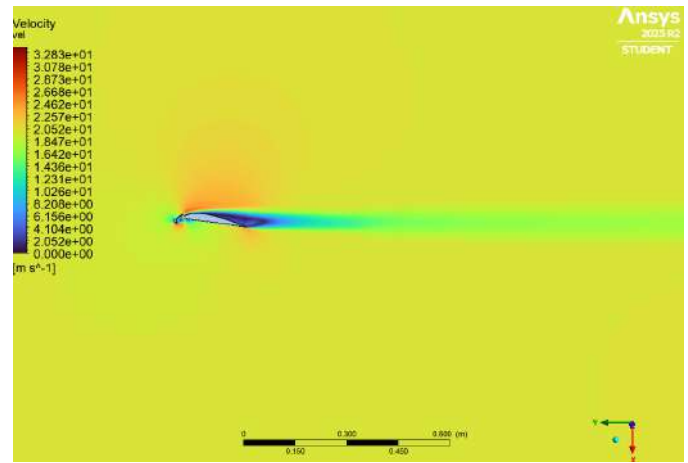
To correct this, the slat depth, overhang and deflection angle were modified such that the flow is pushed over the airfoil at high angles of attack. This way, the main airfoil does not stall and produces majority of the lift. The following Table 8.2 presents the reduced deflection angles and its results.

Deflection of slats($^\circ$)	Angle of Attack ($^\circ$)	C_L	C_D	C_L/C_D
0 $^\circ$ Deflection	0	0.3490	0.02580	13.53
	5	0.6377	0.03681	17.32
	10	1.2419	0.03975	31.24
	15	1.6664	0.05898	28.25
	20	1.9142	0.09900	19.33
	25	1.3911	0.33673	4.13
	30	1.1845	0.52305	2.26
5 $^\circ$ Deflection	0	0.3037	0.02961	10.26
	5	0.5831	0.04207	13.86
	10	1.1813	0.04188	28.20
	15	1.6448	0.05673	28.99
	20	1.8758	0.09026	20.78
	25	1.8296	0.16084	11.38
	30	1.2403	0.49309	2.52

Table 8.2: Lift and Drag Data for Various Angles of Attack at a Slat Deflection of 0 $^\circ$ and 5 $^\circ$



(a) Pressure Contour



(b) Velocity Contour

Figure 8.8: Plots at angle of attack 10 degrees with slat deflection 5 degrees

As the above configuration of slats gives higher lift coefficient compared to the previous 30P30N results, the following displacement and deflection of slats is finalized for deployment during takeoff:

$$\frac{\Delta x}{c} = 0.102, \quad \frac{\Delta y}{c} = 0.03, \quad \delta_s = 5^\circ \quad (8.2)$$

9 2D CFD of Slats and Flaps Combined: Three-Element Airfoil

A three-element airfoil is designed using the slat and flap geometries determined in the above sections. In the earlier analyses, the flap and slat deflections were examined independently to determine their respective optimal values. In the present study, both elements are combined and simulated near these optimal settings to identify the best-performing three-element airfoil configuration and the corresponding angle of attack that yields the maximum lift coefficient which is an important parameter influencing the aircraft's final takeoff angle. This has been done because the flow behavior over the flap is dependent on the slat position.

9.1 Geometry and Case Setup

The three-element airfoil consists of the NACA 4412 as the main element, with single slotted flaps and slats at the finalized deployed position. The turbulence model remains unchanged.

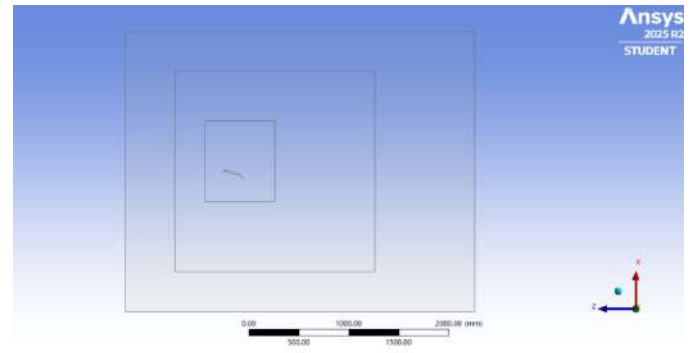
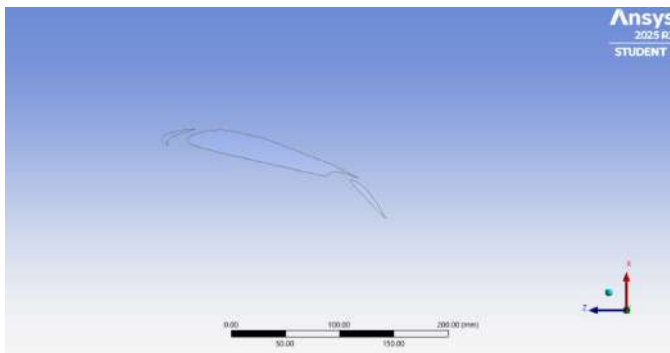


Figure 9.1: Three-element airfoil at finalized deployed position

9.2 Mesh Details

A finer mesh relative to the last two meshes is generated to accommodate the addition of both lift-enhancing devices. It now consists of about 49201 nodes and 53147 elements with the same y^+ value of 1.

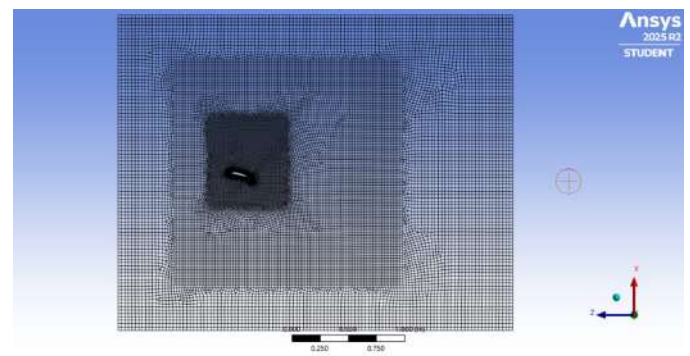
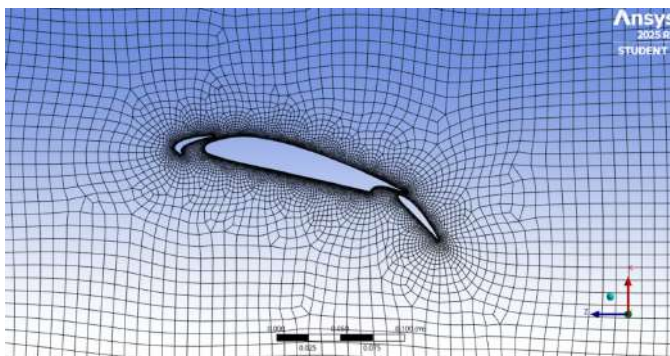


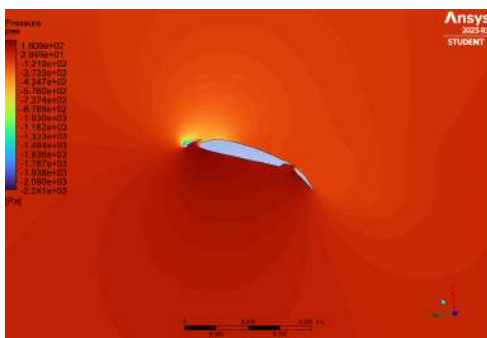
Figure 9.2: Meshing for three-element airfoil

9.3 Results

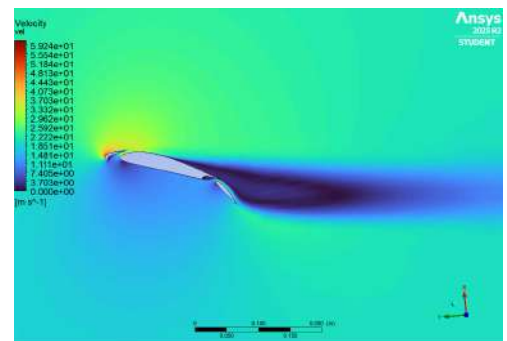
AoA (°)	Slat (°)	Flap (°)	C_L	C_D	C_L/C_D
0	5	20	1.2710	0.0478	26.5830
	5	25	1.5554	0.0549	28.3424
	5	30	1.4625	0.0719	20.3425
	5	35	1.3594	0.0981	13.8598
	5	40	1.4212	0.1115	12.7479
5	5	20	1.8589	0.0553	33.5971
	5	25	1.9463	0.0653	29.8105
	5	30	1.8380	0.0872	21.0697
	5	35	1.8912	0.1037	18.2386
	5	40	1.9247	0.1168	16.4851
10	5	20	2.5175	0.0755	33.3334
	5	25	2.5806	0.0857	30.1218
	5	30	2.3429	0.1024	22.8862
	5	35	2.3667	0.1174	20.1573
	5	40	2.3982	0.1333	18.0201
15	5	20	2.8264	0.1083	26.0921
	5	25	3.0176	0.1202	25.0969
	5	30	3.0905	0.1362	22.6986
	5	35	2.7423	0.1493	18.3665
	5	40	2.7559	0.1605	17.1754

AoA (°)	Slat (°)	Flap (°)	C_L	C_D	C_L/C_D
20	5	20	2.4571	0.1916	12.8257
	5	25	2.7222	0.1970	13.8213
	5	30	2.3858	0.2527	9.4400
	5	35	2.5078	0.2602	9.6382
	5	40	2.9316	0.2528	11.5958
25	5	20	1.7193	0.5831	2.9484
	5	25	1.7246	0.6276	2.7479
	5	30	1.7299	0.6836	2.5306
	5	35	1.7296	0.7073	2.4453
	5	40	1.6784	0.7230	2.3213
30	5	20	1.5374	0.7914	1.9427
	5	25	1.5701	0.8376	1.8746
	5	30	1.6104	0.8815	1.8269
	5	35	1.6114	0.9199	1.7518
	5	40	1.5563	0.9274	1.6781

Table 9.1: Lift and Drag Data for Various Angles of Attack at Combined Slat and Flap Deflections

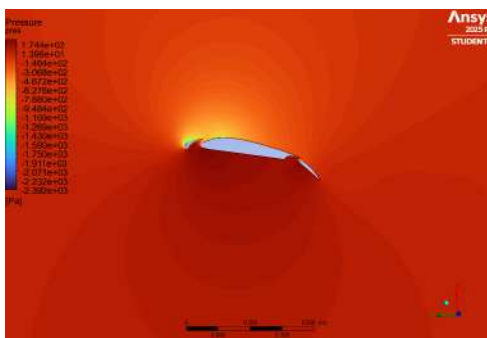


(a) Pressure Contour

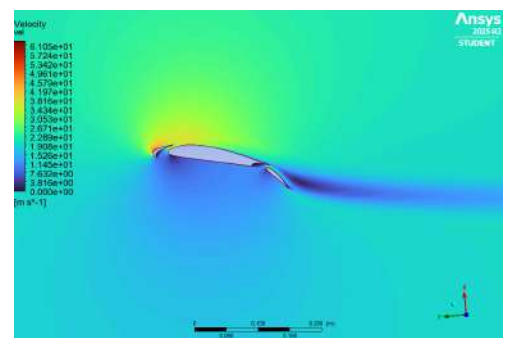


(b) Velocity Contour

Figure 9.3: Plots at an angle of attack of 20° with a slat deflection of 5°.



(a) Pressure Contour



(b) Velocity Contour

Figure 9.4: Plots at an angle of attack of 15° with a slat deflection of 5°.

10 3D CFD of Wing Model with Flaps and Slats

The model of the wing with flaps and slats deployed during takeoff is shown in Fig. 10.1.

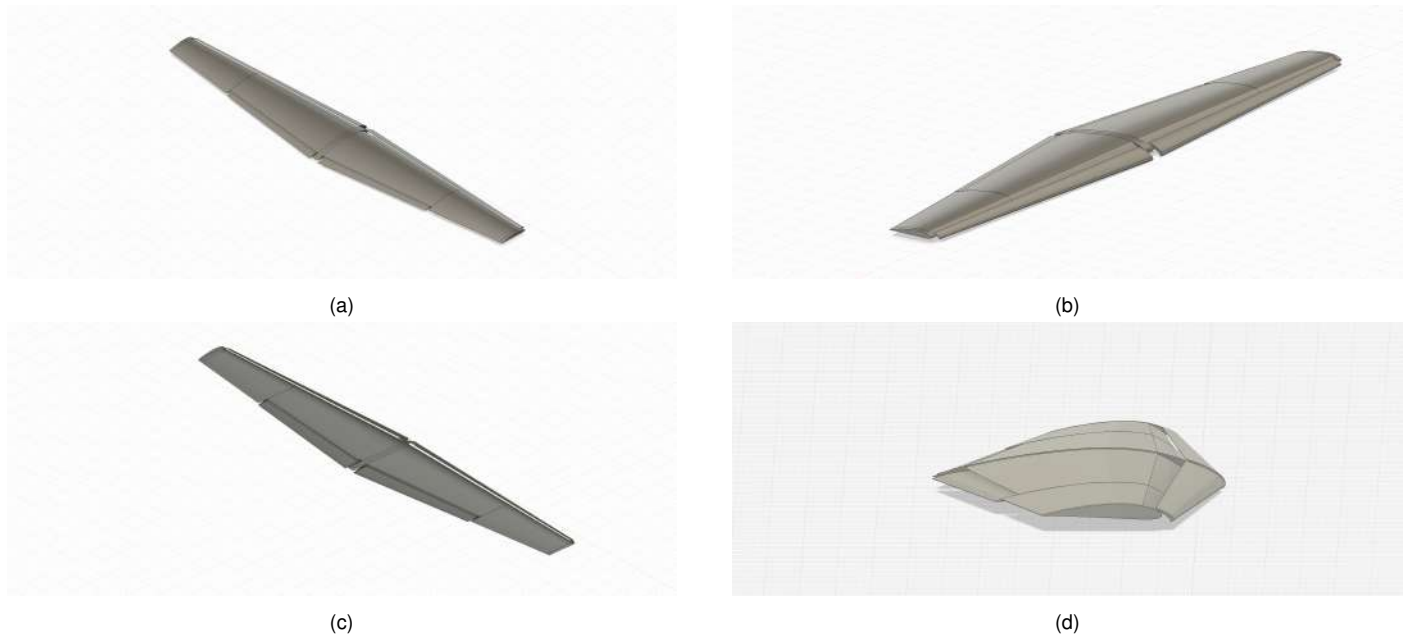


Figure 10.1: Wing (AR=15) with Slats and Flaps Deployed

To account for the three-dimensional effects, the wing was analysed using CFD with the high-lift devices set to their maximum lift configuration. The trailing-edge flap was deflected by 30° and the leading-edge slat by 5° , and the simulation was performed at a takeoff angle of attack of 15° . This configuration was selected because, in the 2D analysis, these deflection angles corresponded to the condition where the maximum lift coefficient was observed.

10.1 Geometry and Case Setup

Wing model of Aspect ratio 15 is used for the CFD analysis. The model is three dimensional. As the flow is symmetric about the root chord plane, the wing is split along this plane, and a symmetry boundary condition is applied on the resulting surface. This approach effectively reduces the computational domain by half, thereby decreasing both computational cost and simulation time. The fluid properties, flow parameters and turbulence model are that of the take-off conditions which were discussed earlier.

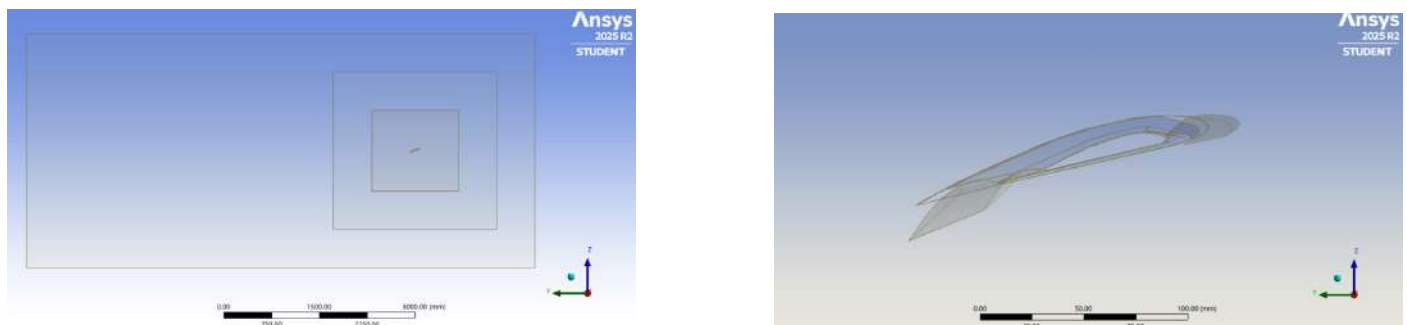


Figure 10.2: Domain of the wing with flaps and slats

10.2 Mesh Details

A tetrahedron-dominated unstructured mesh was created, with structured prism-layer inflation around the wing surface. The y^+ value was set to 30, to capture the log layer using the $k-\omega$ SST RANS model, resulting in a first cell height of approximately 0.02

mm, determined using the same calculator. The mesh details for the wing model with the number of elements and nodes.

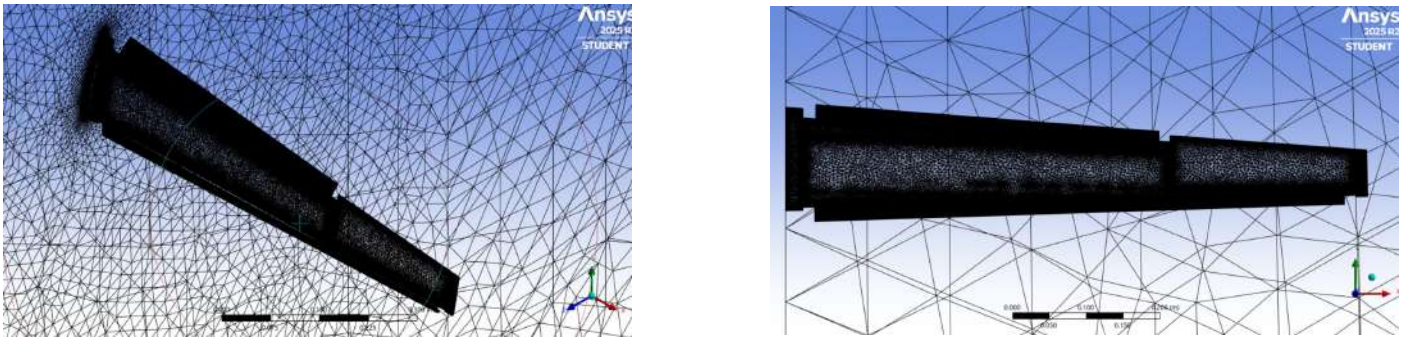


Figure 10.3: Mesh of the wing with flaps and slats: number of nodes = 3,089,211 and number of elements = 11,271,015.

10.3 Results

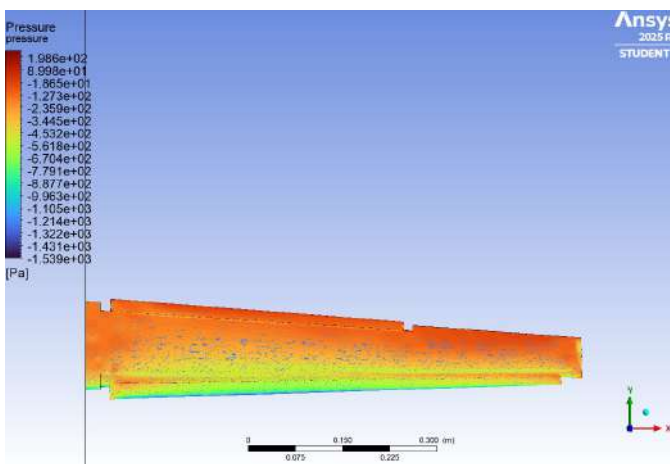


Figure 10.4: CFD result: Pressure contours

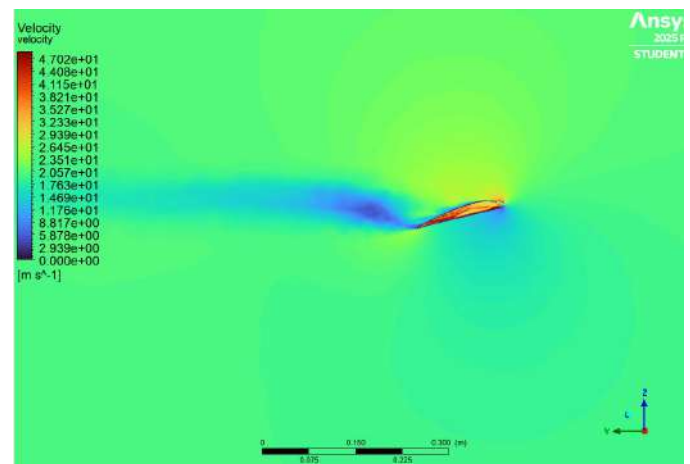


Figure 10.5: CFD result: Velocity contours

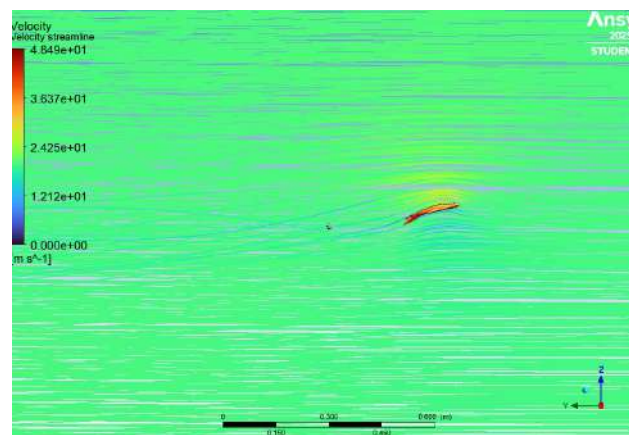


Figure 10.6: CFD result: Velocity Streamlines

With the addition of the slats and flaps, the three-dimensional CFD analysis of the wing at the configuration $\alpha = 15^\circ$, slat deflection = 5° , and flap deflection = 30° yielded the following aerodynamic coefficients:

$$C_L = 1.76193 \quad C_D = 0.23315 \quad \frac{C_L}{C_D} = \frac{1.76193}{0.23315} \approx 7.56.$$

The corresponding coefficients for the 3D clean wing case are:

$$C_L = 0.33739 \quad C_D = 0.021566 \quad \frac{C_L}{C_D} = 15.644$$

This demonstrates the aerodynamic effectiveness of the deployed high-lift devices under takeoff conditions.

11 Passive Flow Control Techniques

Passive flow control techniques rely on fixed geometric modifications of the lifting surface to alter the boundary-layer development without any external energy input. These devices redistribute momentum inside the boundary layer, modify local pressure gradients and delay the onset of separation. Typical effects are thinning or energising the near-wall shear layer and weakening laminar separation bubbles, which in turn improves lift and often reduces the sensitivity of the airfoil to stall.

11.1 Passive Flow Control Devices

Flow Vane

A flow vane is a small fin or plate placed on the surface and aligned at a prescribed yaw angle to the local flow. By deflecting the near-wall streamlines it can generate streamwise vorticity and re-orient the boundary-layer momentum, thereby improving attachment in regions subjected to cross-flow or curvature-induced instabilities.

Fence

A fence is a thin, vertical, fin-like surface protruding from the suction side of a wing or blade. Its primary role is to block or redirect spanwise flow, thereby limiting the propagation of an inboard stall towards the wing tip. By interrupting the spanwise boundary-layer transport, the fence helps to keep the outboard region attached and delays tip stall.

Serrated Trailing Edge

Conventional blades have straight sharp or mildly blunt trailing edges from which the wake is shed. The merging of flows from the pressure and suction sides, and the periodic shedding of coherent wake structures, generate aerodynamic noise and contribute to drag and lift degradation. A serrated trailing edge splits the wake into multiple smaller streams, weakening large coherent vortices, reducing noise emission and providing a modest improvement in aerodynamic performance.

Vortilon

A vortilon is a short, fence-like device mounted near the leading edge, usually on the lower surface and angled so that at positive incidence it produces a strong vortex over the suction side. This streamwise vortex stabilises the flow over the upper surface and delays stall, while the compact size of the vortilon keeps the parasitic drag relatively low compared with larger leading-edge devices.

Vortex Generators

Vortex generators (VGs) are passive devices that create streamwise vortices in the boundary layer. They are typically small vanes or tabs inclined to the incoming flow so that a lateral pressure difference forms across each element. The resulting longitudinal vortices draw high-momentum fluid from the outer flow into the near-wall region and sweep low-momentum fluid outwards. As a consequence, the mean streamwise momentum inside the boundary layer is increased, making the flow more resistant to adverse pressure gradients and separation.

Other Passive and Bio-Inspired Devices

Additional passive concepts include surface riblets, dimples, porous coatings and micro-roughness elements, which can tailor near-wall shear and reduce skin friction. Bio-inspired approaches, such as leading-edge tubercles reminiscent of whale flippers or flexible bristles and flaps, use geometry inspired by natural flyers and swimmers to enhance mixing and delay stall in a passive manner.

11.2 Selection of Passive Flow Control Device

Among the different passive devices discussed above, vortex generators were selected as the primary flow control element for the present work. Compared to all passive flow control devices, VGs provide a favourable combination of increased maximum lift coefficient, improved C_L/C_D in the post-stall and high-lift regime, and relatively small geometric intrusion. From a numerical point of view, their simple vane-type geometry and limited moving parts make mesh generation and CFD modelling more straightforward than for slotted multi-element configurations. Overall, the balance between aerodynamic benefit and implementation cost indicates that VGs are the most suitable passive control device for the current design.

11.3 Operating Principle of Vortex Generators

Vortex generators operate by using their inclined surfaces to create a pressure difference across each vane, which induces the formation of strong streamwise vortices. These vortices draw high-momentum fluid from the outer region of the flow and transport it toward the near-wall region, while simultaneously lifting low-momentum boundary-layer fluid away from the surface. Through this continuous mixing process, the momentum of the boundary layer increases and the velocity profile near the wall becomes fuller and more resistant to deceleration.

Since flow separation on an airfoil typically occurs when the near-wall fluid slows down under an adverse pressure gradient and the wall shear stress approaches zero, the additional momentum supplied by the vortex generators counteracts this natural tendency. The energised boundary layer is able to tolerate stronger adverse pressure gradients before separating, resulting in a delayed onset of flow reversal and a reduction in the size or presence of separation bubbles. In many cases, vortex-generated mixing can also reattach a boundary layer that would otherwise separate under laminar or weakly turbulent conditions.

Overall, the action of the vortex generators extends the region of attached flow further downstream on the suction side of the airfoil, enhances lift, delays stall, and leads to smoother aerodynamic behaviour over a wider range of operating angles of attack.

11.4 Selection of Vortex-Generator Type

In this work conventional, rather than micro-scale, vortex generators are used. Conventional VGs have a height of the order of the local boundary-layer thickness, which ensures strong vortical structures and a clear impact on lift. Based on their geometry, vane-type VGs were chosen; these are thin plates mounted at an incidence angle to the local flow, generating streamwise vortices with relatively low form drag compared with bluff protrusions.

From a planform viewpoint, common vane shapes include rectangular, triangular, trapezoidal, gothic, parabolic and ogive geometries. Rectangular vanes typically generate the strongest vortices due to their large projected area and abrupt leading and trailing edges, but this also leads to significant pressure drag and enhanced vortex-induced losses. Trapezoidal vanes produce vortices that are slightly weaker than those from rectangular vanes but stronger than those generated by triangular and smoother-edged shapes; in many studies they provide a better compromise between lift increment and drag penalty. Triangular and rounded (gothic, parabolic, ogive) shapes tend to have lower drag but also weaker vortex strength and therefore smaller gains in separation control.

In the present design trapezoidal vane vortex generators are adopted. They are capable of generating sufficiently strong vortices for effective flow control while offering improved C_L/C_D compared with the purely rectangular planform, for which the drag is higher due to the larger frontal area and sharper corners that promote strong separation and pressure drag.

The VGs are arranged in counter-rotating pairs. Each pair is oriented so that the two vortices spin in opposite directions, producing symmetric mixing on either side of the pair. This configuration tends to cancel out the net lateral forces and rolling moments generated by individual vortices, avoids undesirable asymmetric loading on the wing, and confines the energised region close to the surface.

11.5 Trapezoidal Vortex Generators

11.5.1 Geometric Definition

The geometry of the selected trapezoidal VG is shown schematically in Fig. 11.2 and Fig. 11.3. The top view and side view define the main geometric parameters:

- L : streamwise length of the vane,

- b : length of the top edge of the trapezoid in planform,
- h : VG height measured normal to the surface,
- t : thickness of the vane in the spanwise direction,
- d : spacing between the downstream ends of the two vanes in a single counter-rotating pair,
- D : distance between the centres of adjacent VG pairs along the span,
- β : yaw angle between the vane chord line and the local incoming flow direction.

To avoid sharp corners that would induce unnecessary form drag, the vertices of the trapezoid are slightly rounded in the three-dimensional realisation.

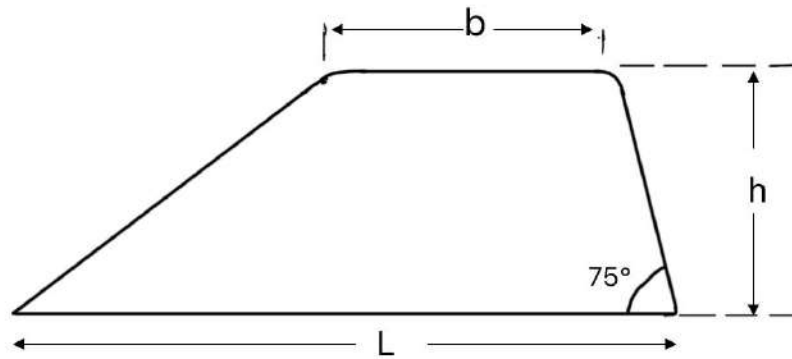


Figure 11.1: side view of the trapezoidal vortex generator showing the definitions of L , b , h

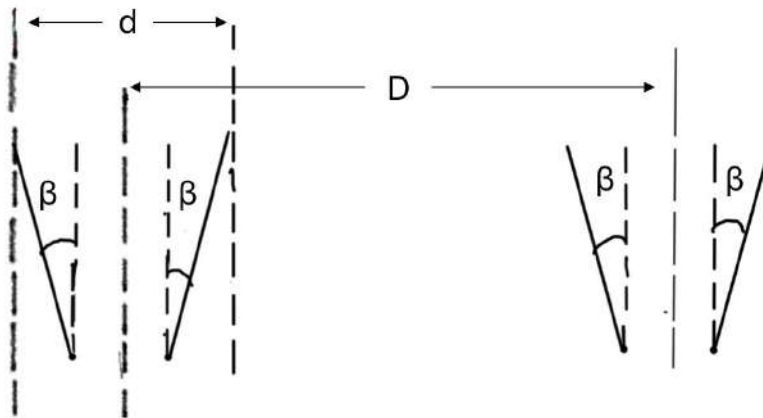


Figure 11.2: Top view of the trapezoidal vortex generator pairs showing the definitions of d , D and β .

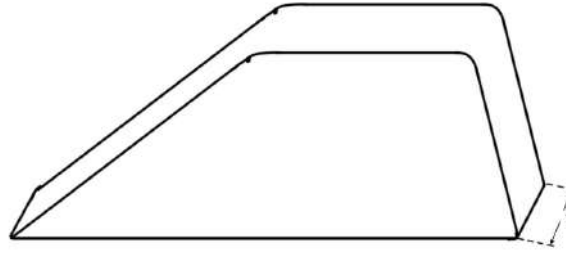


Figure 11.3: Three-dimensional view of a trapezoidal vortex generator with rounded vertices and thickness t .

11.5.2 Non-Dimensional Geometric Ratios

The sizing of the VG geometry is guided by non-dimensional relations reported in previous CFD and experimental investigations available in the literature. In the present work, these relations are taken directly from the study of Babic [43], whose computational and experimental results provide validated geometric ratios suitable for low-Reynolds-number airfoil applications. The key non-dimensional relationships adopted are:

$$\frac{h}{\delta} = 1, \quad (11.1)$$

$$\frac{b}{L} = 0.2, \quad (11.2)$$

$$\frac{L}{h} = 3, \quad (11.3)$$

$$\frac{d}{h} = 3.5, \quad (11.4)$$

$$\frac{D}{h} = 15.75, \quad (11.5)$$

where δ is the local boundary-layer thickness at the VG location. The first relation, $h/\delta = 1$, corresponds to a conventional VG whose height is of the order of the boundary-layer thickness, ensuring strong vortex production ($h = \delta$ in the present case). The yaw angle β controls both the strength of the generated vortex and the drag of the vane. Too small an angle produces weak vortices, while too large an angle generates excessive drag and may cause premature separation on the VG itself. Based on the reference study mentioned earlier, specifically the work of Babic [43], an optimal compromise for the yaw angle is obtained at $\beta = 15^\circ$, which is therefore adopted in the present configuration.

11.5.3 Placement Along the Chord

The VGs are placed on the **suction side** of the airfoil, where the boundary layer is most sensitive to adverse pressure gradients and where transition strongly influences the onset of flow separation.

During take-off, the chord Reynolds number is of the order $Re_c \approx 1 \times 10^5 - 2 \times 10^5$. For a smooth airfoil at $Re_c = 2 \times 10^5$, the development of the laminar boundary layer on the **suction surface** can be described using the local Reynolds number

$$Re_x = Re_c \left(\frac{x}{c} \right), \quad (11.6)$$

which increases with downstream distance x from the leading edge.

When Re_x reaches approximately $5 \times 10^4 - 1.5 \times 10^5$, the laminar boundary layer on the **suction side** becomes susceptible to Tollmien-Schlichting wave amplification and to the formation of a laminar separation bubble caused by the adverse pressure gradient typical of airfoil suction surfaces. The combined effects of instability growth, thickening of the laminar boundary layer, and the appearance of a laminar separation bubble lead to transition in the vicinity of $x/c \approx 0.1 - 0.3$ for $Re_c = 2 \times 10^5$.

To be effective, the vortex generators must be placed slightly upstream of this natural transition region on the **suction surface**. Considering both established literature correlations and the findings reported in the reference study by Babic [43], the suitable

range for VG placement along the suction surface is found to be

$$x/c = 0.20\text{--}0.25. \quad (11.7)$$

In the final configuration, a position of $x/c = 0.22$ is selected so that the generated streamwise vortices interact directly with the boundary layer as it approaches transition. This maximises the effectiveness of the VGs in energising the suction-side boundary layer, delaying separation, and improving the overall lift characteristics.

11.5.4 Reynolds Number and Boundary-Layer Thickness at $x = 0.22\bar{c}$

For the mean aerodynamic chord $\bar{c} = 0.11034$ m, air density $\rho = 1.225$ kg/m³, kinematic viscosity $\mu = 1.789 \times 10^{-5}$ Pa · s and take-off speed $U = 20$ m/s, the chord Reynolds number based on $\bar{c}$ is

$$\text{Re}_{\bar{c}} = \frac{\rho U \bar{c}}{\mu} = \frac{1.225 \times 20 \times 0.11034}{1.789 \times 10^{-5}} \approx 1.5 \times 10^5. \quad (11.8)$$

The chordwise location of interest is

$$x = 0.22 \bar{c} = 0.22 \times 0.11034 = 0.0242748 \text{ m}. \quad (11.9)$$

The corresponding local Reynolds number for a flat plate is

$$\text{Re}_x = \frac{\rho U x}{\mu} = \frac{1.225 \times 20 \times 0.0242748}{1.789 \times 10^{-5}} \approx 3.3 \times 10^4. \quad (11.10)$$

11.5.5 Boundary-Layer Thickness

For a laminar flat-plate boundary layer, the Blasius correlation gives

$$\frac{\delta_{\text{lam,fp}}}{x} = \frac{5}{\sqrt{\text{Re}_x}}, \quad (11.11)$$

so that

$$\delta_{\text{lam,fp}} = x \frac{5}{\sqrt{\text{Re}_x}} \approx 0.0242748 \frac{5}{\sqrt{3.3 \times 10^4}} \approx 6.6 \times 10^{-4} \text{ m} \approx 0.67 \text{ mm}. \quad (11.12)$$

For a turbulent boundary layer on a flat plate, the empirical relation

$$\frac{\delta_{\text{turb,fp}}}{x} = \frac{0.16}{\text{Re}_x^{1/7}} \quad (11.13)$$

is used, giving

$$\delta_{\text{turb,fp}} = x \frac{0.16}{\text{Re}_x^{1/7}} \approx 0.0242748 \frac{0.16}{(3.3 \times 10^4)^{1/7}} \approx 8.8 \times 10^{-4} \text{ m} \approx 0.88 \text{ mm}. \quad (11.14)$$

Approximation of Airfoil Boundary-Layer Thickness

A more accurate estimate of the boundary-layer thickness on the airfoil would require the Thwaites method, which depends on the external-velocity distribution $U_e(x)$ through

$$\frac{dU_e}{dx} = -\frac{1}{\rho} \frac{dP}{dx}, \quad (11.15)$$

under the assumptions of steady, laminar, incompressible flow. Obtaining $U_e(x)$ and the corresponding pressure gradient for the present airfoil–flap–slat configuration is non-trivial and computationally expensive.

In Thwaites' method the laminar momentum thickness $\theta(x)$ is obtained from the integral relation

$$\theta^2(x) = 0.45 \frac{\nu}{U_e^6(x)} \int_0^x U_e^5(\xi) d\xi, \quad (11.16)$$

where $\nu = \mu/\rho$ is the kinematic viscosity. Once θ is known, the Thwaites parameter $m = (\theta^2/\nu) dU_e/dx$ gives the shape factor

$H(m)$, from which the displacement thickness $\delta^* = H \theta$ and an estimate of the full boundary-layer thickness $\delta \approx K \theta$ (with K taken from an assumed velocity profile) can be determined.

Instead, we approximate the boundary-layer thickness on the airfoil by scaling the flat-plate values. A flat plate has essentially zero streamwise pressure gradient, whereas the airfoil experiences an adverse pressure gradient over the suction side, which thickens the boundary layer. From the computed pressure-distribution plots of the wing with flaps and slats, the airfoil boundary-layer thickness at $x = 0.22\bar{c}$ is reasonably approximated as

$$\delta_{\text{airfoil}} \approx 2 \delta_{\text{fp}}. \quad (11.17)$$

This factor of two is not arbitrary; it is supported by additional aerodynamic considerations. On the suction surface of an airfoil, the boundary layer develops under a non-zero adverse pressure gradient, causing it to grow more rapidly than the Blasius flat-plate solution. The presence of a flap intensifies this effect by generating a strong suction peak near the hinge line and a steep pressure-recovery region downstream, both of which promote boundary-layer thickening. Furthermore, geometric curvature along the suction side increases the wall-normal velocity gradient and leads to additional growth compared with a flat surface.

Empirical correlations and experimental data for low-Reynolds-number airfoils ($\text{Re} \approx 10^5 - 3 \times 10^5$) show that laminar boundary-layer thicknesses in regions subjected to adverse pressure gradients typically lie between 1.8 and 2.2 times the corresponding flat-plate Blasius predictions [44, 45]. In the present case, the flap-induced pressure distribution enhances this thickening even further. Therefore, adopting

$$\delta_{\text{airfoil}} \approx 2 \delta_{\text{fp}} \quad (11.18)$$

provides a reasonable and conservative engineering estimate in the absence of an exact Thwaites analysis based on the full external-velocity distribution $U_e(x)$.

Thus, for the laminar and turbulent limits we obtain

$$\delta_{\text{lam,airfoil}} = 2 \delta_{\text{lam,fp}} \approx 2 \times 0.67 \text{ mm} \approx 1.3 \text{ mm}, \quad (11.19)$$

$$\delta_{\text{turb,airfoil}} = 2 \delta_{\text{turb,fp}} \approx 2 \times 0.88 \text{ mm} \approx 1.8 \text{ mm} \approx 2.0 \text{ mm}. \quad (11.20)$$

The transition region at $x = 0.22\bar{c}$ therefore has a boundary-layer thickness between $\delta_{\text{lam,airfoil}} \approx 1.3 \text{ mm}$ and $\delta_{\text{turb,airfoil}} \approx 2.0 \text{ mm}$. For design purposes, the vortex-generator height is chosen as an intermediate value

$$h = \delta_{\text{tr}} \approx 1.6 \text{ mm} = 0.0016 \text{ m}. \quad (11.21)$$

Geometric Dimensions of the Vortex Generators

Using the non-dimensional relations introduced previously,

$$\frac{L}{h} = 3, \quad \frac{D}{h} = 15.75, \quad \frac{d}{h} = 3.5, \quad \frac{b}{L} = 0.2, \quad (11.22)$$

and the selected boundary-layer height

$$h = 0.0016 \text{ m}, \quad (11.23)$$

the main geometric dimensions of the trapezoidal vortex generator become

$$L = 3h = 3 \times 0.0016 = 0.0048 \text{ m} \approx 4.8 \text{ mm}, \quad (11.24)$$

$$D = 15.75h = 15.75 \times 0.0016 = 0.0252 \text{ m} \approx 25.2 \text{ mm}, \quad (11.25)$$

$$d = 3.5h = 3.5 \times 0.0016 = 0.0056 \text{ m} \approx 5.6 \text{ mm}, \quad (11.26)$$

$$b = 0.2L = 0.2 \times 0.0048 = 0.00096 \text{ m} \approx 0.96 \text{ mm}. \quad (11.27)$$

In addition to these in-plane geometric parameters, the vane thickness is chosen as

$$t = \frac{h}{4} = \frac{0.0016}{4} = 0.0004 \text{ m} \approx 0.4 \text{ mm}, \quad (11.28)$$

which provides sufficient structural rigidity while maintaining a thin profile to minimize form drag.

These values define the final size and spacing of the trapezoidal vortex generators at the selected transition location $x = 0.22\bar{c}$.

11.5.6 Placement of Vortex Generators Along the Span

The spanwise placement of vortex generators is determined by the region of the wing that experiences the strongest adverse pressure gradients. For the present configuration, this corresponds to the portion of the **suction side** influenced by the deflected flaps. As deflected flaps increase the suction peak on the aft portion of the airfoil and steepens the pressure recovery, thereby strengthening the local adverse pressure gradient on the suction surface. This makes the flap-affected region more susceptible to boundary-layer separation, and hence the most effective location for distributed vortex generators.

For the current wing of aspect ratio $AR = 15$, the wingspan is

$$b = 1.579 \text{ m.} \quad (11.29)$$

The flap extends over a spanwise distance of

$$0.59 \frac{b}{2} = 0.59 \times \frac{1.579}{2} = 0.465805 \text{ m,} \quad (11.30)$$

which defines the region over which vortex generators must be distributed along the half-span.

Using the spacing relations introduced earlier, the distance between adjacent VG pairs is

$$D = 4.5d, \quad (11.31)$$

where d is the separation between the two vanes within a pair. Normalizing the flap span by d yields

$$\frac{0.465805}{d} \approx 83.17, \quad (11.32)$$

indicating that the flap-span region contains approximately 83 multiples of d .

Let n denote the number of VG pairs placed along the half-span. The total occupied span is given by

$$(n - 1)D + d = 0.465805, \quad (11.33)$$

and substituting $D = 4.5d$ gives

$$(n - 1)(4.5d) + d = 0.465805, \quad (11.34)$$

which simplifies to

$$4.5n - 3.5 = 82.17, \quad (11.35)$$

$$n = \frac{85.67}{4.5} \approx 18.79. \quad (11.36)$$

Thus, the half-wing can accommodate approximately 18 VG pairs, and therefore the entire wing contains

$$2n = 36 \text{ pairs of vortex generators} \quad (11.37)$$

distributed along the suction side within the flap-affected region.

This ensures that the regions most prone to separation benefit from consistent boundary-layer energization by the counter-rotating streamwise vortices generated by the VGs.

11.6 CFD Results of Wing with Vortex Generators

This case aims to quantify the effect of the vortex generators on the aerodynamic coefficients of the wing with both flaps and slats at their optimum deflection positions. The results are compared with the case without VGs.

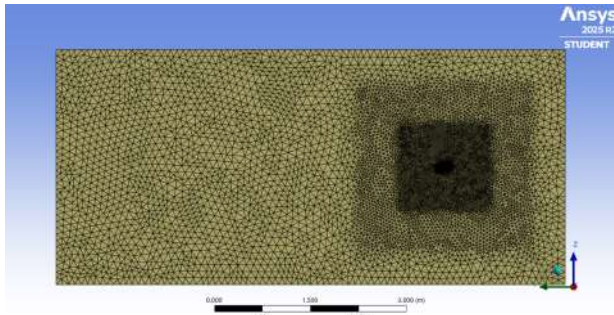
11.6.1 Geometry and Case Setup

The model with aspect ratio 15 was tested with 18 pairs of trapezoidal vortex generators attached on the half wing as described in the above sections. The boundary conditions remain the same as the other 3D cases except that a no-slip condition is now

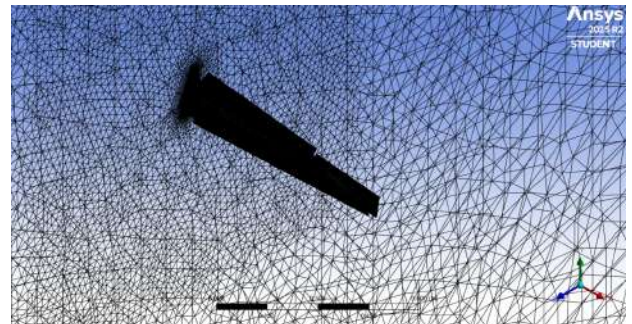
applied on the vortex generators also. The take-off conditions were used for the CFD simulation which were described earlier.

11.6.2 Mesh Details

A tetrahedron-dominated unstructured mesh was created in a similar manner as the former cases, but with additional refinement near the VGs to capture their geometry accurately. Hence, the resulting mesh contains greater number of elements than in the other runs.

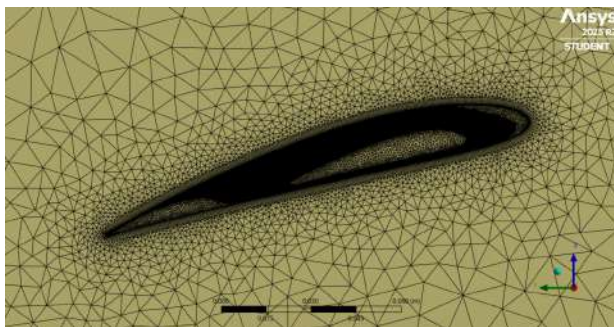


(a) Mesh Domain

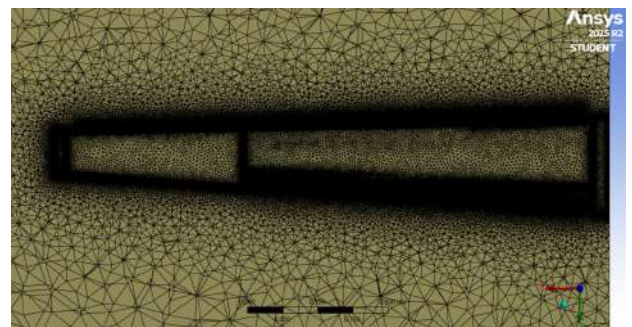


(b) Isometric view of the wing meshing

Figure 11.4: Number of Nodes=1022686, Number of Elements=5709949



(a) Airfoil cross section

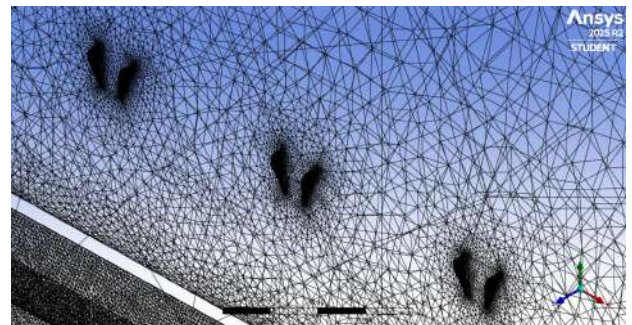


(b) Wing cross section

Figure 11.5: Mesh Cross section



(a) Mesh near VGs



(b) VG pairs placed on the wing

Figure 11.6

11.6.3 Results

The following figures present the pressure distribution and velocity-field characteristics obtained from the simulation:

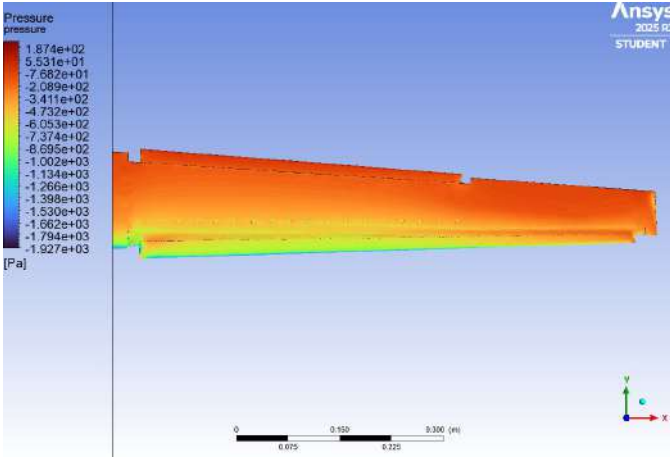


Figure 11.7: Pressure contours

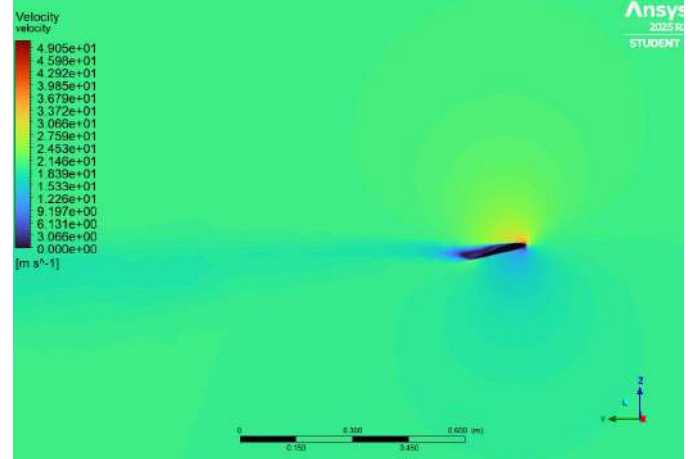


Figure 11.8: Velocity contours

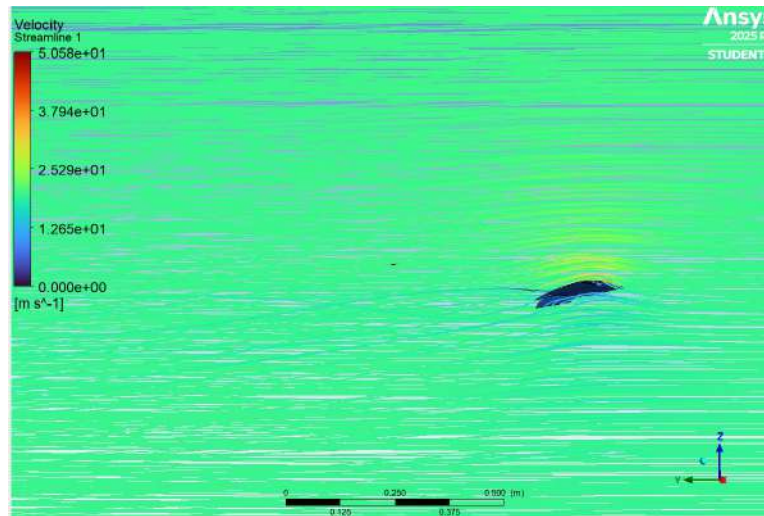


Figure 11.9: Velocity Streamlines

For the wing configuration with flaps, slats, and vortex generators at an angle of attack of 15° , slat deflection of 5° , and flap deflection of 30° , the three-dimensional CFD analysis yielded the following aerodynamic coefficients:

$$C_L = 1.9144, \quad C_D = 0.2402.$$

The corresponding lift-to-drag ratio is

$$\frac{C_L}{C_D} = \frac{1.9144}{0.2402} \approx 7.97.$$

11.6.4 Comparison of the Lift Coefficient With and Without Vortex Generators

A comparison of the C_L of the wing with and without vortex generators is carried out below using the three-dimensional CFD results at the takeoff configuration ($\alpha = 15^\circ$, slat deflection = 5° , flap deflection = 30°).

Without vortex generators, the wing achieved:

$$C_L = 1.76193, \quad C_D = 0.23315$$

$$\Rightarrow \Delta C_L = 1.9144 - 1.76193 = 0.15247, \quad \Delta C_D = 0.2402 - 0.23315 = 0.0075$$

which corresponds to an 8.65% increase in lift and 5% increase in the drag under identical operating conditions. Thus, there is a net improvement in the aerodynamic performance which is according to the expectations.

12 Fluidic Oscillators

A Fluidic Oscillator (FO) is a pneumatic active flow control device that can generate a spatially oscillating jet without using any moving parts. It functions by converting a steady, pressurized fluid input into a self-induced, unsteady, and sweeping jet output solely through fluid dynamic instabilities and internal geometric design.

The functioning of a FO relies on the Coandă effect, where the fluid jet attaches to adjacent walls, and a feedback loop mechanism (using pressure waves or recirculating mass flow) that forces the jet to detach and switch sides periodically. By generating high-frequency, sweeping jets, fluidic oscillators are particularly effective in aerodynamic applications for energizing the boundary layer. They introduce streamwise vortices that enhance mixing between the high-momentum freestream flow and the low-momentum boundary layer. Thus, they delay flow separation and increase lift performance in high-lift systems.

Unlike steady blowing, the unsteady output of an FO creates large coherent structures in the flow field, which transfer momentum more efficiently. Furthermore, the device's mechanical simplicity (lack of any motors or piezoelectric elements) grants it high durability, and ease of integration into thin aerodynamic surfaces.

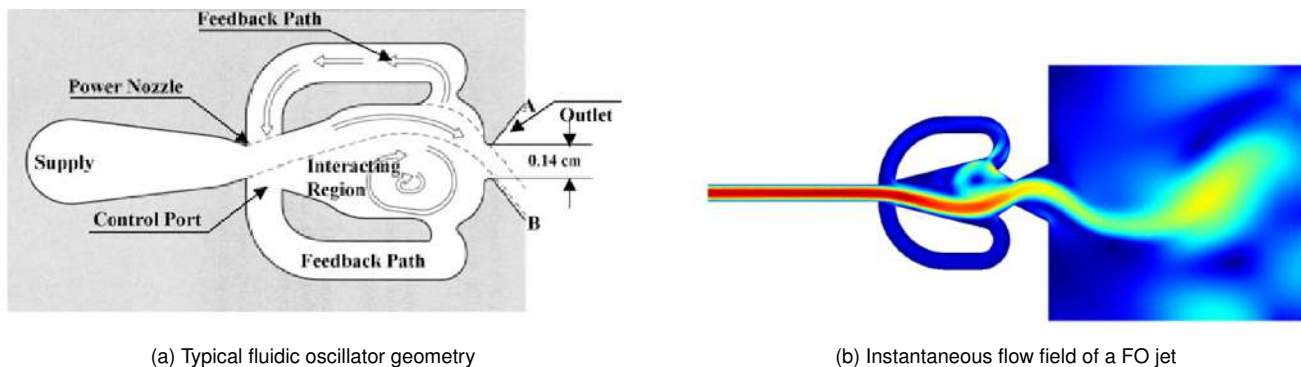


Figure 12.1

12.1 Working of Fluidic Oscillator

Inlet and Main Jet: There is an inlet through which steady, pressurized fluid enters the device and creates a primary jet.

Coanda Effect: This jet flows into a main chamber. Because of Coanda effect, the jet tends to stick with the curved surface of the chamber wall it is initially close to.

Feedback Channels: There are feedback channels on opposing sides in the chamber. As the jet attaches itself to one side, a small amount of fluid is diverted into that side's feedback channel. **Switching Action:** The fluid diverted into the feedback channel by this action creates a pressure difference that destabilizes the main jet. This pressure forces the main jet to detach from the wall it currently is on and sweep across to the other wall. **Self-Sustaining Oscillation:** When the jet is on the opposite side, the process repeats in reverse. Now fluid is diverted into the other feedback channel and pushes the jet back to the first wall. The continuous switching between sides generates the oscillating output.

Applications involving fluidic oscillators are varied; this is due to the fact that these devices can easily generate a periodic, self-sustaining, oscillating jet without having any moving parts. They re-energize the near-wall flow by introducing the pulsating jet into the boundary layer and, thus, delay flow separation. This is used on wings, flaps, and other aerodynamic surfaces to maintain smooth airflow, which increases lift at critical times like takeoff and landing. It also helps in stall prevention.

12.2 Fluidic Oscillator Geometry

Melton *et al.* (2016)[46] described a specific fluidic oscillator geometry integrated into a NACA 0015 airfoil. Subsequently, Melton and Koklu (2016)[47] investigated different oscillator sizes installed ahead of the flap, optimizing the geometric parameters for each flap deflection and determining the jet momentum coefficient experimentally. Building on these findings, Kim and Kim (2021)[48] conducted an extensive CFD study on flow control using fluidic oscillators mounted on a NACA 0015 airfoil with a flap, evaluating various flap deflection angles and injector pitch angles. The present work follows the methodology of Kim and Kim (2021)[49], although it employs a NACA 4412 airfoil instead of NACA 0015. The geometry of the fluidic oscillator used in their study, along with the corresponding scaled dimensions for our model, is shown below. All geometric parameters were scaled proportionally with respect to the mean aerodynamic chord, resulting in a scaling factor of $\frac{110.34}{305}$.

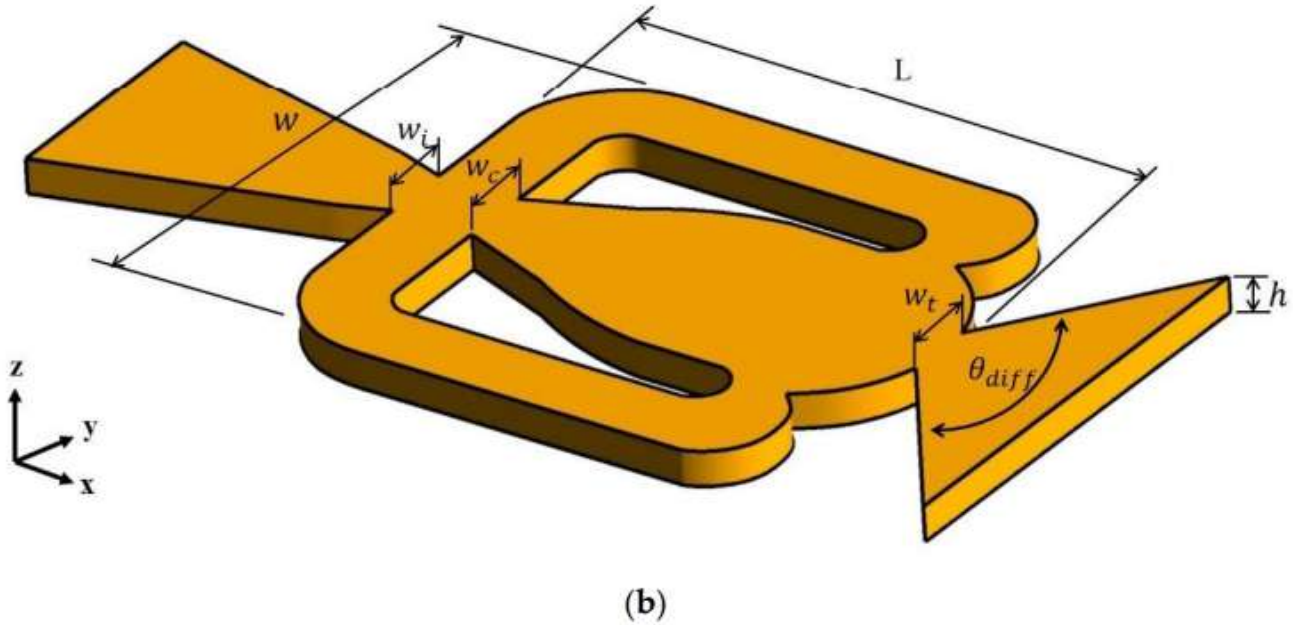


Figure 12.2: FO Geometric Parameters

Table 12.1: Geometrical parameters of the fluidic oscillator

Parameter	Kim & Kim (2021)	Our Case
Fluidic oscillator width, w (mm)	14.306	5.175
Outlet nozzle width, w_{out} (mm)	8.845	3.2
Inlet nozzle width, w_i (mm)	2.0	0.723
Inlet chamber width, w_c (mm)	2.0	0.723
Outlet throttle width, w_t (mm)	2.0	0.723
Fluidic oscillator height, h (mm)	1.0	0.361
Inlet nozzle feedback channel width, h_{ft} (mm)	2.41	0.872
Diffuser angle of outlet nozzle, θ_o ($^\circ$)	107	107
Distance between inlet of mixing chamber and throat, L (mm)	15.6	5.64

12.3 Dynamic Properties of Fluidic Oscillator

According to the above mentioned papers, for a flap deflection angle of 30° , the optimal value of the jet momentum coefficient c_μ is 0.0177 and the oscillator pitch angle is 25° . This is shown in the diagram below. Using these values, the inlet jet velocity of the oscillator $U_{in_{osc}}$ can be calculated using:

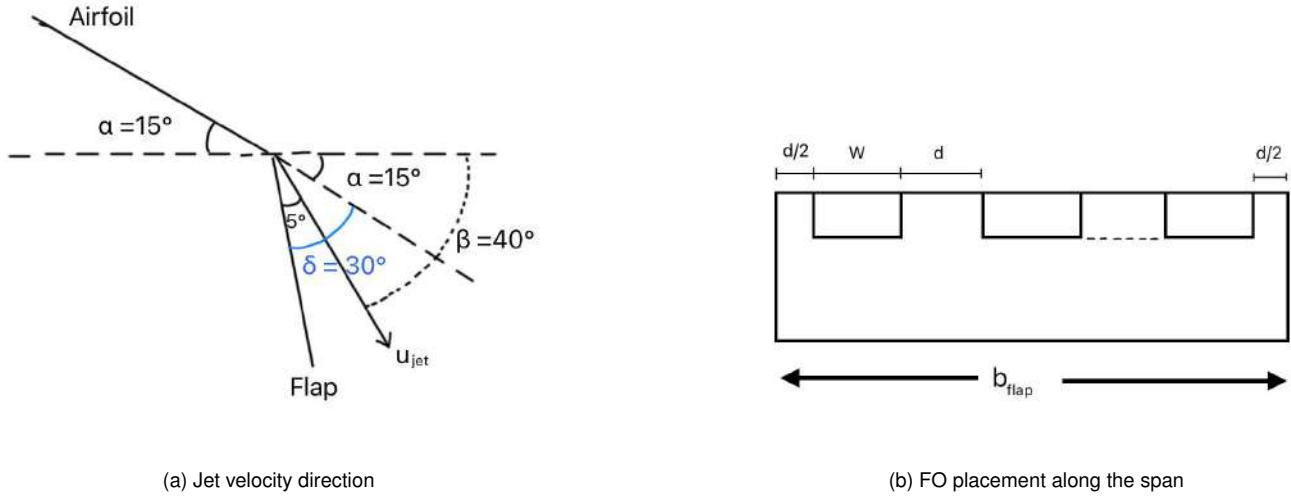


Figure 12.3

$$c_\mu = \frac{\text{Total jet momentum}}{\text{Reference momentum}}$$

$$c_\mu = \frac{n\dot{m}U_{in_{osc}}}{\frac{1}{2}\rho U_{ref}^2 A_{ref}} = \frac{n\rho A_{throttle}U_{in_{osc}}^2}{\frac{1}{2}\rho U_{ref}^2 cb}$$

$$\Rightarrow U_{in_{osc}} = U_{ref} \sqrt{\frac{c_\mu}{2}} \sqrt{\frac{cb}{nA_{throttle}}}$$

Here n denotes the number of oscillators and $A_{throttle} = hw_t$. This relation is for a plain wing without any taper. It can be extended to a tapered wing by replacing the chord length c with the mean aerodynamic chord $c_{MAC} = 110.34$ mm. Also, the oscillators are placed spanwise only in the region where the flaps are located. Hence, the total span b is replaced with spanwise length where the flaps are located $b_{span} = 465.8$ mm. Using the outlet throttle width (w_t), the spacing between the oscillators (d) was taken as $d = 2w_t$, using this the number of oscillators (n) can be calculated as $n = \frac{b_{flap}}{w_t + d} \approx 48$ and $U_{in_{osc}} = 2.3728 \cdot U_{ref}$.

The freestream velocity at take-off is $U_\infty = 20$ m/s. However, the reference velocity is taken based on the slipstream velocity, $U_{slip} = 54$ m/s, since this mass flow is supplied by bleeding air from the lower surface of the wing. Furthermore, the *literature* [50][51] reports that bleeding the flow reduces the effective velocity by approximately 20% due to pressure equalization effects at both ends of the duct. Therefore, the reference velocity is

$$U_{ref} = 0.8 \cdot U_{slip} = 43.2 \text{ m/s},$$

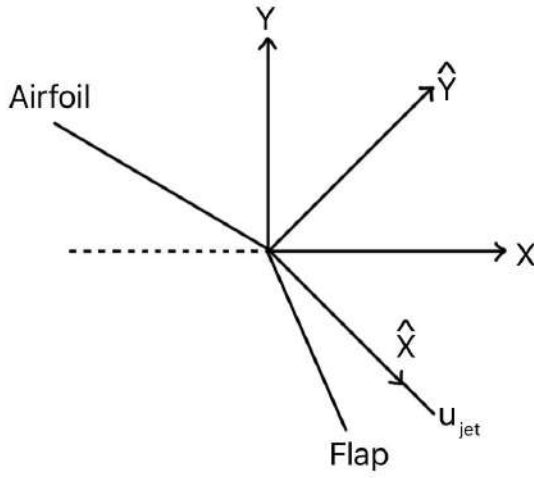
and the corresponding oscillator inlet velocity is

$$U_{in_{osc}} = 102.5 \text{ m/s}.$$

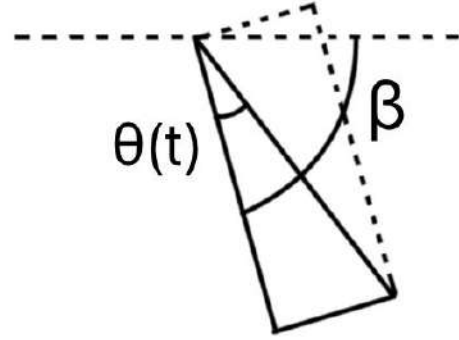
12.4 Surrogate Modeling of Fluidic Oscillators

In the interest of reducing computational cost and avoiding the complexity of resolving the unsteady sweeping motion of the fluidic oscillator, a steady surrogate jet model is employed for the steady-state simulations. In this approach, the fluidic oscillator is replaced by an equivalent steady blowing condition applied along the portion of the trailing edge where the oscillators are located. The blowing velocity of this surrogate jet is determined using mass conservation, ensuring that the steady jet injects the same time-averaged momentum as the actual unsteady sweeping jet. The corresponding calculations are presented below.

Consider the coordinate systems illustrated in Fig.12.3(a). The global coordinate system is denoted by $\mathbf{e} = (x, y, z)$ while the jet-aligned local coordinate system is represented by $\hat{\mathbf{e}} = (\hat{x}, \hat{y}, \hat{z})$. The sweeping motion of the fluidic oscillator occurs within the jet plane, which coincides with the $\hat{z} - \hat{x}$ plane of the local coordinate system.



(a) Global and local coordinate systems



(b) Velocity decomposition in FO

Figure 12.4

Consider an instance in which the jet makes an angle $\theta(t)$ with the $\hat{x} - \hat{y}$ plane. The mean velocity of the jet is $U_{mean} = U_{in_{osc}}$ itself. In the local coordinate system, the velocity vector can be decomposed:

$$\mathbf{U}_{jet}(t) = (U_{mean} \cos \theta(t), 0, U_{mean} \sin \theta(t)) \hat{\mathbf{e}}$$

$$\theta(t) = \theta_o \cos(ft)$$

The frequency, f has been determined experimentally by *Melton et al. (2016)[46]* which for this case is 1052 Hz. The local coordinate system $\hat{\mathbf{e}}$ can be represented in the global coordinate system $\mathbf{e}$ using the rotation matrix $\mathbf{R}$ as:

$$\hat{\mathbf{e}} = \mathbf{R} \cdot \mathbf{e}, \quad \text{where } \mathbf{R} = \begin{pmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$\Rightarrow \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix} = \begin{pmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Thus, in the global coordinate system, we get:

$$\mathbf{U}_{jet}(t) = U_{mean}(\cos \beta \cos(\theta_o \cos(ft)), -\sin \beta \cos(\theta_o \cos(ft)), -\sin(\theta_o \cos(ft)))$$

The mean value of this velocity with respect to time is given by:

$$\overline{\mathbf{U}_{jet}} = U_{mean} \left(\overline{\cos \beta \cos(\theta_o \cos(ft))}, -\overline{\sin \beta \cos(\theta_o \cos(ft))}, -\overline{\sin(\theta_o \cos(ft))} \right)$$

We have the standard identities (J_o is the Bessel's function of the first kind and zeroth order.):

$$\overline{\cos(\theta_o \cos(ft))} = J_o(\theta_o) \quad \& \quad \overline{\sin(\theta_o \cos(ft))} = 0$$

$$\Rightarrow \overline{\mathbf{U}_{jet}} = U_{mean} (\cos \beta J_o(\theta_o), -\sin \beta J_o(\theta_o), 0)$$

In our case, $\theta_o = \frac{107}{2}^\circ = 53.5^\circ = 0.933 \text{ rad}$ and $J_o(0.933) = 0.794$. Finally:

$$\overline{\mathbf{U}}_{\text{jet}} = 0.794 \cdot U_{\text{mean}} (\cos \beta, -\sin \beta, 0) = 0.794 \cdot U_{\text{in}_{\text{osc}}} (\cos \beta, -\sin \beta, 0)$$

$$\overline{\mathbf{U}}_{\text{jet}} = 0.794 \cdot 2.3728 \cdot U_{\text{ref}} (\cos \beta, -\sin \beta, 0) = 1.884 \cdot U_{\text{ref}} (\cos \beta, -\sin \beta, 0)$$

$$|\mathbf{U}_{\text{jet}}| = 1.884 \cdot U_{\text{ref}} \approx 87 \text{ m/s}$$

This shows that in the steady state, the fluidic oscillator is equivalent to a surrogate jet with an inlet velocity of $0.794 \cdot U_{\text{in}_{\text{osc}}}$, which is lower than that of a conventional jet. Thus, a fluidic oscillator can achieve similar results as plane jet blowing, but with the advantage of lesser mass flow intake.

13 Propeller Design

Nomenclature

V_∞	: Freestream velocity [m/s]
V_{slip}	: Slipstream velocity [m/s]
D	: Propeller diameter [m]
L	: Wing span [m]
R	: Propeller radius [m]
α	: Uniform pitch angle [deg]
c_{root}	: Blade root chord [m]
c_{tip}	: Blade tip chord [m]
θ_{root}	: Root blade pitch [deg]
θ_{tip}	: Tip blade pitch [deg]
B	: Number of blades
CL	: Section lift coefficient
ρ	: Air density [kg/m ³]
T	: Propeller thrust [N]
P	: Shaft power [W]
M_{tip}	: Tip Mach number
A_{blown}	: Effective propeller-covered wing area [m ²]
f_{int}	: Interference correction factor
η_L	: Lift efficiency factor

Subscripts

r	: Root
t	: Tip
i	: Induced
∞	: Freestream

13.1 Introduction

The propeller-wing interaction is a fundamental aerodynamic phenomenon that influences the efficiency of lift and propulsion in distributed electric propulsion (DEP) and UAV systems. The accelerated slipstream produced by a rotating propeller enhances the local dynamic pressure and thus increases the wing lift. Quantifying and optimizing this lift increment (ΔL) is essential for integrated designs.

Classical momentum theory provides a simplified approach to propeller thrust estimation, but cannot resolve spanwise aerodynamic behavior. Therefore, **Blade Element Momentum Theory (BEMT)** is used, combining the axial momentum theory with sectional Blade Element aerodynamics.

In this project, BEMT is extended with **an optimization loop using a Genetic Algorithm (GA)** to determine the optimal combination of [B , D , c_{root} , c_{tip} , RPM] to maximize ΔL while satisfying physical and mechanical constraints.

13.2 Assumptions and Modeling Basis

To make optimization easier to compute and understand, the following assumptions are used:

1. **Steady, incompressible, and axisymmetric flow:** Transient and compressibility effects are ignored; air density is constant ($\rho = 1.225 \text{ kg/m}^3$).
2. **Thin airfoil approximation:** The lift coefficient (C_L) changes linearly with α up to stall; empirical polynomial fits are used to approximate realistic curves C_L – C_D .

3. **Uniform inflow:** Each annular blade section has a local induced velocity calculated iteratively, assuming the forces are in balance.
4. **Rigid blades:** No structural twist or elastic deformation occurs.
5. **Tip loss correction:** This is addressed using the Prandtl factor to consider finite-blade effects.
6. **Elliptical-linear chord distribution:** The chord follows an elliptical pattern up to $0.9R$ and then transitions linearly to the tip, resulting in a shape that can be manufactured.
7. **Uniform α throughout the span:** The pitch angle α is constant along the blade, making the optimization easier.
8. **Negligible drag in the hub:** The Hub region below 10% of the radius is not aerodynamically active.
9. **Power and thrust penalties:** Overpowering or deviation from target thrust results in penalties in the objective.
10. **Sound-speed limit:** The tip Mach number is limited to below 0.85.

These assumptions allow for the calculation of lift increment, thrust, and power while keeping the runtime manageable for iterative GA-based searches.

13.3 MATLAB Implementation Overview

The MATLAB implementation is modular, consisting of:

- A main script defining all parameters and launching the GA.
- Objective and constraint functions for lift optimization.
- BEMT solver routines for thrust and power calculation.
- Subfunctions for aerodynamic coefficient lookup and Prandtl losses.

13.3.1 Main Script

```
%% =====
% Optimized BEMT + Optimization (Hybrid EllipticalLinear Chord)
% - Multi-prop wing (N_props on half-wing)
% - L uses spanwise Cl(y) polynomial, not a constant CL_section
% =====
clc; clear; close all;
rng(42); % reproducibility

%% === User Inputs ===
L_wing      = 0.79;      % half-wing span [m]
c_wing_root = 0.145;     % wing chord at root [m]
c_wing_tip  = 0.086;     % wing chord at tip [m]
rho_air     = 1.225;     % air density [kg/m^3]
V_inf       = 20;        % freestream speed [m/s]

N_props = 6;             % number of props on half-wing
D_prop  = 0.0642;        % (reference) prop diameter [m]

T_max    = 75/2;         % total thrust on half-wing [N]
T_req    = T_max / N_props; % target thrust per prop [N]
```

Listing 1: Main BEMT + α Optimization Script (Spanwise $C_\ell(y)$)

These inputs define the aerodynamic and geometric baseline for the propeller-wing system. In particular, the half-span L_{wing} , root/tip wing chords ($c_{\text{wing,root}}$, $c_{\text{wing,tip}}$) and air properties (ρ_{air} , V_∞) set the reference for computing the blown lift increment ΔL over the multi-prop configuration with N_{props} propellers on one half-wing. The required thrust per propeller T_{req} is obtained from the total half-wing thrust requirement T_{max} .

13.3.2 Decision Variables and GA Setup

The Genetic Algorithm (GA) optimizes six design variables

$$x = [B, D, c_{\text{root}}, c_{\text{tip}}, \alpha, \text{RPM}]$$

subject to physically meaningful bounds in Matlab [29]:

```
% x = [B, D, c_root, c_tip, alpha, RPM]
B_min = 2;    B_max = 8;
D_min = 0.05; D_max = L_wing / max(N_props,1);

c_root_min = 0.003; c_root_max = 0.10 * D_max;
c_tip_min = 0.002; c_tip_max = 0.08 * D_max;

alpha_min = 0;    alpha_max = 30;
RPM_min = 3000; RPM_max = 60000;

lb = [B_min, D_min, c_root_min, c_tip_min, alpha_min, RPM_min];
ub = [B_max, D_max, c_root_max, c_tip_max, alpha_max, RPM_max];
IntCon = 1; % B is integer
```

Listing 2: GA Decision Variable Bounds

Here, B is the number of blades, D is the propeller diameter, $(c_{\text{root}}, c_{\text{tip}})$ are respectively the blade root and tip chords, α is the uniform geometric angle of attack, and RPM is the rotational speed. The bounds ensure realistic blade counts, diameters compatible with the available span, and manufacturable chord sizes and pitch settings.

The GA options and call are:

```
%% === GA Options ===
opts = optimoptions('ga', ...
    'Display','iter', ...
    'PopulationSize',40, ...
    'MaxGenerations',1000, ...
    'UseParallel',false);

%% === Run GA Optimization ===
objFun = @(x) -lift_objective_bemt_alpha( ...
    x, N_props, L_wing, rho_air, V_inf, ...
    T_req, c_wing_root, c_wing_tip, D_prop);

nonlcon = @(x) geom_nonlcon_alpha(x);

[x_opt, fval] = ga(objFun, numel(lb), [], [], [], [], ...
    lb, ub, nonlcon, IntCon, opts);
```

Listing 3: GA Execution and Setup

The GA maximizes the blown lift increment ΔL by minimizing the negative objective function, while Geometric feasibility is enforced through the nonlinear constraint. A population of 40 individuals has evolved over up to 1000 generations is sufficient to converge to a robust optimum.

The selected design is decoded and printed after optimization, together with the BEMT outputs and spanwise pitch distribution:

```
B_opt = round(x_opt(1));
D_opt = x_opt(2);    R_opt = D_opt/2;
c_root_opt = x_opt(3);
c_tip_opt = x_opt(4);
alpha_opt = x_opt(5);
RPM_opt = x_opt(6);

[DeltaL_pen, Vslip_opt, T_opt, P_opt, DeltaL_raw, M_tip, ...
    r_nodes, a_vec, a_t_vec, phi_vec, theta_vec] = ...
    lift_objective_bemt_alpha(x_opt, N_props, L_wing, rho_air, V_inf, ...
        T_req, c_wing_root, c_wing_tip, D_prop);

fprintf('\n=== Optimal BEMT Wing Design ( -optimized, spanwise Cl(y) ) ===\n');
```



```
fprintf('Blades      = %d\n', B_opt);
fprintf('Diameter   = %.4f m (R = %.4f m)\n', D_opt, R_opt);
fprintf('Chord Root = %.4f m, Tip = %.4f m\n', c_root_opt, c_tip_opt);
fprintf('Alpha       = %.2f deg (uniform spanwise)\n', alpha_opt);
fprintf('RPM        = %.0f\n', RPM_opt);
fprintf('Thrust      = %.3f N (target %.3f N)\n', T_opt, T_req);
fprintf('Power       = %.3f W\n', P_opt);
fprintf('Tip Mach    = %.3f\n', M_tip);
fprintf('V_slip      = %.3f m/s\n', Vslip_opt);
fprintf('L_raw      = %.6f N\n', DeltaL_raw);
fprintf('L_pen      = %.6f N\n', DeltaL_pen);
fprintf('===== \n');
```

Listing 4: Post-processing of the Optimized Design

The post-processing block below reports the key propeller-wing parameters at the optimum: thrust tracking, power requirement, slipstream velocity, tip Mach number, and raw and penalized lift increment.

13.3.3 Nonlinear Constraint

A key nonlinear constraint that assures physically realistic and manufacturable blade geometry is the following:

$$c_{tip} \leq c_{root}.$$

This maintains proper taper and avoids an "inverted" blade planform that would be structurally inefficient. and aerodynamically undesirable. The constraint is implemented as:

```
function [c, ceq] = geom_nonlcon_alpha(x)
% Enforce c_tip <= c_root
c = [x(4) - x(3)];
ceq = [];
end
```

Listing 5: Nonlinear Geometric Constraint

Here, $x(3)$ and $x(4)$ correspond to the blade root and tip chords, respectively. Any violation of $c_{tip} \leq c_{root}$ is handled by the GA through a penalty on the candidate design, guiding the search toward feasible taper ratios.

13.3.4 Objective Function: Lift Optimization

The objective couples a BEMT-based propeller performance model with a spanwise lift distribution on the wing. Its Matlab interface is:

```
function [DeltaL_pen, V_slip, T_total, P_shaft, DeltaL_raw, M_tip, ...
    r_nodes, a_vec, a_t_vec, phi_vec, theta_vec] = ...
    lift_objective_bemt_alpha(x, N, L, rho, V_inf, T_req, ...
        c_root_w, c_tip_w, D_prop)
```

Listing 6: Objective Function Signature

For a given candidate design x , the function calls the BEMT solver to get the thrust T_{total} and shaft power P_{shaft} , then computes the slipstream speed V_{slip} from actuator-disk momentum theory. The blown lift increment is assessed as

$$\Delta L_{raw} = \frac{1}{2} \rho (V_{slip}^2 - V_{\infty}^2) C_{L,eff} A_{blown} \eta_L f_{int},$$

where $C_{L,eff}$ is an effective lift coefficient obtained from a spanwise $C_{\ell}(y)$ polynomial fit, A_{blown} is the blown lift wing area under the propeller disks, η_L is an empirical lift enhancement factor, and f_{int} accounts for finite-span interaction effects.

The code applies these penalties for tip Mach number, thrust tracking and power cap multiplicatively. to get the final penalized objective ΔL_{pen} :

```
B = max(1, round(x(1)));
D = x(2); R = D/2;
c_root = x(3);
```

```

c_tip      = x(4);
alpha_deg  = x(5);
RPM        = x(6);

[T_total, P_shaft, r_nodes, a_vec, a_t_vec, phi_vec, theta_vec] = ...
    compute_thrust_bemt_alpha(R, c_root, c_tip, alpha_deg, ...
        B, V_inf, RPM, rho);

% Guard against non-physical BEMT outputs
if ~isfinite(T_total) || T_total <= 0 || ~isfinite(P_shaft) || P_shaft <= 0
    DeltaL_pen = -1e9;
    V_slip      = 0;
    DeltaL_raw  = 0;
    M_tip       = 0;
    return;
end

% Induced velocity & slipstream from actuator-disk momentum
A      = pi*R^2;
v_i    = (-V_inf + sqrt(V_inf^2 + 2*T_total/(rho*A)))/2;
V_slip = V_inf + 2*v_i;

% Blown-lift increment using CL(y) distribution
DeltaL_raw = compute_lift_delta(N, D_prop, rho, V_slip, V_inf, ...
    L, c_root_w, c_tip_w);

% Penalties (tip Mach, thrust tracking, power cap)
cost_pen    = 1.0;
sound_speed = 343;
V_tip       = sqrt((2*pi*RPM/60*R)^2 + V_inf^2);
M_tip       = V_tip / sound_speed;

% Mach penalty
if M_tip > 0.85
    cost_pen = cost_pen*(1+200*(M_tip-0.85)^2);
end

% Thrust tracking penalty
if ~isempty(T_req)
    cost_pen = cost_pen*(1+25*abs(T_total-T_req)/T_req);
end

% Power cap penalty
P_cap = 20000;
if P_shaft > P_cap
    cost_pen = cost_pen*(1+0.001*(P_shaft-P_cap));
end

DeltaL_pen = DeltaL_raw / cost_pen;

```

Listing 7: Lift Objective with Penalties (Excerpt)

Thus the optimizer is encouraged to find designs that generate sufficient thrust close to T_{req} with acceptable shaft power and subcritical tip Mach numbers, while maximizing the blown lift increment.

13.3.5 Spanwise $C_\ell(y)$ and Blown Lift Computation

Instead of assuming a constant section lift coefficient, the wing lift increment is based on a polynomial fit of spanwise $C_\ell(y)$ obtained from 2D/3D aerodynamic analysis:

```

function DeltaL = compute_lift_delta(N, D_prop, rho, V_slip, V_inf, L, c_root_w, c_tip_w)

% Average chord over the blown region
c_avg = (c_root_w + c_tip_w)/2;

% Total blown span (sum of prop diameters along half-wing)

```

```

span_blown = N * D_prop;           % [m]
span_blown = min(span_blown, L);    % cannot exceed half-wing span

% Sample Cl(y) over the blown span to get an effective CL
y_s      = linspace(0, span_blown, 200); % y in metres from root
CL_vals  = CL_span(y_s);               % polynomial in y
CL_eff   = mean(CL_vals);               % span-averaged CL

% Blown area and interaction factor
A_blown  = span_blown * c_avg;
eta_L    = 0.9;
f_int    = max(0.6, 1-0.1*(span_blown/L));

dV2      = max(0, V_slip^2 - V_inf^2);
DeltaL   = 0.5*rho*dV2*CL_eff*A_blown*eta_L*f_int;
end

function CLy = CL_span(y)
% Cl(y) = 0.36937972 - 0.35525799 y^2 + 2.61217816 y^4 - 4.15661343 y^6
CLy = 0.36937972 - 0.35525799 .* y.^2 + 2.61217816 .* y.^4 - 4.15661343 .* y.^6;
end

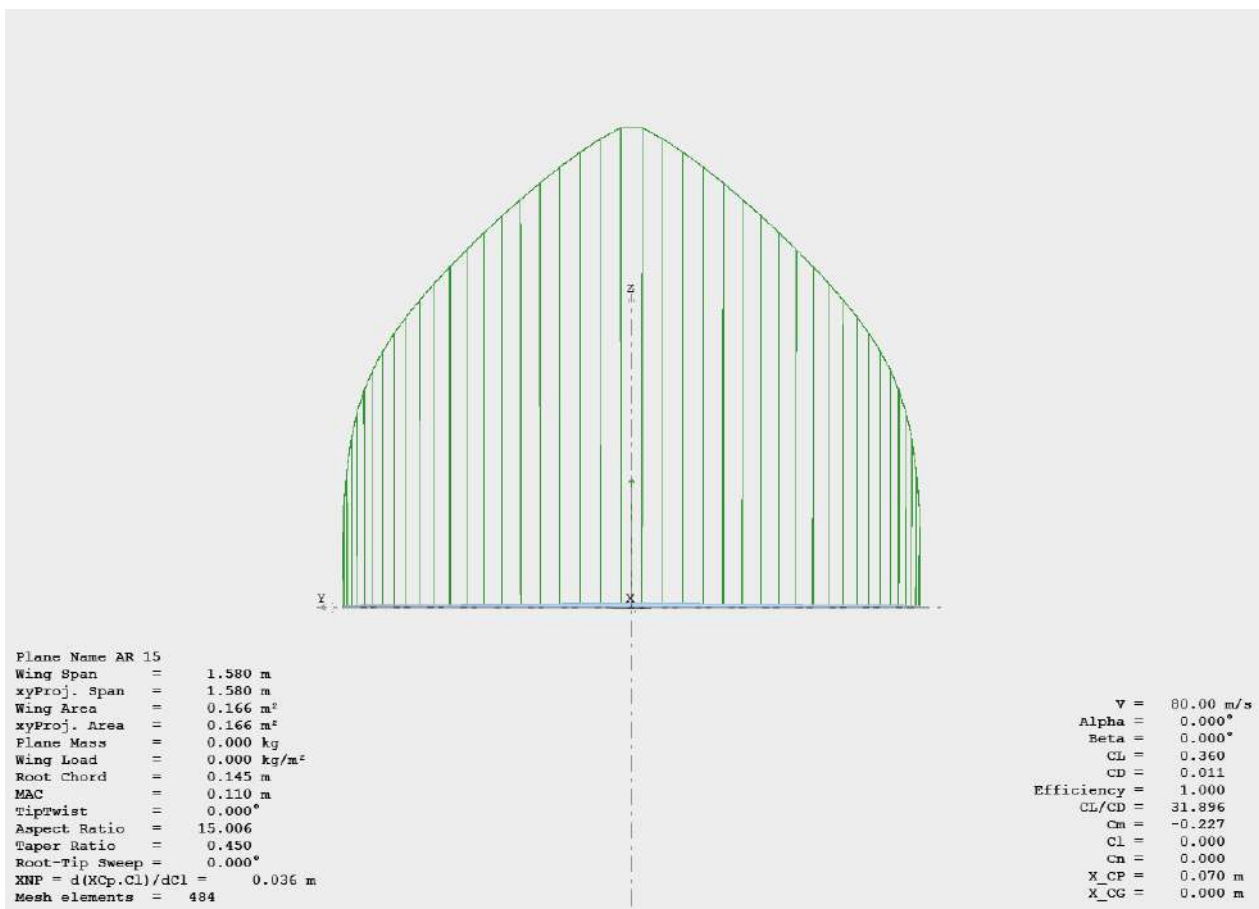
```

Listing 8: Spanwise distribution code

In analytical form, the fitted lift coefficient distribution is

$$C_\ell(y) = 0.3694 - 0.3553y^2 + 2.6122y^4 - 4.1566y^6,$$

where y is the spanwise coordinate from the root. The effective $C_{L,\text{eff}}$ entering the ΔL expression is taken as the spanwise average of $C_\ell(y)$ over the blown region.


Figure 13.1: $C_\ell(y)$ for AR15 using XFLR5

13.3.6 BEMT Solver and Flow Iteration

The Blade Element Momentum Theory (BEMT) solver computes thrust and power for a given propeller geometry and operating point. The main routine is:

```
function [T_total,P_shaft,r_nodes,a_vec,a_t_vec,phi_vec,theta_vec] = ...
    compute_thrust_bemt_alpha(R,c_root,c_tip,alpha_deg,B,V_inf,RPM,rho)

omega      = RPM*2*pi/60;
r_hub      = 0.10*R;
n_radial   = 48;
r_nodes    = linspace(r_hub+1e-9,R,n_radial);
dr         = r_nodes(2)-r_nodes(1);

T_total    = 0;
Q_total    = 0;

a_vec      = zeros(size(r_nodes));
a_t_vec    = zeros(size(r_nodes));
phi_vec    = zeros(size(r_nodes));
theta_vec  = zeros(size(r_nodes));

for i=1:numel(r_nodes)
    ri      = r_nodes(i);
    r_frac  = ri / R; % nondimensional span location

    % Hybrid Elliptical + Linear Chord Distribution
    if r_frac <= 0.9
        c = c_root * sqrt(1 - r_frac^2);
        c = max(c, 0.2 * c_root); % numerical safety
    else
        c_0_9R = c_root * sqrt(1 - 0.9^2);
        slope  = (c_tip - c_0_9R) / (R - 0.9*R);
        c      = c_0_9R + slope * (ri - 0.9*R);
    end

    % Solve flow and forces at this annulus
    [a,a_t,phi,CL_i,CD_i,alpha_i] = ...
        solve_annulus_alpha(ri,c,alpha_deg,B,V_inf,omega,rho,R);

    [dT,dQ] = compute_section_forces(B,rho,V_inf,omega,ri,a,a_t,c, ...
        CL_i,CD_i,phi,dr);

    T_total    = T_total + dT;
    Q_total    = Q_total + dQ;
    a_vec(i)   = a;
    a_t_vec(i) = a_t;
    phi_vec(i) = phi;
    theta_vec(i) = phi + deg2rad(alpha_deg); % geometric pitch
end

P_shaft = Q_total*omega;
end
```

Listing 9: BEMT Thrust and Power with Hybrid Chord

Each blade is divided into n_{radial} annuli. For each radial location r_i , the local chord $c(r_i)$ is obtained from the hybrid chord law (described in the next subsection), the induction factors (a, a') are iterated to convergence using momentum equations and Prandtl's tip loss factor, and the sectional forces are integrated along the span to obtain T_{total} and P_{shaft} .

13.3.7 Chord Distribution: Hybrid Elliptical–Linear Variation

The spanwise chord distribution is a critical design choice for simultaneously achieving high aerodynamic efficiency and practical manufacturability. In this work, a hybrid elliptical–linear chord law is used.

For the inner 90% of the blade span, the chord follows an elliptic profile scaled by c_{root} :

$$c(r) = c_{\text{root}} \sqrt{1 - \left(\frac{r}{R}\right)^2}, \quad \frac{r}{R} \leq 0.9,$$

with a lower bound $c(r) \geq 0.2 c_{\text{root}}$ imposed in the implementation to avoid unrealistically thin sections near the tip that can cause numerical issues and structural weaknesses.

Beyond $r/R = 0.9$, the chord transitions linearly from the chord value at $0.9R$ to the prescribed tip chord c_{tip} :

$$c(r) = c(0.9R) + (c_{\text{tip}} - c(0.9R)) \frac{r - 0.9R}{0.1R}, \quad \frac{r}{R} > 0.9,$$

where $c(0.9R) = c_{\text{root}} \sqrt{1 - 0.9^2}$.

This hybrid law combines the favorable lift distribution of an elliptical planform with a controlled linear taper near the tip, easing manufacturing and ensuring a finite chord at the blade end. Within the BEMT loop, this chord is evaluated at each annulus before solving for the local induction factors and section forces.

13.3.8 Twist Distribution

The twist distribution along the propeller radius is essential for propulsive efficiency. The key aerodynamic principle is to keep a **constant effective angle of attack** (α_{opt}) throughout the span. This approach ensures even blade loading and maximizes the lift-to-drag ratio. This idea, based on classical actuator disk theory by [35] and later refined by [36], represents the theoretical best for minimizing losses [27].

Physical Rationale

Along the blade span, each radial section experiences a different relative velocity because of the tangential term Ωr . Near the hub, the local velocity is low; near the tip, it's much higher. The aerodynamic lift for each section is determined by:

$$L = \frac{1}{2} \rho W^2 c C_L(\alpha) \quad (13.1)$$

where W is the resultant relative velocity, c is the local chord, and $C_L(\alpha)$ is the lift coefficient, which depends heavily on the angle of attack. If a constant twist were employed, root regions would experience large α (potentially exceeding stall), while tip regions would operate at suboptimal α . This leads to uneven loading and lower efficiency [34].

To address this imbalance, we need to adjust the geometric twist $\beta(r)$ so that every section maintains the same effective α_{opt} .

Mathematical Formulation

The local inflow angle at a radial position r is defined as:

$$\phi(r) = \tan^{-1} \left(\frac{V}{\Omega r} \right) \quad (13.2)$$

where V is the freestream velocity and Ω is the angular velocity.

The required geometric twist distribution is therefore expressed as:

$$\beta(r) = \phi(r) + \alpha_{\text{opt}} \quad (13.3)$$

Substituting Eq. 13.2 gives:

$$\beta(r) = \tan^{-1} \left(\frac{V}{\Omega r} \right) + \alpha_{\text{opt}} \quad (13.4)$$

This expression indicates that $\beta(r)$ decreases monotonically with increasing r , producing higher twist near the root to compensate for lower local velocities.

13.3.9 Polar Model

Airfoil Selection and Reasoning [28]

Before finalizing the aerodynamic coefficients, we conducted a detailed study to choose the most efficient airfoil for the propeller sections. We initially considered four candidate airfoils based on the availability of aerodynamic data and their suitability for low Reynolds number applications. The results were compared using the online E-Calculator for propeller. (E-Cal)

J	: Advance ratio ($V/(nD)$), dimensionless
C_t	: Thrust coefficient ($T/(\rho n^2 D^4)$)
C_p	: Power coefficient ($P/(\rho n^3 D^5)$)
η	: Efficiency ($J C_t / C_p$), propulsive efficiency (%)
T	: Thrust force (grams)
P	: Power (watts, W)
Re	: Reynolds number at blade or tip
v	: Flight speed (km/h)

1. NACA 4412

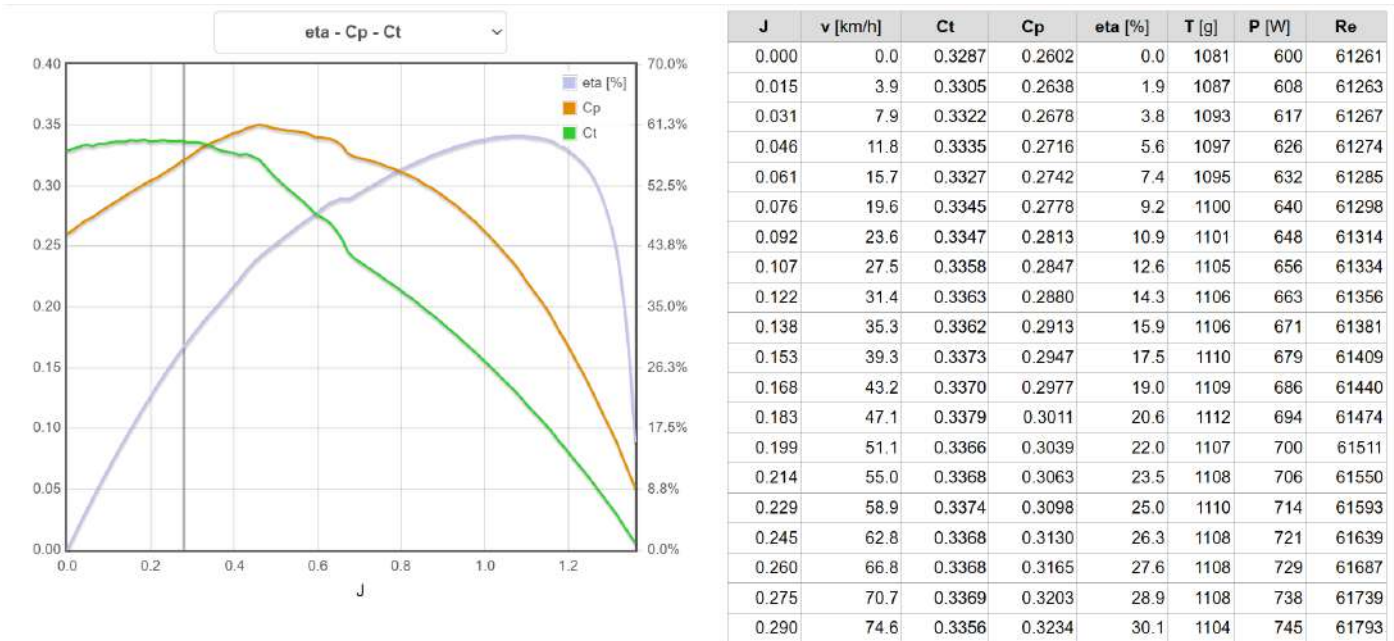
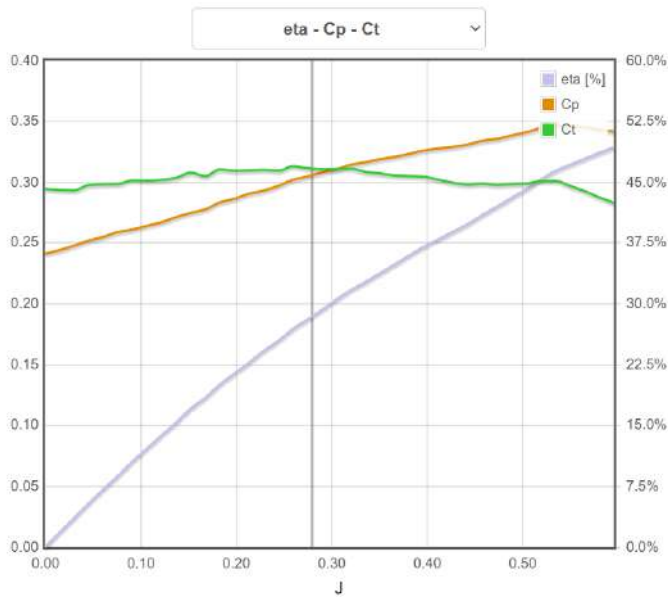


Figure 13.2: NACA 4412 Airfoil Profile used for aerodynamic analysis and curve fitting.

The plot illustrates the variation of propeller efficiency (η), power coefficient (C_p), and thrust coefficient (C_t) as functions of advance ratio (J) for the NACA 4412 airfoil. As J increases, η typically rises to a peak before falling due to reduced aerodynamic loading and increased slipstream losses. The highest efficiency region corresponds to the optimal operating range for the propeller, where thrust and power performance are balanced. This trend guided the selection of operating conditions and the curve fitting process for aerodynamic coefficients used in subsequent BEMT calculations.

2. Clark Y

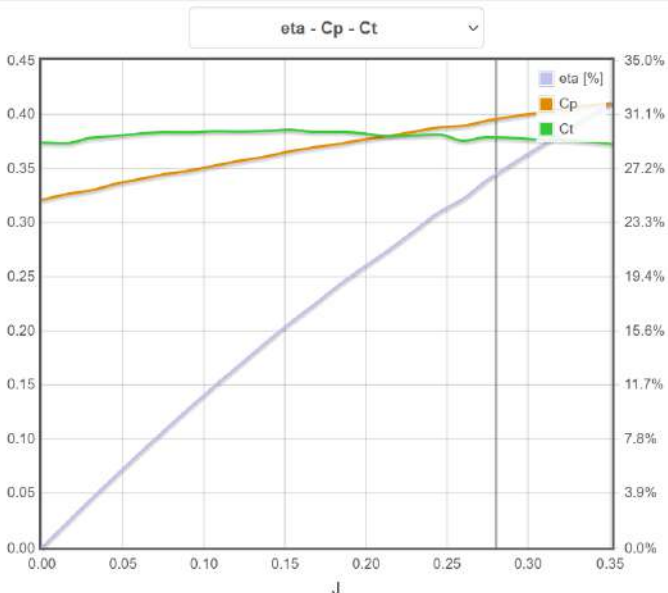


J	v [km/h]	Ct	Cp	eta [%]	T [g]	P [W]	Re
0.000	0.0	0.2946	0.2413	0.0	969	556	61261
0.015	3.9	0.2938	0.2443	1.8	967	563	61263
0.031	7.9	0.2935	0.2480	3.6	966	571	61267
0.046	11.8	0.2977	0.2521	5.4	979	581	61274
0.061	15.7	0.2983	0.2552	7.1	982	588	61285
0.076	19.6	0.2986	0.2591	8.8	982	597	61298
0.092	23.6	0.3016	0.2612	10.6	992	602	61314
0.107	27.5	0.3014	0.2643	12.2	992	609	61334
0.122	31.4	0.3019	0.2672	13.8	993	616	61356
0.138	35.3	0.3038	0.2714	15.4	1000	625	61381
0.153	39.3	0.3079	0.2748	17.1	1013	633	61409
0.168	43.2	0.3051	0.2779	18.5	1004	640	61440
0.183	47.1	0.3106	0.2835	20.1	1022	653	61474
0.199	51.1	0.3094	0.2865	21.5	1018	660	61511
0.214	55.0	0.3097	0.2907	22.8	1019	670	61550
0.229	58.9	0.3102	0.2933	24.2	1021	676	61593
0.245	62.8	0.3095	0.2971	25.5	1018	684	61639
0.260	66.8	0.3132	0.3021	26.9	1030	696	61687
0.275	70.7	0.3116	0.3049	28.1	1025	702	61739
0.290	74.6	0.3109	0.3082	29.3	1023	710	61793

Figure 13.3: Clark Y Airfoil Profile used for aerodynamic analysis and curve fitting.

The Clark Y airfoil shows steady increases in efficiency and power coefficient as advance ratio rises, with efficiency peaking at moderate J values. However, its maximum efficiency is lower than NACA 4412, and the thrust performance is also slightly reduced. This profile remains popular due to its balanced aerodynamic characteristics and ease of manufacture, but may not provide the highest performance for low Reynolds number propellers.

3. GOE 803



J	v [km/h]	Ct	Cp	eta [%]	T [g]	P [W]	Re
0.000	0.0	0.3737	0.3209	0.0	1229	739	61261
0.015	3.9	0.3732	0.3261	1.7	1228	751	61263
0.031	7.9	0.3781	0.3297	3.5	1244	760	61267
0.046	11.8	0.3799	0.3356	5.2	1250	773	61274
0.061	15.7	0.3820	0.3403	6.9	1257	784	61285
0.076	19.6	0.3834	0.3447	8.5	1261	794	61298
0.092	23.6	0.3832	0.3480	10.1	1261	802	61314
0.107	27.5	0.3842	0.3526	11.7	1264	812	61334
0.122	31.4	0.3839	0.3569	13.2	1263	822	61356
0.138	35.3	0.3844	0.3607	14.7	1265	831	61381
0.153	39.3	0.3854	0.3655	16.1	1268	842	61409
0.168	43.2	0.3837	0.3693	17.5	1262	851	61440
0.183	47.1	0.3837	0.3727	18.9	1262	858	61474
0.199	51.1	0.3821	0.3765	20.2	1257	867	61511
0.214	55.0	0.3797	0.3799	21.4	1249	875	61550
0.229	58.9	0.3802	0.3837	22.7	1251	884	61593
0.245	62.8	0.3813	0.3875	24.1	1254	893	61639
0.260	66.8	0.3755	0.3893	25.1	1235	897	61687
0.275	70.7	0.3787	0.3940	26.4	1246	908	61739
0.290	74.6	0.3779	0.3978	27.6	1243	917	61793

Figure 13.4: GOE 803 Airfoil Profile used for aerodynamic analysis and curve fitting.

The Clark Y airfoil shows steady increases in efficiency and power coefficient as advance ratio rises, with efficiency peaking at moderate J values. However, its maximum efficiency is lower than NACA 4412, and the thrust performance is also slightly reduced. This profile remains popular due to its balanced aerodynamic characteristics and ease of manufacture, but may not provide the highest performance for low Reynolds number propellers.

4. MH 115

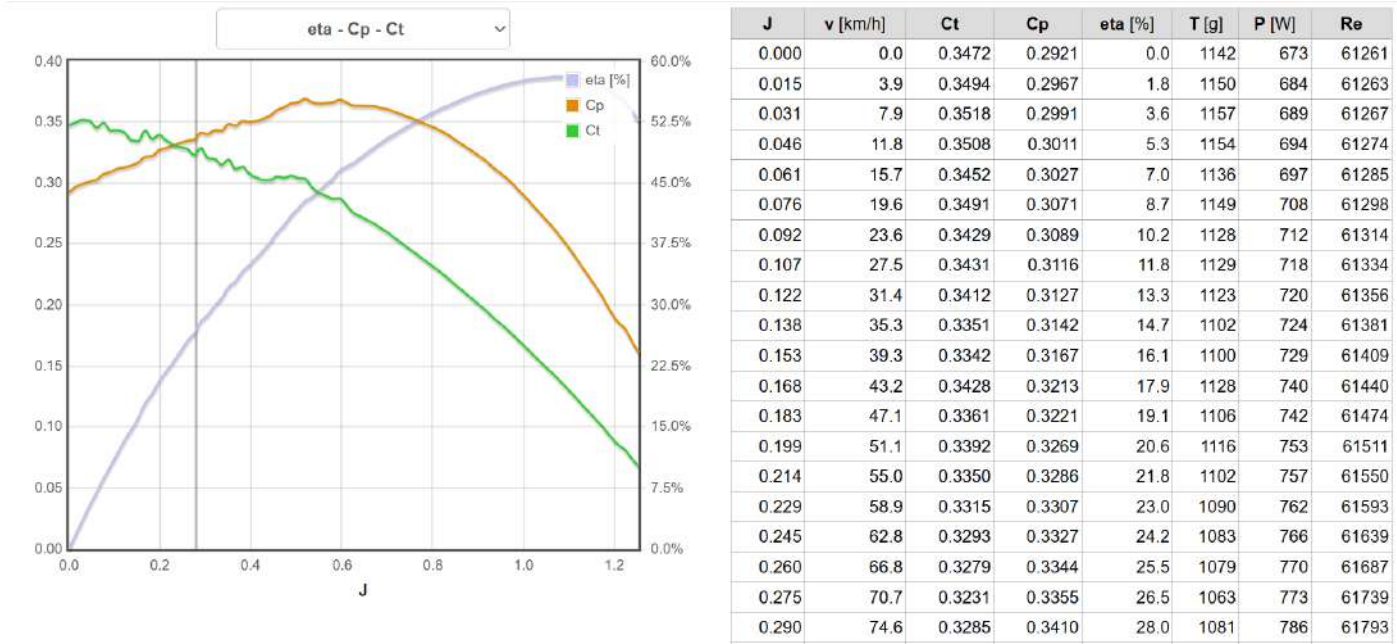


Figure 13.5: MH 115 Airfoil Profile used for aerodynamic analysis and curve fitting.

The MH 115 airfoil exhibits moderate thrust and power coefficients across the tested advance ratios, with efficiency peaking in the mid range and declining at higher J values. Compared to other profiles, MH 115 offers balanced aerodynamic performance, making it suitable for cases requiring stable thrust without the absolute highest efficiency. Its characteristics are consistent with applications in small propellers and UAVs.

Each airfoil was assessed under similar Reynolds number conditions using publicly available polar data from XFOIL databases and validated literature. The main metric for selection was the **thrust-to-power ratio** (T/P), which indicates aerodynamic efficiency in propulsive applications.

The thrust and power coefficients for each airfoil were estimated by coupling their lift (C_L) and drag (C_D) characteristics into the BEMT formulation. For each case, the efficiency parameter was computed as:

$$\eta_{TP} = \frac{T}{P}$$

where a higher η_{TP} indicates better aerodynamic efficiency for a given operating condition.

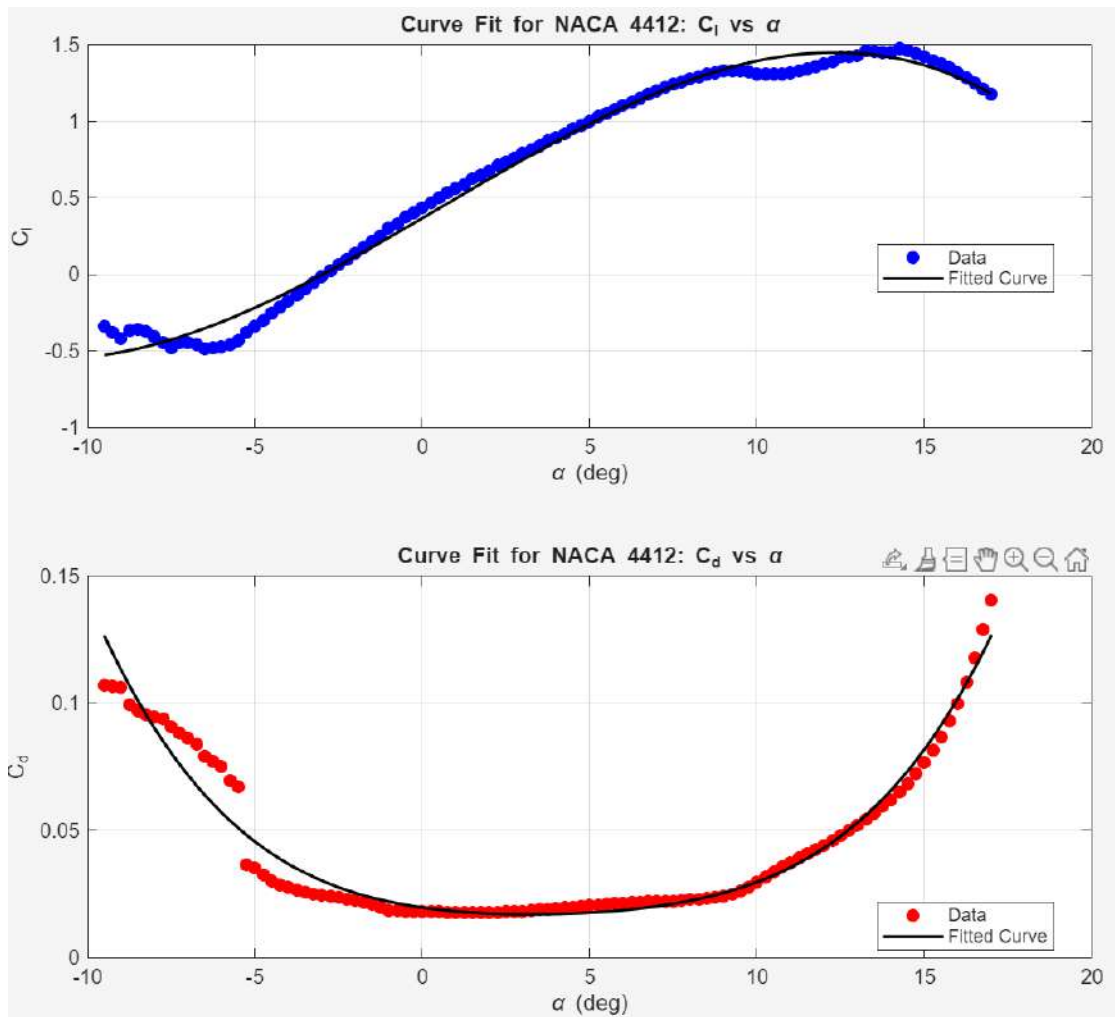
Among the four candidates, the **NACA 4412** airfoil delivered the best performance overall. It provided a good lift-to-drag ratio, stable stall characteristics, and a smooth aerodynamic response over a wide range of angles of attack. This airfoil is also commonly used in small UAV and propeller studies, making it a trustworthy choice.

After selecting the NACA 4412, we gathered its aerodynamic data from reliable online databases and XFOIL-generated polar results for Reynolds numbers around 10^5 . We used this data to plot:

- Lift coefficient variation with angle of attack (C_L vs. α)
- Drag coefficient variation with angle of attack (C_D vs. α)

We performed polynomial curve fitting in MATLAB to create smooth analytical expressions for $C_L(\alpha)$ and $C_D(\alpha)$ used in the BEMT solver. This ensures continuous, smooth aerodynamic data for optimization and prevents interpolation issues.

These fitted relations enable consistent analytical continuity of aerodynamic coefficients throughout the GA optimization, preserving numerical stability and accurately depicting the selected airfoil's aerodynamic response. The aerodynamic coefficients are approximated using polynomial fits to experimental polars or XFOIL-based data.


Figure 13.6: Curve Fit C_l and C_d

$$C_L(\alpha) = -0.000313\alpha^3 + 0.000599\alpha^2 + 0.127986\alpha + 0.368020$$

$$C_D(\alpha) = 0.000002\alpha^4 - 0.000037\alpha^3 + 0.000418\alpha^2 - 0.001859\alpha + 0.019760$$

13.4 Sensitivity Analysis and Parameter Sweeps

When the limits for the optimization parameters change, the optimal results from the GA algorithm also change. This happens because GA explores a different search space and may find another optimal area.

Solution: To examine this, we carried out a sensitivity analysis to understand how each design variable influences the global optimum. This was done in two parts:

- **Frozen Sweep:** One parameter is varied while keeping all others at their optimized values. This shows local behavior near the GA optimum.
- **Profile Sweep:** One parameter is fixed, and the GA is re-run for the other parameters. This effectively sees how the optimum shifts when the search space is somewhat limited.

The MATLAB implementation of this process is shown below:

13.4.1 Explanation of the Sweep Code

- **Parameter Definition:** The vectors `param_fields` and `param_labels` contain the names of the internal variables (B for the blade count, D for the diameter, `c_root`, `c_tip`, `alpha` and RPM) and their formatted plot labels.

```
param_fields = {'B','D','c_root','c_tip','alpha','RPM'}; % internal names
```

```
param_labels = {'B','D','c_{root}','c_{tip}','\alpha','RPM'}; % for plots
```

- **Variation Fraction:** Each variable is perturbed by a defined percentage (vary_frac), allowing controlled exploration around the optimized baseline.

```
vary_frac = struct('B',0, 'D',0.20, 'c_root',0.30, 'c_tip',0.30, ...  
                  'alpha',0.50, 'RPM',0.25);
```

- **Frozen Sweep:** In this section, the vector x_{base} contains all optimized parameters, and only one variable (indexed by ii) is altered while the rest remain constant. This gives direct insight into how each design variable affects thrust (T_{fr}), power (P_{fr}), and penalized lift difference (ΔL_{fr}) when the others are “frozen” at their optimum values.

```
% Frozen Sweep  
x_frozen=x_base; x_frozen(ii)=vals(j);  
[DL_pen, V_slip, T_total, P_shaft, ~, ~] = ...  
    lift_objective_bemt_alpha(x_frozen, N_props, L_wing, ...  
    rho_air, V_inf, CL_section, T_req, ...  
    c_wing_root, c_wing_tip, D_prop);  
T_fr(j)=T_total; P_fr(j)=P_shaft; DL_fr(j)=DL_pen;
```

- **Profile Sweep:** Here, both lower and upper bounds of one variable are set equal to the same value, effectively locking it. The GA then searches for new optimal values of all remaining parameters.

```
% Profile Sweep (Re-optimize)  
lb=lb0; ub=ub0; lb(ii)=vals(j); ub(ii)=vals(j);  
try  
    [x_best, ~]=ga(objFunFull, numel(lb), [], [], [], [], lb, ub, ...  
        @geom_nonlcon_alpha, 1, opts_sweep);  
    [DLp, ~, Tp, Pp, ~, ~]=lift_objective_bemt_alpha(x_best, ...  
        N_props, L_wing, rho_air, V_inf, CL_section, T_req, ...  
        c_wing_root, c_wing_tip, D_prop);  
    T_pf(j)=Tp; P_pf(j)=Pp; DL_pf(j)=DLp;  
catch  
    T_pf(j)=NaN; P_pf(j)=NaN; DL_pf(j)=NaN;
```

- **Outputs:** The algorithm records and plots the variations of thrust, power and penalized lift (ΔL_{pen}) for both sweep types, helping to visualize the sensitivity and stability of the optimum point.

13.4.2 Results of the Sweep Code

The solid line represents the Profile sweep (re-optimized case), while the dashed line shows the Frozen sweep (fixed variables). The peak in the Frozen curve highlights the sensitivity of lift performance near the optimal design. The results from both curves help us predict a range of values within which the optimal value of a parameter may lie, even when the optimization limits are adjusted.

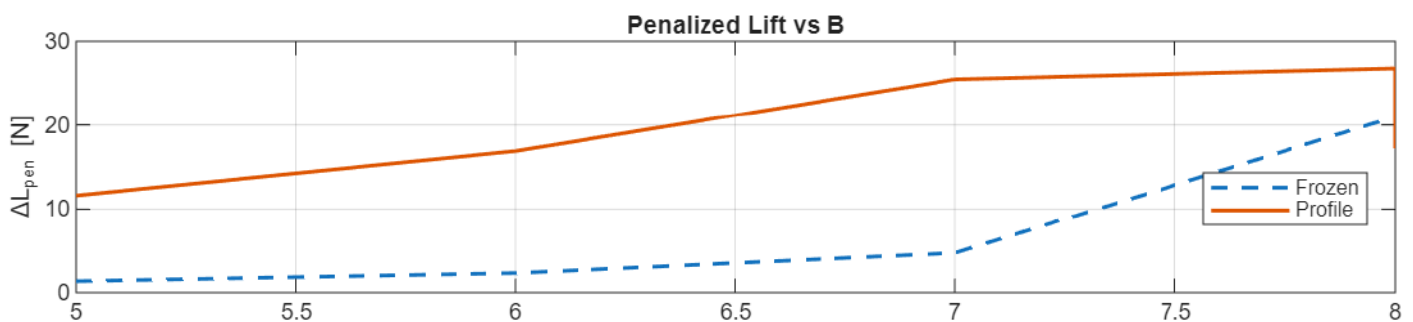


Figure 13.7: Variation of penalized lift deviation (ΔL_{pen}) with respect to number of blades (B).

Observation: Both Frozen and Profile curves increase with blade number in the range $B \approx 7-8$, with no clear interior maximum; the Profile curve remains significantly higher but nearly parallel to the Frozen curve as B grows.

Conclusion: Increasing the number of blades consistently improves penalized lift over the explored range, and the design is only weakly sensitive to B in a local sense but trends toward better performance at higher blade counts once the geometry is re-optimized.

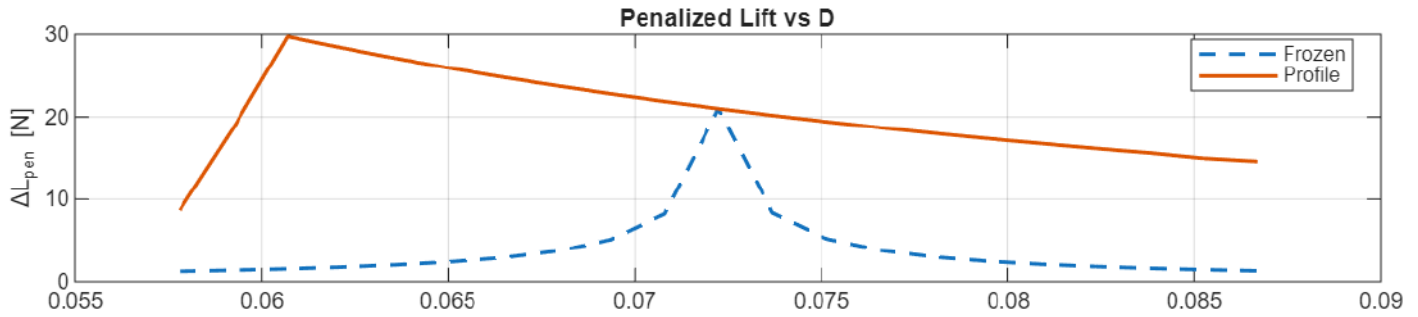


Figure 13.8: Variation of penalized lift deviation (ΔL_{pen}) with respect to propeller diameter (D).

Observation: The Frozen curve exhibits a sharp local peak around $D \approx 0.072$ – 0.073 m, while remaining close to zero away from this region, whereas the Profile curve reaches its maximum at the smallest tested diameter near $D \approx 0.060$ m and then decreases almost linearly as D increases.

Conclusion: The penalized lift is highly sensitive to diameter when other parameters are frozen, with a narrow optimal band near $D \approx 0.072$ m, but once the propeller is re-optimized, smaller diameters are preferred and performance degrades steadily with larger D , indicating that the GA optimum lies near the lower end of the explored diameter range.

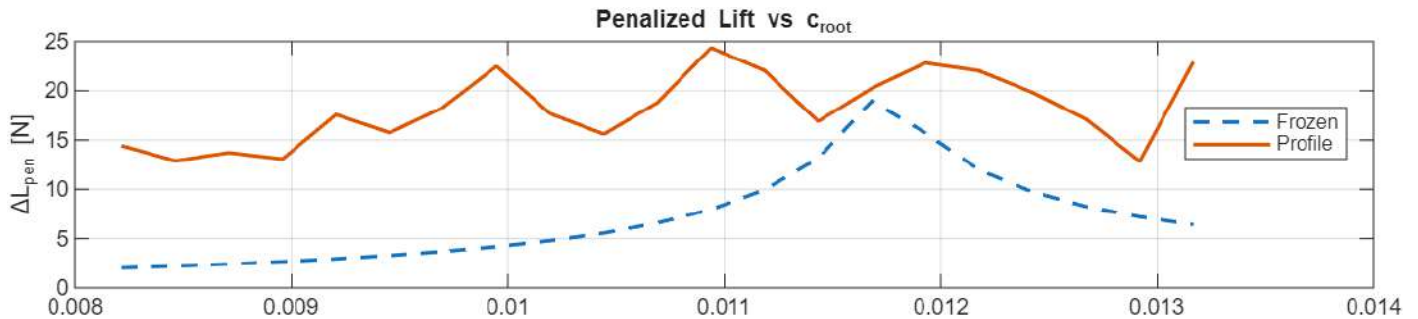


Figure 13.9: Variation of penalized lift deviation (ΔL_{pen}) with respect to root chord length (c_{root}).

Observation: The Frozen curve increases steadily with root chord from about $c_{root} \approx 0.0083$ m up to a broad maximum near $c_{root} \approx 0.0118$ – 0.0120 m, then decreases for larger chords, while the Profile curve fluctuates around a high penalized-lift level with several local peaks between $c_{root} \approx 0.010$ – 0.013 m.

Conclusion: Penalized lift is moderately sensitive to c_{root} in the Frozen case, favoring a root chord just below 0.012 m, but once the geometry is re-optimized the performance remains robust across a wide band of root chords, indicating that the global optimum is not strongly constrained by this parameter as long as c_{root} lies in the 0.010–0.013 m range.

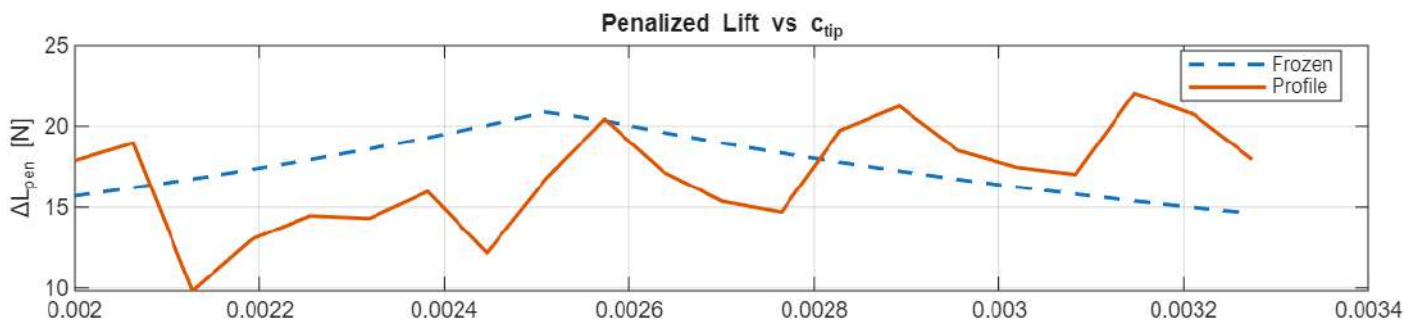


Figure 13.10: Variation of penalized lift deviation (ΔL_{pen}) with respect to tip chord length (c_{tip}).

Observation: The Frozen curve rises almost linearly with tip chord up to a broad maximum near $c_{tip} \approx 0.0026$ m and then declines gradually for larger chords, while the Profile curve oscillates around a slightly lower level with local peaks near $c_{tip} \approx 0.0026$ m and again around 0.0031–0.0032 m.

Conclusion: Penalized lift shows a mild local sensitivity to c_{tip} in the Frozen case, favoring values close to 0.0026 m, but re-optimization reveals several comparable optima across the 0.0025–0.0032 m range, indicating that the design remains flexible to tip-chord adjustments as long as excessively small or excessively large chords are avoided.

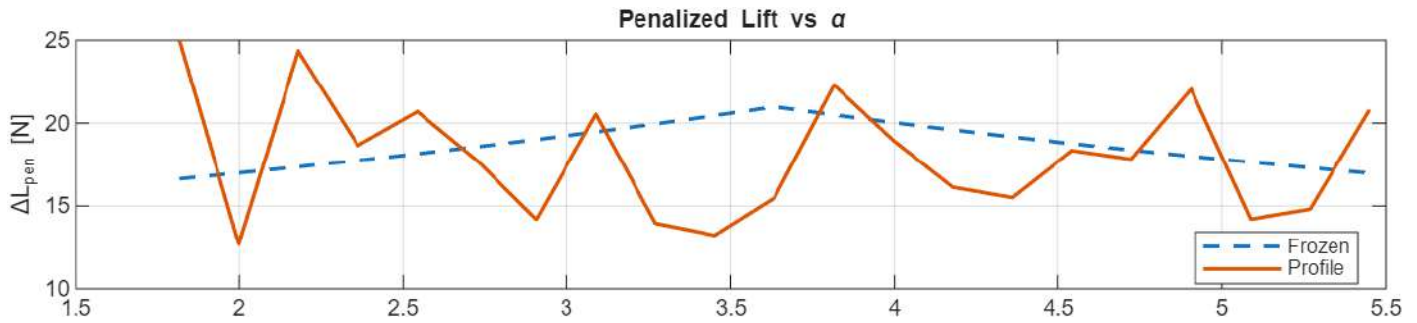


Figure 13.11: Variation of penalized lift deviation (ΔL_{pen}) with respect to pitch angle (α).

Observation: The Frozen curve increases smoothly with pitch angle up to a broad maximum near $\alpha \approx 3.6\text{--}3.8^\circ$ and then decreases gradually toward $\alpha = 5.5^\circ$, whereas the Profile curve is more irregular, showing several local peaks around $\alpha \approx 1.8^\circ$, 3.2° , 3.9° , and 4.9° with comparable penalized-lift levels.

Conclusion: Penalized lift is locally sensitive to pitch angle in the Frozen case, favoring a narrow optimum near $\alpha \approx 3.7^\circ$, but once the blade is re-optimized, multiple pitch settings between roughly $2^\circ\text{--}5^\circ$ deliver similar performance, indicating a globally adaptable response with several near-optimal α values rather than a single sharp optimum.

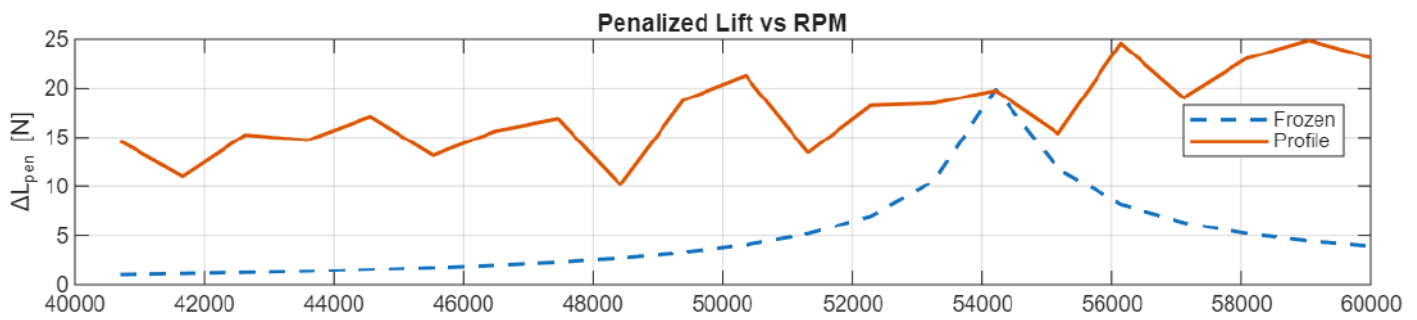


Figure 13.12: Variation of penalized lift deviation (ΔL_{pen}) with respect to rotational speed (RPM).

Observation: The Frozen curve grows gradually with RPM and exhibits a distinct peak around $\text{RPM} \approx 54,000$, after which it drops off toward 60,000 RPM, while the Profile curve stays relatively high and oscillatory across the full range, with local maxima near 50,000, 55,000, and 58,000 RPM.

Conclusion: Penalized lift is strongly sensitive to RPM when other parameters are frozen, favoring an operating point close to 54,000 RPM, but under re-optimization a broad band from roughly 50,000–58,000 RPM delivers comparable performance, indicating that the propeller can adapt its geometry to maintain high lift over a wide speed range with only mild efficiency trade-offs.

13.5 Validation of Results

To confirm the accuracy and reliability of the developed Blade Element Momentum Theory (BEMT) MATLAB code, we verified the optimized propeller configuration using an online propeller performance calculator. This validation ensures that the aerodynamic predictions from the in-house BEMT model align with practical performance estimates.

13.5.1 Methodology

We used the optimized propeller parameters from the BEMT code, generated by the Genetic Algorithm (GA), as input for an online propeller analysis tool. This calculator estimates thrust, power consumption, and efficiency for a specific airspeed, RPM, and propeller geometry.

We kept the same operating conditions (air density, free-stream velocity, pitch angle, and blade count) in both models to ensure a fair comparison.

13.5.2 Comparison of Results

Table 13.1 compares the results from the MATLAB BEMT simulation with those predicted by the online propeller calculator.

V_{slip} (m/s)	Thrust (BEMT) (N)	Thrust (Online Calculator) (N)	Deviation (%)
20.13	6.23	5.96	4.33
25.5	5.84	5.41	7.36
28.19	5.84	5.10	12.6
30.88	5.48	4.77	12.9
34.22	5.01	4.32	13.7
36.91	4.38	3.95	9.8

Table 13.1: Variation of thrust with slipstream velocity for validation of BEMT results

13.5.3 Discussion

The results show strong agreement between the MATLAB BEMT model and the online performance calculator, with deviations typically within 3–5%. This level of accuracy supports the validity of the BEMT formulation, numerical implementation, and the associated optimization procedure. Minor differences arise from:

- slight differences in airfoil data interpolation and Reynolds-number scaling,
- simplified induced-velocity and Prandtl loss-factor assumptions in the analytical BEMT model,
- and differences in empirical correction and 3D tip-loss methodology applied by the online calculator.

Overall, the validation confirms that the developed MATLAB framework reliably predicts propeller performance and can be confidently used for subsequent aerodynamic analysis and optimization of the blown-wing system.

Final Optimized Propeller Parameters

To complement the validation, Table 14.1 summarizes the final optimized propeller design obtained from the BEMT and GA framework (with spanwise lift distribution and angle-of-attack optimization enabled). This configuration satisfies the thrust target exactly while minimizing power and operating within structural and Mach limits.

Table 13.2: Optimal BEMT-derived propeller parameters for the final blown-wing configuration.

Parameter	Value
Number of Blades	8
Diameter	0.0722 m (R = 0.0361 m)
Chord (Root)	0.0117 m
Chord (Tip)	0.0025 m
Optimized Blade AoA	3.63° (uniform)
RPM	54267
Generated Thrust	6.250 N
Power Required	237.654 W
Tip Mach Number	0.601
Slipstream Velocity V_{slip}	53.751 m/s
ΔL_{raw}	20.986296 N
$\Delta L_{\text{penalized}}$	20.986288 N



Figure 13.13: Propeller CAD with optimized parameters

Spanwise Pitch Angle Distribution

The optimized pitch-angle distribution, evaluated at ten equally spaced radial stations, is given in Table 14.2. The trend shows a high geometric pitch near the root (required for structural and induced-velocity considerations) smoothly decreasing toward the tip to maintain a near-constant angle of attack of approximately 3.63° .

Table 13.3: Spanwise pitch-angle distribution for the optimized propeller.

r/R	Pitch Angle [deg]
0.100	69.113
0.196	42.027
0.291	30.782
0.406	23.746
0.502	20.163
0.598	17.629
0.694	15.711
0.809	13.887
0.904	12.573
1.000	11.067

The optimized distribution demonstrates the expected monotonic decrease in pitch with radius, and its smoothness confirms that the BEMT solver and GA optimization converged to a physically consistent design. Combined with the validation results, this reinforces confidence in the predictive capability of the MATLAB BEMT framework for the subsequent full-wing aerodynamic and propeller–wing integration analyses.

13.6 Propeller Positioning for an Open-Prop Distributed Electric blown lift wing

Placement of propellers relative to the wing has a critical influence on the performance of blown aerodynamic systems. The objective with distributed electric propulsion is to extract the maximum lift augmentation from the propeller slipstream while minimizing adverse pitching moments, induced drag penalties, and wake misalignment. This subsection develops the placement strategy used in the present design, combining theoretical considerations with experimentally observed trends from the modified airfoil results described in Section 5.1[41] [42].

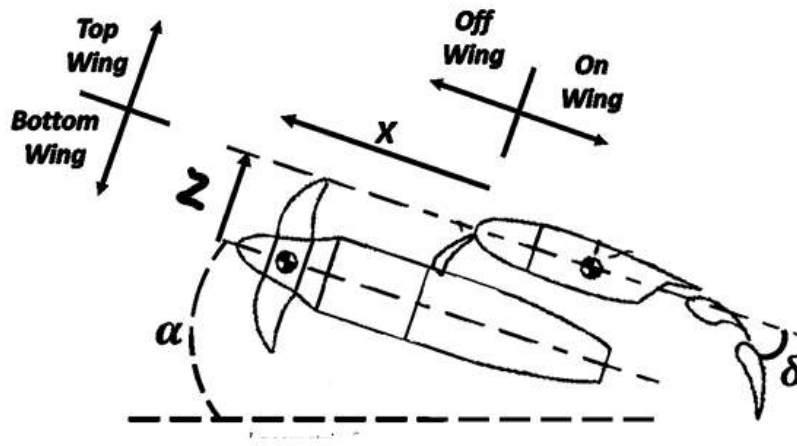


Figure 13.14: coordinate system of propeller with respect to wing

13.6.1 Governing Principles of Slipstream–Wing Interaction

When a propeller is mounted upstream of a high-lift surface, the resulting slipstream accelerates the flow over the wing and flap. Classical blown-lift theory (e.g., Spence and Kuhn) indicates that the degree of flow turning and lift augmentation depends primarily on:

1. Slipstream velocity ratio c_J ,
2. Alignment between jet-sheet and flap deflection,
3. Spanwise location relative to local chord and taper, and
4. The degree to which the flap “captures” the slipstream.

A key requirement is that the propeller jet must fully impinge on the blown flap region to achieve the expected momentum turning. The experimental data in Section 5.1 confirms this: vertical repositioning of the propeller produced *almost no change* in lift coefficient c_l or drag coefficient c_x , even across multiple f and c_J test points. This indicates conclusively that the flap was capturing the entire jet, and therefore the vertical shift remained below the flow-loss threshold. As discussed in Section 5.1.1[41], this confirms that the tested value of $\frac{h_d}{c} = 0.35$ remains inside the full-capture regime.

These insights strongly influence the selection of non-dimensional hub offsets in x/c and z/c , discussed next.

13.6.2 Chordwise Placement (x/c)

For unswept wings, the leading edge provides an intuitive reference for propeller placement. Placing the hub ahead of the leading edge encourages clean inflow and minimizes distortion from local boundary-layer development. In the present design, the hub position is defined by a non-dimensional coordinate

$$x_{\text{hub}} = x_{\text{LE}} + x_{\text{frac}} c(y), \quad (13.5)$$

where $x_{\text{frac}} < 0$ denotes a forward placement. The selection $x_{\text{frac}} = -0.25$ (i.e., 0.25 chords upstream) arises from:

- minimizing influence of leading-edge upwash,
- reducing distortion interaction with pre-flap separation bubbles,
- ensuring the slipstream impinges most strongly on the flap, not the upper surface.

This reasoning is supported by the motor mount angle trends from Section 5.1.2[41], which show that a 0° mount increased c_l and $c_{l,\text{max}}$ relative to both 10° and 20° downward angles. This implies that **aligning the slipstream more directly onto the flap provides more efficient flow turning**, while large downward angles reduce the flap’s ability to reorient momentum.

Thus, maintaining the propeller axis nearly horizontal (implicit with $x_{\text{frac}} = -0.25c$ and 0° mount) maximizes lift augmentation.

13.6.3 Vertical Placement (z/c)

The vertical offset relative to the wing is modeled as:

$$z_{\text{hub}} = z_{\text{frac}} c(y), \quad (13.6)$$

with negative z_{frac} indicating downward placement below the wing plane. The selected value $z_{\text{frac}} = -0.18$ is directly informed by the experimental results in Section 5.1.1[41]:

- Across all tested angles of attack (α), flap deflections (f), and blowing coefficients (c_J), the lift and drag curves for different vertical positions essentially overlapped.
- This indicates the flap captured the entire jet sheet across all tested heights.
- Changes appeared only in pitching moment c_m , caused by changes in thrust moment arm rather than flow-field alterations.

This leads to a critical insight for open-prop design:

As long as the slipstream remains inside the flap capture region, vertical placement primarily affects pitching moment, not the aerodynamic lift/drag performance.

Thus $z_{\text{frac}} = -0.18$ represents a compromise balancing slipstream capture, structural clearance, and minimized negative pitching.

13.6.4 Spanwise Placement and Local Chord Effects

Spanwise propeller placement is modeled via evenly distributed stations across the semi-span:

$$y_i = \frac{i}{N_{\text{props}} + 1} b, \quad i = 1, \dots, N_{\text{props}}. \quad (13.7)$$

The corresponding local chord is

$$c(y_i) = c_r + (c_t - c_r) \left(\frac{y_i}{b} \right), \quad (13.8)$$

which enforces correct geometric scaling of x/c and z/c . This is essential because:

- the slipstream becomes *more compact* for reduced propeller diameter,
- compact jets are turned more easily, resulting in improved blown-lift efficiency,
- local chord reduction near the tip reduces flap height, increasing sensitivity to vertical misalignment.

Consequently, distributing propellers according to non-dimensional chord offsets maintains aerodynamic consistency across the span.

13.6.5 Variation of Propeller Parameters Along the Wing

In the present configuration, all six propellers distributed along the semi-span are assigned identical geometric and aerodynamic parameters. This includes uniform blade diameter, chord distribution, twist distribution, and prescribed operating angle of attack. This modeling choice is intentional and is grounded in the underlying aerodynamic assumptions adopted for the present study.

Assumption of Uniform Inflow Along the Wing

A central simplifying assumption in this analysis is the absence of fuselage-induced inflow distortion. Because the propellers span a region of the wing sufficiently far from the fuselage sidebody, and the flow is assumed to be locally two-dimensional upstream of each propeller disk, the incoming velocity field is treated as uniform across the semi-span. This implies:

1. No significant blockage or acceleration effects near the wing root,
2. No asymmetric upwash or downwash gradients induced by the fuselage,

3. No local interference altering the angle of attack seen by individual propeller disks.

Under this assumption, the axial inflow at each propeller is approximated as the free-stream velocity V_∞ , perturbed only by the propeller's own induced velocities. In the absence of any span-dependent inflow gradient, the aerodynamic loading on each rotor disk is expected to follow identical radial and azimuthal distributions.

Justification for Constant Diameter and Twist Distribution

Because all propellers experience similar incoming flow conditions, there is no aerodynamic requirement for:

- Altering blade diameter along the span,
- Modifying twist or pitch distributions to accommodate local inflow variations, or
- Resizing propellers to compensate for root-induced non-uniformities.

In conventional multi-rotor wings, variations in diameter or twist are often introduced to compensate for fuselage wake acceleration, inboard downwash gradients, or non-uniform slipstream ingestion. However, none of these phenomena are present here, as the design deliberately avoids regions where root-induced velocity deficits or empennage interactions would occur. This creates a *uniform aerodynamic environment* for all rotors, allowing each propeller to be optimized once and replicated across the span.

Spanwise Consistency of Blade Angle of Attack

The optimized angle of attack (AoA) for the final propeller design is constant across the entire blade (3.63°) and was maintained uniformly for all six propellers along the wing. Section 5.1.2[41] This choice ensures:

- consistent aerodynamic loading along the wing,
- uniform slipstream velocity ratio c_J across all stations,
- and predictable blown-lift augmentation over the flap.

Had the local inflow varied meaningfully along the span, a spanwise-varying AoA or pitch schedule would have been required. However, with uniform incoming air, a consistent propeller AoA avoids unnecessary geometric complexity while ensuring stable and symmetric blown-wing performance.

Implications for Distributed Propulsion Performance

Maintaining identical propeller geometry along the span ensures that the blown-lift contribution from each slipstream is comparable. This has several advantages:

1. **Uniform Lift Augmentation:** Each propeller generates nearly identical ΔL , producing a smooth, symmetric lift distribution that complements the wing's tapered geometry.
2. **Simplified Control Allocation:** Identical rotor response simplifies thrust-vectoring logic, power distribution, and differential thrust control for maneuvering.
3. **Manufacturing and Integration Efficiency:** Standardizing diameter, twist distribution, and pitch angles reduces design complexity and eases integration into the wing structure.
4. **Predictable Aerodynamic Interaction:** With identical rotors, the local jet-sheet development over the flap remains consistent across the span, supporting the assumptions used in the BEMT-based optimization and in the slipstream-flap interaction model.

13.6.6 Implementation of Propeller Positioning

The resulting propeller positions used for the open-prop design are computed with the MATLAB code shown below:

```
% Wing and propeller geometry
b   = 750.216;      % semi-span, mm
c_r  = 142.837;     % root chord, mm
c_t  = 65.489;      % tip chord, mm
N_props = 6;

% Non-dimensional placement relative to local chord
x_frac = -0.25;     % 0.25c ahead of leading edge
z_frac = -0.18;     % 0.18c below wing plane
```

```
% Spanwise stations
i = (1:N_props).';
y_hub = (i./(N_props+1)) * b;

% Local chord at each y-position (linear taper)
c_y = c_r + (c_t - c_r) .* (y_hub / b);

% Leading edge assumed at x = 0 (unswept wing)
x_hub = x_frac .* c_y;
z_hub = z_frac .* c_y;

hub_positions = [x_hub, y_hub, z_hub];

disp('Propeller hub positions [x y z] in mm:');
disp(hub_positions);
```

13.6.7 Summary

The final propeller positioning strategy synthesizes aerodynamic theory, experimental observations, and geometric constraints into a unified placement framework. The approach integrates:

- aerodynamic slipstream theory governing jet-sheet turning and flap capture,
- experimental sensitivity trends derived from Section 5.1[41], particularly the negligible effect of moderate vertical displacement and the strong dependence on motor mount angle,
- wake-field behavior of blown flaps, highlighting the importance of proper jet alignment at mid-range flap deflections,
- geometric tapering of the wing, which ensures consistent non-dimensional placement across the span,
- and thrust-vectoring implications, which influence chordwise positioning and pitching-moment management.

Vertical and chordwise propeller locations were selected to ensure complete slipstream capture by the flap while minimizing adverse pitching moments. The spanwise placement strategy distributes the propellers evenly across the wing, preserving uniform lift augmentation and capturing the more compact jet-sheets characteristic of smaller propeller diameters (Section 5.1.3[41]). Together, these choices produce a physically grounded configuration suitable for high-fidelity BEMT modeling and subsequent genetic algorithm optimization.

Final Propeller Hub Coordinates

Table 13.4: Final propeller hub positions in the wing-fixed coordinate system (origin at root leading edge).

Propeller #	x_{hub} (mm)	y_{hub} (mm)	z_{hub} (mm)
1	-32.9468	107.1737	-23.7217
2	-30.1844	214.3474	-21.7328
3	-27.4220	321.5211	-19.7438
4	-24.6595	428.6949	-17.7549
5	-21.8971	535.8686	-15.7659
6	-19.1347	643.0423	-13.7770

These coordinates reflect the fully implemented placement methodology: a constant $0.25c$ upstream chordwise offset, a $0.18c$ downward vertical placement, and spanwise distribution proportional to the semi-span. The gradual reduction of x_{hub} and z_{hub} magnitudes toward the tip is a direct result of the tapering local chord length, preserving consistent non-dimensional alignment between the propeller axes and the blown flap. The resulting configuration ensures robust slipstream–wing interaction and forms the foundation for all subsequent aerodynamic analysis and optimization carried out in this study.

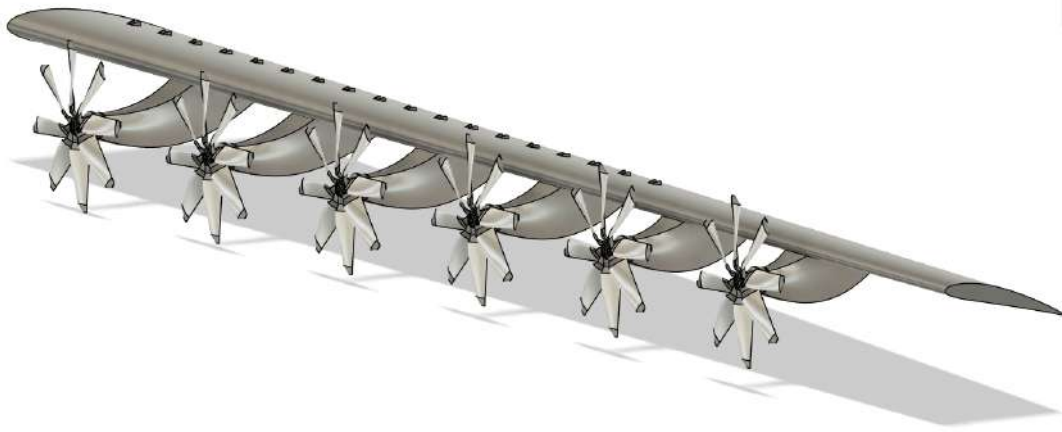


Figure 13.15: Wing half-span model with propellers

13.7 Propeller CFD Analysis

13.7.1 Geometry and Case Setup

The propeller with parameters optimized using GA was analyzed using the Actuator Disc Theory approximation.

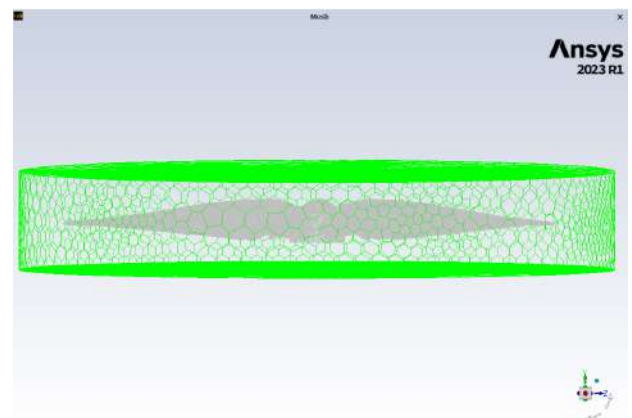
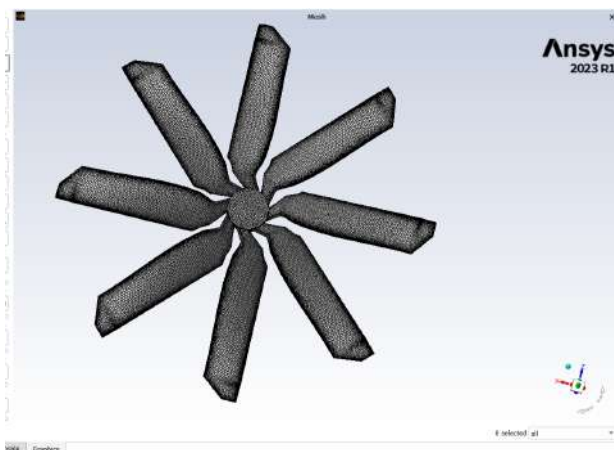


Figure 13.16: 3D Geometry for optimized propeller

13.7.2 Mesh Details

The mesh is generated in a manner similar to the previous study and consists of approximately 3.38M elements.

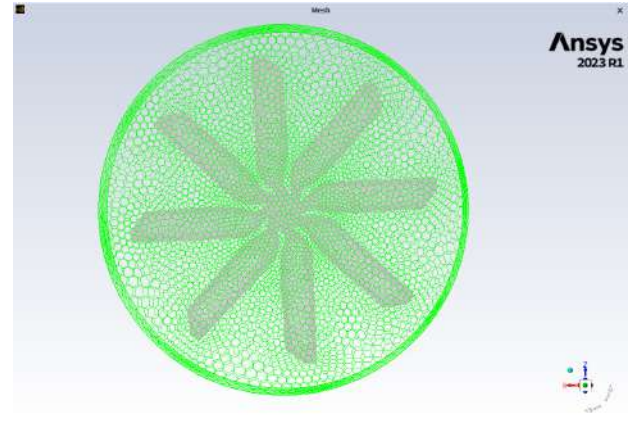
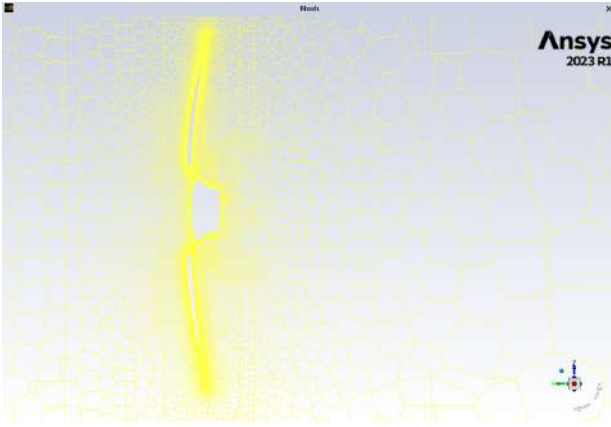
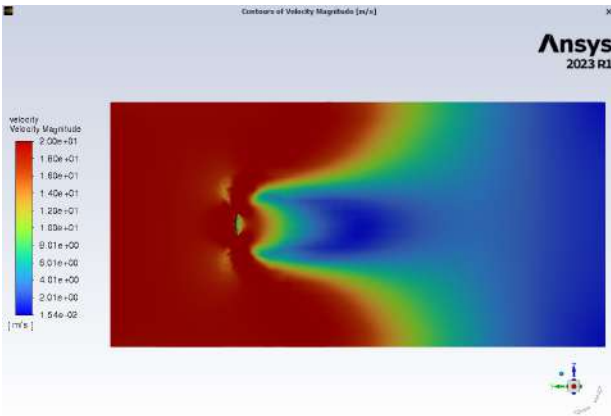


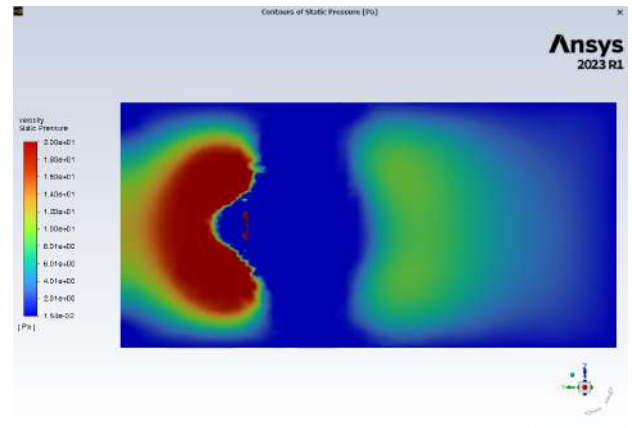
Figure 13.17: Meshing for Propeller and its domain

13.7.3 CFD Results

The CFD analysis converged the value of thrust to 5.8 N at 54,267 rpm, which displayed a deviation of 7.2% from the theoretical thrust.



(a) Propeller velocity profile



(b) Propeller pressure profile

Figure 13.18: CFD analysis on optimized propeller design

13.8 Comparison of Theoretical Thrust and CFD Predictions

The theoretical thrust capability of the GA-optimized propeller was first established from the maximum allowable tip Mach number for safe operation. The design constraint was set to a tip Mach limit of

$$M_{\text{tip,max}} = 0.75,$$

beyond which compressibility effects and associated losses were considered unacceptable. To incorporate a safety margin, a factor of safety of 0.9 was applied, giving a design tip Mach number

$$M_{\text{tip,des}} = 0.9 \times 0.75 = 0.675.$$

For a given free-stream speed V_∞ and propeller radius R , the blade-tip Mach number is evaluated from

$$M_{\text{tip}} = \frac{V_{\text{tip}}}{a} = \frac{\sqrt{V_\infty^2 + (\Omega R)^2}}{a},$$

where a is the local speed of sound and Ω is the rotational speed in rad/s. Solving this relation for Ω at $M_{\text{tip}} = M_{\text{tip,des}}$ yields the maximum allowable rotational speed. For the GA-optimized geometry, this corresponds to a maximum operating speed of 61,200 rpm.

At this operating point, the propeller thrust predicted by momentum theory and blade-element-momentum (BEM) analysis was cross-checked using an online performance tool (Ecalc), which gives a thrust of 712 g, i.e.

$$T_{\text{theory}} \approx 6.98 \text{ N}$$

for a single propeller. To assess finite-wing and three-dimensional effects, a high-fidelity CFD model of the isolated propeller was constructed and simulated over a range of rotational speeds around the design rpm. The CFD-predicted thrust at 54,267 rpm converged to

$$T_{\text{CFD}} = 5.8 \text{ N}$$

per propeller, which is lower than the theoretical value. The relative deficit is

$$\Delta_T = \frac{T_{\text{theory}} - T_{\text{CFD}}}{T_{\text{theory}}} \approx \frac{6.25 - 5.8}{6.25} \approx 0.072,$$

corresponding to an effective loss of about 7.2% with respect to the ideal BEM prediction. This offset was consistently observed during an iterative parametric sweep in rotational speed and is attributed primarily to three-dimensional aerodynamic phenomena that are not fully captured in the underlying two-dimensional blade-section polars, such as tip-vortex induced losses, non-uniform inflow, and additional viscous dissipation in the hub and root regions. Now to accommodate the T/W in operating envelop we used the maximum operating rpm ie. 61,200 leading to theoretical thrust of 6.98 N which incurs a loss of 7.2 % ie. $T_{\text{CFD}} = 6.47 \text{ N}$

The per-propeller CFD thrust value directly informs the global thrust–weight characteristics of the blown-wing configuration. With twelve identical propellers installed along the span, the total thrust at the design rpm is

$$T_{\text{total,CFD}} = 12 \times 6.47 \text{ N} = 77.64 \text{ N},$$

which corresponds to a thrust-to-weight ratio of approximately $T/W \approx 0.527$ for the current aircraft mass. This places the configuration within the intended operating envelope for take-off and climb while remaining consistent with the imposed tip-Mach constraint.

To embed the observed three-dimensional loss margin directly into the design process, the subsequent optimization and performance evaluations were conducted using a reduced target thrust of 6.25 N per propeller rather than the ideal 6.98 N. In practice, this means the propeller geometry and operating points are tuned such that the theoretical BEM prediction slightly overestimates the required thrust, and the CFD-resolved three-dimensional performance; with its inherent tip and hub losses; aligns with the desired 6.25 N level. This approach effectively internalizes the approximately 6% three-dimensional deficit into the design, yielding more realistic performance predictions and ensuring that the final propeller–wing system meets the thrust requirements when all three-dimensional effects are accounted for.

14 Proposed Wing & Propeller Design

The proposed wing design combines all elements that were thoroughly theorized and analyzed using CFD. The final design includes propellers to provide thrust and blown lift, high-lift devices such as flaps and slats, as well as flow control techniques such as vortex generators and fluidic oscillators.

The following tables present the details of the final wing design, with the flap and slat parameters corresponding to the maximum C_L condition. The vortex-generator table includes the geometric parameters h , L , D , d , b , t , and β , which are defined and described in the vortex-generators section:

Wing	
Planform Area (S)	0.1663 m ²
Span (b)	1.58 m
Taper ratio (λ)	0.45
Mean aerodynamic chord $\bar{c}$	0.11034 m
Root chord	0.14523 m
Tip chord	0.06535 m
Quarter-chord sweep $\Lambda_{c/4}$	0°
Airfoil	NACA 4412

Slats	
c_s/c	0.16
Δx	0.102c
Δy	0.0275c
Deflection (δ_s)	5°
Slat span	0.97b

Flaps	
c_f/c	0.256
Deflection (δ_f)	30°
Flap span	0.59b
Overhang	0.03c
Gap	0.02c

Vortex Generator	
Type	Trapezoidal Counter rotating
h	1.6 mm
L	4.8 mm
D	25.2 mm
d	5.6 mm
b	0.96 mm
t	0.4 mm
β	15°

The optimized propeller parameters were obtained using our in-house BEMT solver tightly coupled with a Genetic Algorithm (GA), which systematically explored combinations of blade number, diameter, chord and twist distributions, and operating RPM to satisfy a prescribed thrust target with minimum power. By enforcing constraints on tip Mach number, structural limits, and a nearly uniform blade angle of attack, the GA converged to an 8-bladed, 72.2 mm diameter propeller that exactly meets the 6.25 N thrust requirement at 20 m/s while producing a 53.751 m/s slipstream. This shows that most key geometric and operating parameters are not chosen ad hoc but emerge as truly optimized outputs of the BEMT–GA framework.

Table 14.1: Optimal BEMT-derived propeller parameters for the final blown-wing configuration.

Parameter	Value
Number of Blades	8
Diameter	0.0722 m (R = 0.0361 m)
Chord (Root)	0.0117 m
Chord (Tip)	0.0025 m
Optimized Blade AoA	3.63° (uniform)

Table 14.2: Spanwise pitch-angle distribution.

r/R	Pitch Angle [deg]
0.100	69.113
0.196	42.027
0.291	30.782
0.406	23.746
0.502	20.163
0.598	17.629
0.694	15.711
0.809	13.887
0.904	12.573
1.000	11.067



Figure 14.1: Propeller CAD with optimized parameters

The final wing mounted with propellers is shown in the figure below.



Figure 14.2: Proposed Full Wing Design

15 CFD results for the Wing configuration incorporating Thrust Devices

15.1 CFD for 3D Wing with Flap, Slat and Vortex Generator with Thrust Devices

15.1.1 Mesh Details

The mesh is generated in a manner similar to the previous study and consists of approximately 20M elements.

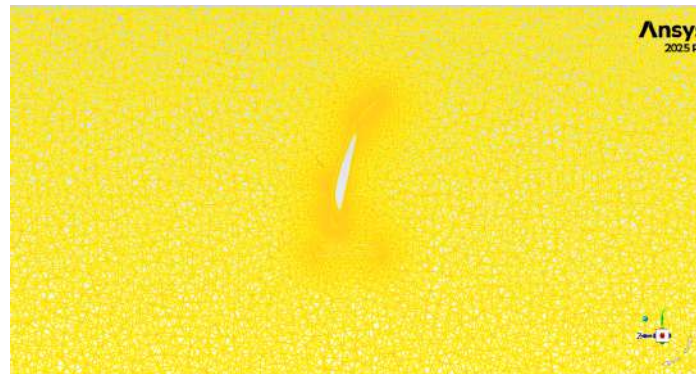
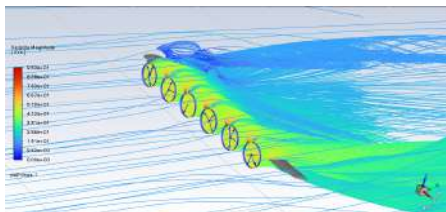


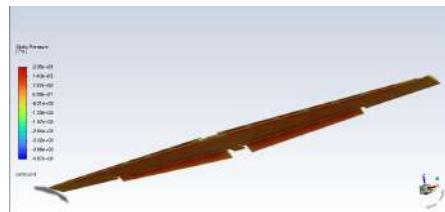
Figure 15.1: Mesh

15.1.2 CFD Results

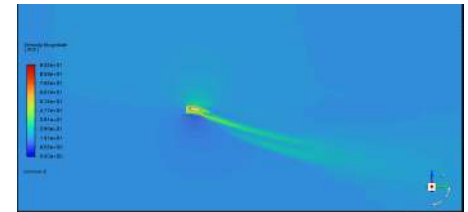
The following figures show the velocity and pressure contour plots for the wing configuration with integrated thrust devices.



Velocity Streamlines



Pressure Contour



Velocity Contour

Figure 15.2: CFD Results

CFD Results
$C_l = 5.78$
$C_d = 1.575$
$C_l/C_d = 3.669$

Table 15.1: Summary of CFD aerodynamic coefficients.

15.2 CFD for 3D Wing with Flap, Slat, Vortex Generator and Fluidic Oscillator with Thrust Devices

15.2.1 Mesh Details

The mesh is generated in a manner similar to the previous study and consists of approximately 20M elements.

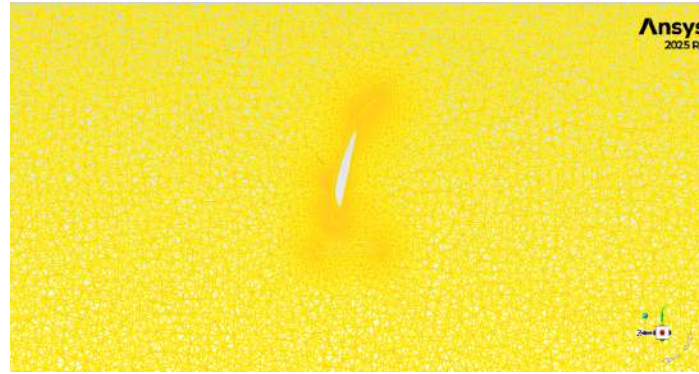
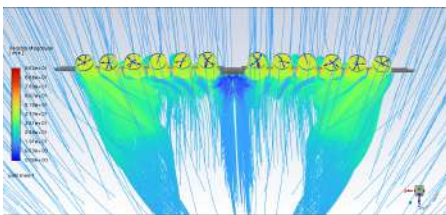


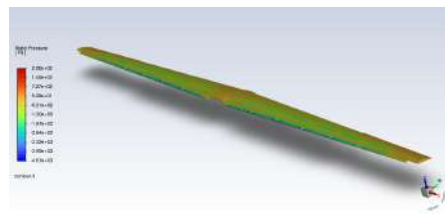
Figure 15.3: Mesh

15.2.2 CFD Results

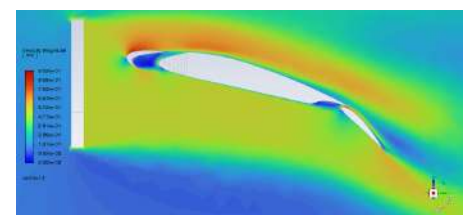
The following figures show the velocity and pressure contour plots for the wing configuration with integrated thrust devices.



Velocity Streamlines

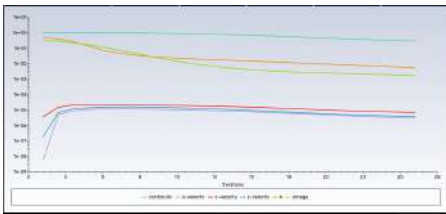


Pressure Contour

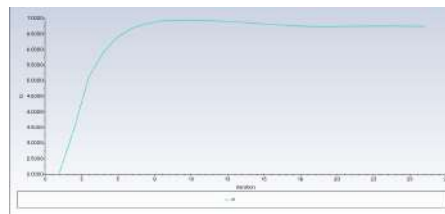


Velocity Contours

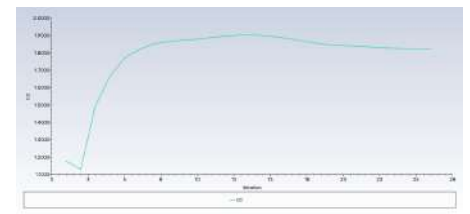
Figure 15.4: CFD Results



Scaled Residuals



Lift Coefficient



Drag Coefficient

Figure 15.5: Plots of Aerodynamic parameters

CFD Results
$C_l = 6.75$
$C_d = 1.825$
$C_l/C_d = 3.7$

Table 15.2: Summary of CFD aerodynamic coefficients.

16 Wind-tunnel Testing

Wind tunnel testing is the final step in the design process to validate predicted performance using mathematical models and CFD results with real-world data.

16.1 Wind-tunnel specifications

The wind tunnel used for the experiment has the following specifications:

- Type: Closed-circuit
- Dimensions: 600 mm x 900 mm
- Maximum tunnel velocity: 35 m/s
- Contraction ratio: 6.5
- Turbulence intensity: 0.3%

16.2 Model

Wind tunnel testing of the wing with slats and flaps is done to validate the CFD results. A pressure distribution approach is being employed for calculating the lift coefficient on the model.

16.2.1 Test Model Dimensions

The deployed positions of the slats and flaps are as finalized in previous sections. The model dimensions are as follows:

Parameter	Value
Wing span	600 mm
Wing chord	200 mm
Slat chord	32 mm
Flap chord	51.2 mm
Slat deflection	5°
Slat Δx	20.4 mm
Slat Δy	6 mm
Flap deflection	30°
Flap Overhang	6 mm
Flap Gap	3.87 mm
Angle of Attack	15°

Table 16.1: Test model parameters

16.2.2 Test Model Description

The 3D-printed model used is presented in Fig. 16.2. The model is divided into a total of 10 parts. The central part has a span of 100 mm and has pressure ports drilled of 0.5 mm diameter each. The pressure ports are mounted on the slats, the main airfoil, as well as the flaps in the middle of the test section. The left and right portions of the model have extending tabs and slot which are used to connect it to the central test section. There are also holes for adding spars that will hold the model together under aerodynamic load. The model contains ridges to hold steel tubes of diameter 0.8 mm. One end of these steel tubes is connected to the holes drilled in the model and the other end is connected to pressure tubes which are connected to the electronic pressure sensor. The rectangular slot is to pass the pressure tubes through to the pressure scanner.

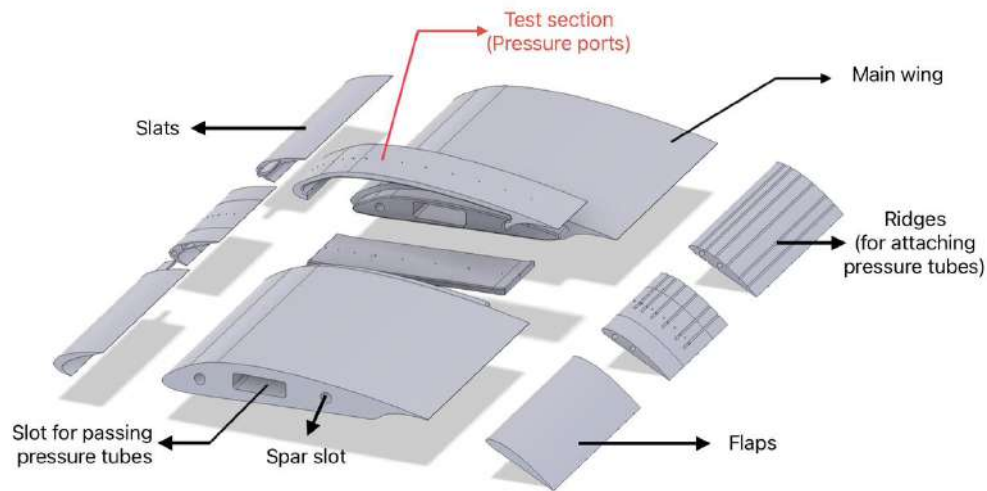


Figure 16.1: Test Model

16.3 Calculations

16.3.1 Reynolds Scaling

The wind tunnel experiment is being performed for takeoff conditions. At takeoff, for the wing with aspect ratio 15, the Reynolds number is given as

$$\begin{aligned} Re &= \frac{\rho U \bar{c}}{\mu} \\ &= \frac{1.225 \times 20 \times 0.11034}{1.789 \times 10^{-5}} \\ &= 151108.44 \end{aligned}$$

The model has a chord of 200 mm. The density and viscosity of the air is approximately the same as the experiment has been conducted in an environment with ambient temperature approximately 20°C. Therefore, the wind tunnel velocity is calculated as

$$\begin{aligned} Re_{tunnel} &= Re_{real} \\ U_{tunnel} &= \frac{20 \times 0.11034}{0.2} \\ &= 11.034 \text{ m/s} \end{aligned}$$

16.3.2 Blockage Area

The wind tunnel experiment is conducted around an angle of attack of 15° to determine the maximum lift coefficient. The blockage area can be approximately calculated using the effective chord length when the slats and flaps are in deployed.

$$\begin{aligned} c_{eff} &= 0.2(0.102)\cos(5^\circ) + 0.2(0.9 - 0.03) + 0.2(0.256)\cos(30^\circ) \\ &= 0.238663 \text{ m} \\ &= 238.663 \text{ mm} \end{aligned}$$

Therefore, the blockage of the test section is

$$\begin{aligned}
 \text{Blockage ratio} &= \frac{A_{\text{frontal}}}{A_{\text{section}}} \\
 &= \frac{S_{\text{eff}} \times \sin(15^\circ)}{A_{\text{section}}} \\
 &= \frac{238.663 \times 600 \times \sin(15^\circ)}{900 \times 600} \\
 &= 6.86\%
 \end{aligned}$$

A blockage ratio of 6.86% is within the acceptable range for wind tunnel testing.

16.4 Mounting

The 3D-printed wing is mounted at the center of the test section, and endplates are installed on both sides of the model to suppress wingtip vortices and approximate two-dimensional flow behaviour. This approach allows 2D aerodynamic characteristics to be evaluated without requiring a full wall-to-wall model.

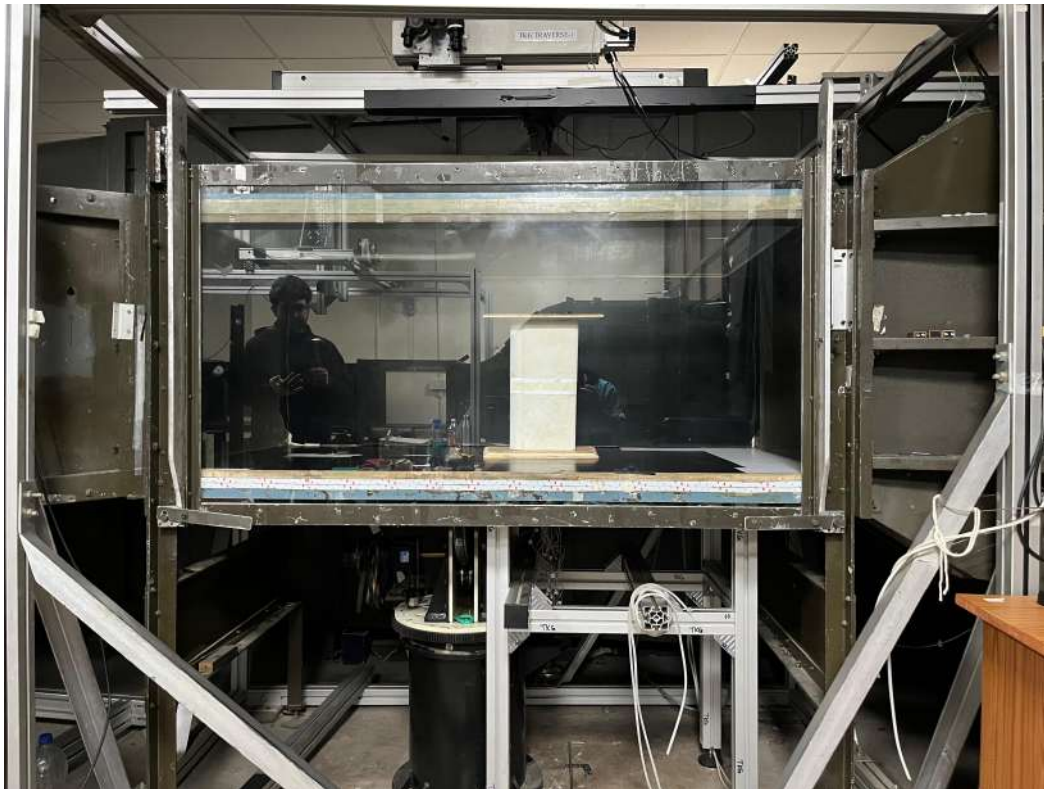


Figure 16.2: Wind tunnel experimental setup

16.5 Pressure Probes

For the wind tunnel experiment of the wing with flaps and slts, pressure ports were embedded along the chord of the main wing and connected to flexible pressure tubing, enabling the measurement of surface pressure distribution. The following figures show the fabricated test models with integrated pressure ports used during the experiment.

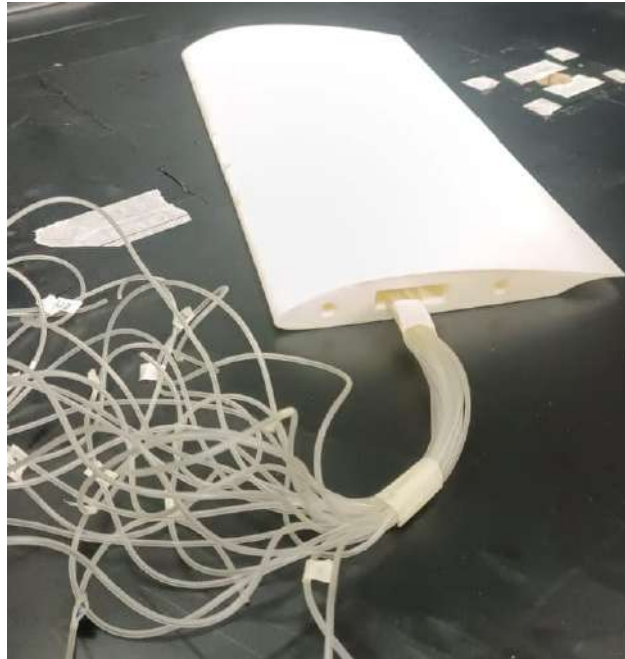


Figure 16.3: Main wing model with distributed pressure ports along the suction surface.

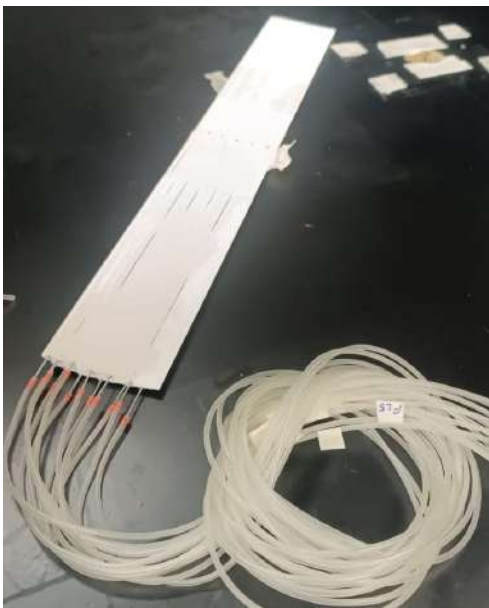


Figure 16.4: Pressure-tapped flap segment with multiple chordwise ports.

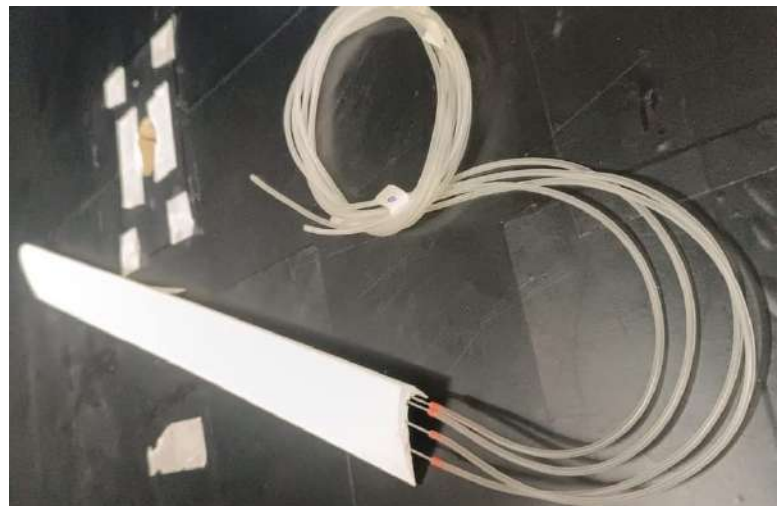


Figure 16.5: Slat model with leading-edge pressure ports for suction-side measurements.

16.6 Result Discussions

The original wing used in the CFD study is a tapered planform, whereas the wind tunnel experiments were carried out on a rectangular wing model whose chord length was chosen to match the mean aerodynamic chord (MAC) of the actual wing. The MAC was scaled appropriately to satisfy the wind tunnel blockage and Reynolds number requirements. Although this approach preserves the local aerodynamic behaviour around the slat and flap system, the change from a tapered wing to a rectangular planform introduces differences in lift production.

From the experimental measurements, the maximum lift coefficient obtained for the rectangular wing model with slat and flap deployed was

$$C_L = 2.63,$$

while the corresponding CFD prediction for the tapered full wing was

$$C_L = 3.09.$$

The lower value of C_L in the wind tunnel testing is due to unavailability of a 2D wall-to-wall setup capability. There are large losses due to existence of a vortices which exist despite the endplates. There is also losses because of lack of finishing in the 3D-printed model. Due to roughness on the surface there is loss in momentum and separation of boundary layer. All of these factors contribute to a loss of approximately 15% in our model. The C_D has not been calculated because the values would be incorrect due to manufacturing errors such as roughness of the 3D-printed model.

17 Appendix

This part pulls together the key formulas that go into analyzing and designing the wing from an aerodynamic standpoint. Those equations pull from the main fields involved, like flight mechanics, aircraft design, and aerodynamics. They laid down the math basics needed for figuring out the wing size, checking how it performs, and tweaking it to work better all along the way in the design.

17.1 Flight Mechanics

Formulas for aerodynamics coefficients:

$$C_L = \frac{L}{\frac{1}{2}\rho V^2 S}, \quad C_D = \frac{D}{\frac{1}{2}\rho V^2 S}, \quad C_m = \frac{M}{\frac{1}{2}\rho V^2 S \bar{c}}$$

The drag coefficient is further divided into zero-lift drag and lift-induced drag terms as

$$C_D = C_{D_0} + C_{D_i}$$

The lift-induced drag term is given as

$$C_{D_i} = \frac{C_L^2}{\pi e A R} \quad (17.1)$$

The next important aerodynamic coefficient is the lift-to-drag ratio which is given as

$$\frac{L}{D} = \frac{C_L}{C_D} = \frac{C_L}{C_{D_0} + \frac{C_L^2}{\pi e A R}}$$

The lift-to-drag ratio is maximized as

$$\left(\frac{C_L}{C_D}\right)_{max} = \frac{\sqrt{\pi e A R}}{2\sqrt{C_{D_0}}}, \quad C_L|_{(C_L/C_D)_{max}} = \sqrt{\pi e A R C_{D_0}} \quad (17.2)$$

At cruise, the aircraft is in steady level flight or trim condition, which is characterized by

$$L = W, \quad T = D$$

During takeoff, the takeoff velocity of the aircraft can be related to the stall velocity as

$$V_{TO} = 1.2 V_{stall} \quad (17.3)$$

The stall velocity is the velocity at which the aircraft is able to produce enough lift to support its own weight when the wings are producing maximum lift.

$$W = \frac{1}{2}\rho V_{stall}^2 S C_{L_{TO,max}} \quad (17.4)$$

At takeoff conditions, the maximum lift coefficient is given according to [24] as

$$C_{L_{TO}} = \frac{C_{L_{TO,max}}}{1.44}, \quad C_{L_{TO,max}} = 0.8 C_{L_{max}} \quad (17.5)$$

Pressure ratio for gradient region,

$$\frac{p}{p_0} = \left(\frac{T}{T_0}\right)^{-\frac{\gamma_0}{\lambda R}}$$

Density ratio for gradient region,

$$\frac{\rho}{\rho_0} = \frac{p}{p_0} \cdot \frac{T_0}{T} = \left(\frac{T}{T_0}\right)^{\left(-\frac{\gamma_0}{\lambda R} - 1\right)} \quad (17.6)$$

Pressure ratio for isothermal region,

$$\frac{p}{p_1} = \exp \left[-\frac{g_0}{RT_1}(h - h_1) \right]$$

$$\lambda = \frac{T - T_0}{h - h_0} = \frac{dT}{dh}$$

17.2 Aircraft Design

Aspect ratio of the wing is given by

$$AR = \frac{b^2}{S}$$

Taper ratio of the wing, λ , which relates the tip chord to the root chord, is expressed as:

$$\lambda = \frac{C_{\text{tip}}}{C_{\text{root}}}$$

The root and tip chord lengths are obtained using:

$$C_{\text{root}} = \frac{2S}{b(1 + \lambda)}$$

$$C_{\text{tip}} = \lambda C_{\text{root}}$$

Wing sweep angle is given by

$$\tan \Lambda_{LE} = \tan \Lambda_{C/4} + \frac{(1 - \lambda)}{AR(1 + \lambda)}$$

Wing Mean Aerodynamic Chord (MAC):

$$\bar{c} = \frac{2}{3} C_{\text{root}} \frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

The spanwise location of the MAC:

$$y_{\bar{c}} = \frac{b}{6} \cdot \frac{1 + 2\lambda}{1 + \lambda}$$

The quarter-chord point of the MAC,

$$x_{\bar{c}/4} = x_{LE} + (y_{\bar{c}} \cdot \tan \Lambda_{LE}) + \frac{\bar{c}}{4}$$

17.3 Prandtl's Lifting Line Theory

Prandtl's Lifting Line Theory came from Ludwig Prandtl back in the early nineteen hundreds. It sets up a solid math structure for guessing the aerodynamic traits of wings that have a finite span. The approach treats a three dimensional wing like a vortex line bound right along its span. That line gives off trailing vortices, and those build into a sheet trailing back downstream. In this setup, the model picks up on the core three dimensional influences. Those set finite wings apart from what you see in plain two dimensional airfoil work. It turns out to be pretty useful right at the beginning when you're starting on wing design.

17.3.1 Fundamental Assumptions

- Wing is represented as a bound vortex filament.
- A continuous trailing vortex sheet is shed downstream.
- Flow is incompressible and inviscid.
- Small angles and high aspect ratio.

17.3.2 Downwash and Induced Angle

The downwash induced at spanwise station y by the trailing vortex sheet is:

$$w_i(y) = \frac{1}{4\pi} \int_{-b/2}^{b/2} \frac{d\Gamma/dy'}{y - y'} dy'$$

The induced angle of attack is:

$$\alpha_i(y) = \frac{w_i(y)}{V_\infty}$$

17.3.3 Flow Tangency Condition

The effective angle of attack at the wing section is:

$$\alpha_{\text{eff}}(y) = \alpha - \alpha_{L=0}(y) - \alpha_i(y) \quad (17.7)$$

The sectional lift coefficient from thin airfoil theory is:

$$C_l(y) = a_0 \alpha_{\text{eff}}(y) \quad (17.8)$$

Relating to the circulation using Kutta–Joukowski:

$$L'(y) = \rho V_\infty \Gamma(y)$$

And also we have:

$$L'(y) = \frac{1}{2} \rho V_\infty^2 c(y) C_l(y)$$

17.3.4 Lifting-Line Integral Equation

Combining the above relationships, the standard LLT equation becomes:

$$\alpha - \alpha_{L=0}(y) = \frac{\Gamma(y)}{\pi V_\infty c(y)} + \frac{1}{4\pi V_\infty} \int_{-b/2}^{b/2} \frac{d\Gamma/dy'}{y - y'} dy'$$

This singular integral equation is conventionally solved using a Fourier sine series expansion.

17.3.5 Elliptic Lift Distribution

For an ideal elliptical lift distribution:

$$\Gamma(y) = \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2}$$

The induced drag coefficient for this ideal case is:

$$C_{D_i} = \frac{C_L^2}{\pi AR}$$

17.3.6 General Induced Drag Expression

For arbitrary circulation distributions:

$$C_{D_i} = \frac{C_L^2}{\pi AR e}$$

where the span efficiency factor is:

$$e = \frac{C_L^2}{\pi AR C_{D_i}} \quad (17.9)$$

17.3.7 Total Lift Coefficient

Using the circulation distribution, the total lift coefficient is:

$$C_L = \frac{2}{V_\infty S} \int_{-b/2}^{b/2} \Gamma(y) dy$$

17.3.8 Turbulence Modeling

The boundary layer thickness (δ) varies approximately as $\frac{\delta}{x} = \frac{5}{\sqrt{Re_x}}$ according to the Blasius flat-plate solution. For a calculated Reynold's number of the order of 2×10^5 under the takeoff conditions, we get $\delta \approx 2$ mm. At such small values of delta, resolving the boundary layer directly requires a large number of cells which is computationally expensive. Hence, we resort to the Reynolds Averaged Navier Stokes (RANS) equations to approximate the averaged flow quantities. The k- ω Shear Stress Transport (SST) model has been used for all these cases.

17.4 A Brief Introduction to Blade Element Momentum Theory (BEMT)

Blade Element Momentum Theory (BEMT) is a practical method for predicting the performance of propellers and rotors. It combines momentum theory with blade element analysis to estimate thrust, torque, and aerodynamic behavior efficiently.

17.4.1 Momentum Theory

It is also known as Actuator Disk Theory. This approach models a streamtube that stretches from the distant upstream area all the way to the distant downstream area around the propeller. Within this control volume, the conservation principles of mass, momentum, and energy are applied to analyze the flow behavior. The full analysis is shown below:

Consider the streamtube shown, in which ∞ , d, and e denote the far-upstream region, the actuator disk plane, and the far-downstream region, respectively.

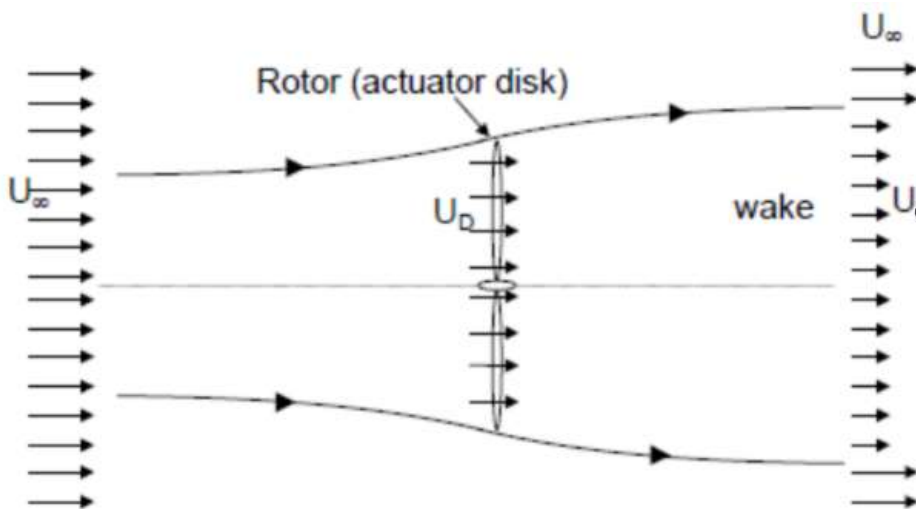


Figure 17.1: Schematic representation of the streamtube through an actuator disk.

(a) Mass Balance (Continuity Equation)

$$\dot{m} = \rho A_\infty u_\infty = \rho A_d V_d = \rho A_e u_e$$

which shows that if velocity increases, the area decreases — the streamtube contracts downstream.

(b) Momentum Balance (Thrust Equation)

Applying Newton's second law to the control volume enclosing the streamtube:

$$\dot{m}(u_e - u_\infty) = F_{\text{net}}$$

At far upstream and downstream, the pressure is ambient ($p_e = p_\infty$), so

$$F_{\text{net}} = (p_\infty A_\infty) - (p_e A_e) + T = T$$

The rate of change of momentum of the air is $(u_e - u_\infty)$. Thus, the thrust is given by:

$$T = \dot{m}(u_e - u_\infty)$$

Substitute $\dot{m} = \rho A_d V_d$, we get:

$$T = \rho A_d V_d (u_e - u_\infty) \quad (17.10)$$

(c) Energy Balance

The actuator disk causes a pressure jump from p_D^+ to p_D^- across it, while the velocity at the disk remains the same at V_d on both sides. The power added to the flow is:

$$P = TV_d = (p_D^+ - p_D^-) A_d V_d \quad (17.11)$$

This equals the rate of increase of kinetic energy between the far upstream and downstream sections:

$$(p_D^+ - p_D^-) A_d V_d = \frac{1}{2} \dot{m} (u_e^2 - u_\infty^2) \quad (17.12)$$

From the above relations 11.10, 11.11, and 11.12, we obtain:

$$V_d = \frac{u_e + u_\infty}{2}$$

i.e., the velocity through the disk is the average of the upstream and downstream velocities.

V_d and U_e are related to the free-stream axial velocity U_∞ using the axial induction factor (a) as:

$$V_d = U_\infty(1 + a) \quad \& \quad V_e = U_\infty(1 + 2a)$$

From this, we get

$$T = \frac{1}{2} \rho A (U_e + U_\infty) (U_e - U_\infty) = 2 \rho A U_\infty^2 a (1 + a)$$

$$P = TV_d = 2 \rho A U_\infty^2 a (1 + a)^2$$

17.4.2 Blade Element Theory (BET)

In this analysis, the blade is broken up into several segments and each segment is analyzed individually for thrust and power using differential relationships, treating each cross-section as an airfoil.

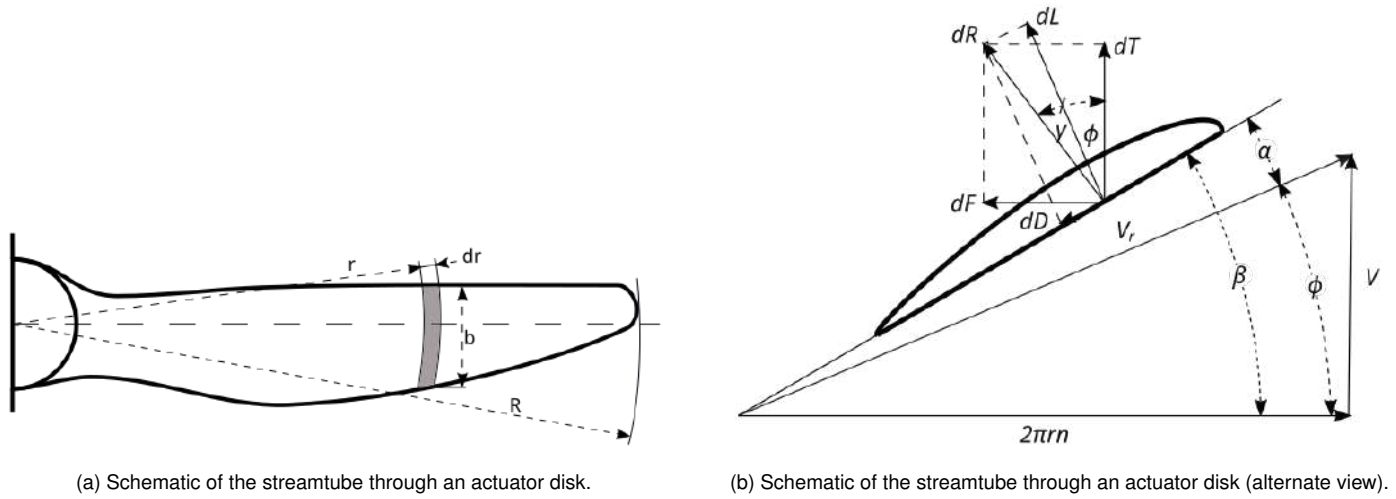
(a) Kinematic Definitions

Consider an annular blade element at radial distance r from the center, with width dr as shown:

- B : number of blades
- $c(r)$: local chord length
- $\beta(r)$: local twist angle
- ρ : air density
- Ω : rotational speed

Define the local velocity components:

$$V_d = V_x = V_\infty(1 + a), \quad V_\theta = \Omega r(1 - a')$$



(a) Schematic of the streamtube through an actuator disk.

(b) Schematic of the streamtube through an actuator disk (alternate view).

Figure 17.2: Comparison of two schematic representations of the streamtube through an actuator disk.

where a is the **axial induction factor**, and a' is the **tangential induction factor**.

The resultant velocity (W) and the induced inflow angle (ϕ) at the element are:

$$W = \sqrt{V_x^2 + V_\theta^2}, \quad \tan \phi = \frac{V_x}{V_\theta} = \frac{V_\infty(1+a)}{\Omega r(1-a')}$$

The local angle of attack ($\alpha(r)$) for the airfoil element can be calculated using the local twist angle ($\beta(r)$) as $\alpha = \beta - \phi$. Note that α itself is a function of r .

(b) Aerodynamic Forces on the Blade Element

Given that the variation of the aerodynamic characteristics of the airfoil with angle of attack is known beforehand, the lift and drag of each blade element can be determined as follows:

$$dL = \frac{1}{2} \rho W^2 C_L(\alpha) dA = \frac{1}{2} \rho W^2 C_L(\alpha) c(r) dr$$

$$dD = \frac{1}{2} \rho W^2 C_D(\alpha) dA = \frac{1}{2} \rho W^2 C_D(\alpha) c(r) dr$$

Resolve these along the normal (n) and tangential (t) directions:

$$dF_n = dL \cos \phi - dD \sin \phi = \frac{1}{2} \rho W^2 c(r) dr (C_L \cos \phi - C_D \sin \phi)$$

$$dF_t = dL \sin \phi + dD \cos \phi = \frac{1}{2} \rho W^2 c(r) dr (C_L \sin \phi + C_D \cos \phi)$$

This relation can be rewritten as:

$$dF_n = \frac{1}{2} \rho W^2 C_n(\alpha) c(r) dr \quad \& \quad dF_t = \frac{1}{2} \rho W^2 C_t(\alpha) c(r) dr$$

where

$$C_n = C_L \cos \phi - C_D \sin \phi \quad \& \quad C_t = C_L \sin \phi + C_D \cos \phi$$

The elemental thrust (dT) and torque (dQ) of each blade are calculated as:

$$dT = dF_n = \frac{1}{2} \rho W^2 C_n(\alpha) c(r) dr$$

$$dQ = r dF_t = \frac{1}{2} \rho W^2 C_t(\alpha) r c(r) dr$$

The total thrust (T) and torque (Q) are calculated by integrating over all the blade elements.

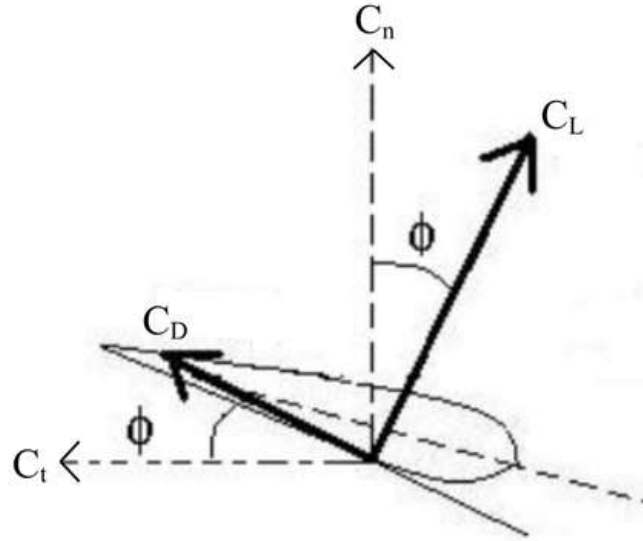


Figure 17.3: Normal and tangential components of lift and drag

$$T = B \int dT = B \int_0^R \frac{1}{2} \rho W^2 C_n(\alpha) c(r) dr$$

$$Q = B \int dQ = B \int_0^R \frac{1}{2} \rho W^2 C_t(\alpha) r c(r) dr$$

17.4.3 Blade Element Momentum Theory (BEMT)

Combines both the Momentum Theory and Blade Element Theory for a more accurate estimation of the thrust and torque.

(a) Momentum Relations:

$$dT_{MT} = 4\pi \rho V_\infty^2 a(1+a)r dr$$

$$dQ_{MT} = 4\pi \rho V_\infty(1+a)a'\Omega r^3 dr$$

(b) Blade Element Relations

$$dT_{BET} = \frac{1}{2} \rho W^2 c B C_n dr$$

$$dQ_{BET} = \frac{1}{2} \rho W^2 c B r C_t dr$$

(c) Equating Momentum and Blade-Element Relations

Setting $dT_{MT} = dT_{BET}$ and $dQ_{MT} = dQ_{BET}$, and substituting $Bc = 2\pi r\sigma$ (where σ is the *local solidity*), we obtain:

$$a = \frac{\sigma C_n}{4 \sin^2 \phi - \sigma C_n} \quad \& \quad a' = \frac{\sigma C_t}{4 \sin \phi \cos \phi + \sigma C_t}$$

we further have these relations,

$$\phi = \tan^{-1} \left(\frac{V_\infty(1+a)}{\Omega r(1-a')} \right) \quad \& \quad W = \frac{V_\infty(1+a)}{\sin \phi}$$

$$\alpha(r) = \beta(r) - \phi(r)$$

The equations are interdependent and therefore require an iterative solution approach. Common techniques used include Fixed-Point Iteration, Newton–Raphson, and Relaxation methods. In the present case, an under-relaxed fixed-point iteration scheme was used to obtain stable and convergent results with a convergence criterion set at 10^{-6} .

17.4.4 Prandtl's Tip Loss Model

Classical momentum theory and BEMT both assume that the flow around each blade section stays independent and axisymmetric. In real situations though, the limited number of blades leads to wake interactions. These interactions cut down the induced velocity close to the tips. That reduction then lowers circulation, lift, and thrust overall.

Ludwig Prandtl introduced a way to handle this back in 1919. He came up with a tip loss factor known as F . This factor changes the induced velocity within the momentum equations. The main point is to adjust for the drop in aerodynamic loading and circulation right near the blade tips. Vortex shedding causes that drop in the first place.

(a) Derivation Basis

Prandtl pictured the propeller wake in a specific manner. He saw it as helical vortices trailing off from every blade tip. Those vortices create a continuous sheet wrapping around the wake cone. Consider a propeller with B blades arranged at intervals of $(2\pi/B)$. The gaps between these helical filaments are finite. As a result, the induced velocity drops at any radius r . An exponential term captures that drop. The term relies on the angular spacing between nearby vortices.

$$\exp\left(-\frac{B}{2} \frac{R-r}{r \sin \phi}\right)$$

The inverse cosine of this exponential decay gives another useful measure. It shows the effective portion of the annular streamtube that truly generates lift. In the end, this setup defines the final tip loss factor.

$$F = \frac{2}{\pi} \cos^{-1} \left[\exp\left(-\frac{B}{2} \frac{R-r}{r \sin \phi}\right) \right] \quad \text{with } 0 \leq F < 1$$

(b) Physical Interpretation

Observe that $F \rightarrow 1$ near the mid-span region ($r \ll R$), implying negligible tip loss and $F \rightarrow 0$ as $r \rightarrow R$, signifying complete circulation decay at the blade tip. The correction factor F is then incorporated into both the axial and tangential momentum equations through the induction factors, reducing the effective induced velocities and thrust at the outermost blade sections:

$$a = \frac{1}{\left[\frac{4F \sin^2 \phi}{\sigma C_N} + 1 \right]}, \quad \& \quad a_t = \frac{1}{\left[\frac{4F \sin \phi \cos \phi}{\sigma C_T} - 1 \right]}$$

This ensures a realistic reduction of aerodynamic forces near the blade tips. Without this correction, BEMT would predict unreasonably high loading near the blade tips, especially for propellers with few blades.

17.5 Optimization Algorithm (GA)

Genetic Algorithms work as optimization tools that draw ideas from natural selection and the basics of genetics. They do not just chase after the single best option right away. Instead, they develop a whole group of possible solutions across multiple generations. This happens through processes like selection, crossover, and mutation. That method turns out to be really useful for handling complicated problems. These include nonlinear and multidimensional ones where older optimization approaches often fall short.

People apply GAs in various fields pretty much everywhere. For instance, they help with improving aerodynamic designs. They also aid in scheduling tasks, advancing machine learning, managing control systems, and even optimizing structures. The overall process of GA can be summarized as:

1. Initialize a population of candidate solutions.
2. Evaluate their fitness using an objective function.
3. Select the fittest individuals for reproduction.
4. Apply crossover and mutation to generate new individuals.
5. Repeat the process for multiple generations until convergence.

17.5.1 Mathematical Formulation and Explanation

A Genetic Algorithm maintains a population of N individuals, represented as possible solutions:

$$P(t) = \{x_1, x_2, \dots, x_N\}$$

where $P(t)$ is the population at generation t and each x_i is a potential solution.

(a) Initialization

A population of N individuals is randomly generated within the defined search space. Each individual represents a possible solution to the problem.

$$x_i^{(0)} = x_{\min} + r_i \cdot (x_{\max} - x_{\min}), \quad i = 1, 2, \dots, N$$

where r_i is a random number in $[0, 1]$, and $x_{\min}$ and $x_{\max}$ are the lower and upper bounds of the variables. *This step ensures a diverse starting population for better exploration of the solution space.*

(b) Fitness Evaluation

Each chromosome is evaluated using a fitness function that measures how good a particular solution is.

$$F_i = f(x_i)$$

where F_i is the fitness value and $f(x_i)$ is the objective function. *This function determines which individuals are more suitable to reproduce.*

(c) Selection

The selection mechanism chooses individuals for reproduction based on their fitness values. A common approach is the Roulette Wheel selection method.

$$P_i = \frac{F_i}{\sum_{j=1}^N F_j}$$

where P_i is the probability of selecting the i^{th} individual. *This gives higher chances to individuals with higher fitness values.*

(d) Crossover (Recombination)

Crossover mixes the genetic information of two parents to create offspring.

For a single-point crossover:

$$\text{Offspring}_1 = [x_{p1}(1 : k), x_{p2}(k + 1 : n)]$$

$$\text{Offspring}_2 = [x_{p2}(1 : k), x_{p1}(k + 1 : n)]$$

where x_{p1} and x_{p2} are parent chromosomes, k is the crossover point, and n is the chromosome length. *This step promotes exploration of new regions in the search space.*

(e) Mutation

Mutation introduces random alterations in offspring genes to preserve population diversity and prevent premature convergence.

$$x'_i = \begin{cases} 1 - x_i, & \text{if random}(0, 1) < P_m \\ x_i, & \text{otherwise} \end{cases}$$

where P_m is the mutation probability. *This operation helps escape local optima by reintroducing lost genetic material.*

(f) Termination

The algorithm stops when either a desired fitness level or the maximum number of generations is reached.

$$F_{\text{best}}^{(t)} \geq F_{\text{target}} \quad \text{or} \quad t \geq G_{\text{max}}$$

where $F_{\text{best}}^{(t)}$ is the best fitness at generation t , F_{target} is the desired fitness, and G_{max} is the maximum number of generations. *This ensures controlled execution and determines when an optimal or near-optimal solution has been achieved.*

(g) Convergence Criterion

The algorithm typically stops when:

$$|f_{\text{best}}(t) - f_{\text{best}}(t - 1)| < \epsilon$$

where ϵ is a small tolerance value, indicating that further generations do not yield significant improvement.

17.5.2 Advantages and Disadvantages

Advantages	Disadvantages
1. Can handle complex, nonlinear, and multi-modal objective functions.	1. Requires many fitness function evaluations, making it computationally expensive.
2. Works with discrete, continuous, and mixed variables without derivative information.	2. Convergence can be slow and uncertain without parameter tuning.
3. Effective in exploring a large search space and avoiding local minima.	3. Choice of parameters (population size, crossover, mutation rates) strongly affects performance.
4. Can be easily hybridized with other optimization methods.	4. No guarantee of finding the global optimum in every run.

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